



Tabular Foundation Models and Synthetic Data for Clinical Prediction

Programme

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Abstract

Background. Pancreatic neuroendocrine tumours are rare, and liver metastasis is associated with more complex diagnosis-time risk assessment and prognosis. This study examined whether published diagnostic and prognostic workflows could be reconstructed using an available Surveillance, Epidemiology, and End Results (SEER) cohort. It also evaluated whether tabular foundation models, Tabular Prior-Data Fitted Network (TabPFN) representations, multilayer perceptrons (MLPs) or synthetic-data augmentation improved prediction.

Methods. The November 2023 SEER release yielded 2,672 eligible diagnostic records, including 462 patients with liver metastasis at diagnosis; those 462 patients formed the prognostic cohort. The reference workflows were reconstructed using the available variables, and the resulting models were applied without modification to an independent University Hospital Erlangen cohort of 30 patients, including 11 with liver metastasis. Supervised TabPFN embeddings and complete synthetic cohorts containing predictors and outcomes were evaluated for predictive information, structural similarity, category support and preservation of clinical relationships. Conventional models, tabular foundation models and prespecified MLPs were compared using the same repeated SEER training and test partitions. Synthetic augmentation added 1,000, 5,000, 10,000 or 15,000 generated records to each real training partition, while performance was measured only in held-out real patients.

Results. The reference workflows recovered strong diagnostic discrimination and meaningful prognostic risk ordering, although the unavailable original SEER release prevented exact replication. In the independent Erlangen cohort, the primary diagnostic nomogram achieved an area under the receiver operating characteristic curve (ROC AUC) of 0.934 (95% confidence interval 0.809–1.000; $n = 29$), and the primary prognostic nomogram achieved a Harrell concordance index of 0.833 (0.625–1.000; $n = 11$). Both estimates were imprecise because of the small sample. In the repeated SEER evaluation, TabPFN-3, TabICLv2 and TabDPT achieved diagnostic ROC AUCs of 0.902–0.903, with improvements of 0.0074–0.0088 over logistic regression on the same patients. The clinical MLP showed no general advantage. No foundation model or MLP showed a reliable prognostic concordance gain. Random survival forest had the lowest integrated Brier score (0.153), improving on Cox elastic net by 0.0071. At the largest augmentation size, diagnostic XGBoost improved ROC AUC by 0.0126 and reduced Brier score by 0.0060; other diagnostic models did not improve consistently, and prognostic augmentation provided no benefit. Embeddings retained predictive information and local-neighbourhood structure, whereas the synthetic cohorts lost grade-4 support and shifted survival and several relationships between variables.

Conclusions. The benefits of foundation models and synthetic data were specific to the task, model and evaluation design. The strongest evidence supports small internal diagnostic gains and a selective augmentation benefit for XGBoost, not a general prognostic or clinical advantage. The external results are encouraging for the reproduced reference models but remain too small to establish subgroup safety or clinical usefulness. Larger independent validation is required before any model can support clinical decisions.

Keywords: pancreatic neuroendocrine tumour; SEER; tabular foundation models; TabPFN; synthetic data; survival analysis; clinical prediction

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1 Introduction

1.1 Clinical setting and study premise

Pancreatic neuroendocrine tumours (pNETs) are uncommon neoplasms with heterogeneous clinical behaviour. Liver metastasis is particularly relevant because it marks advanced disease and is associated with substantially different management and prognosis. This report addresses two related but distinct clinical prediction tasks in registry data. The *diagnostic task* estimates whether liver metastasis is recorded at the time of the initial cancer diagnosis. The *prognostic task* estimates overall survival among the subgroup of patients who already have liver metastasis at diagnosis. The diagnostic outcome therefore does not represent the later development of metastasis, and the prognostic cohort does not represent all patients with pNET.

The starting point is the multicentre study of Li et al. (2023), which developed diagnostic and prognostic nomograms from Surveillance, Epidemiology, and End Results (SEER) data and evaluated them in additional hospital cohorts. SEER is a population-based cancer surveillance programme maintained by the US National Cancer Institute (National Cancer Institute, n.d.a). Reconstructing the published workflows on the available SEER extract establishes clinically interpretable reference analyses, exposes differences between the published description and an executable pipeline, and provides a common baseline for the subsequent experiments. Reproduction in this report means reproducing the analytical logic as closely as the available variables and methodological description allow; it does not imply access to the original patient sample or the external validation cohorts used by the reference study. A separate 30-patient hospital cohort from Universitätsklinikum Erlangen (University Hospital Erlangen, hereafter UK–Erlangen) was subsequently made available. It is used to evaluate, without refitting, the diagnostic and prognostic models trained during the SEER-based reproduction of the reference study. This cohort is independent of the reference study's hospital cohorts and is not presented as their replication.

1.2 Scientific motivation

Modern tabular foundation models offer more than an additional prediction algorithm. A pretrained model may be used directly for in-context prediction, as a representation model that maps patients into a dense embedding space, or as a generator of synthetic tabular records. TabPFN provides a prominent example of this paradigm for small tabular datasets (Hollmann et al., 2025). Each role must be assessed differently. When the model is used for prediction, it must generalise to real patients who were not used for fitting. When it is used to create embeddings, the resulting representation should retain information that is useful for the clinical task. When it is used to generate synthetic records, the generated cohort should reproduce relevant clinical structure, but it must not be treated as independent patient evidence.

Consequently, a higher score from one model is not sufficient to establish that a transformation is faithful or that synthetic augmentation is useful. The evaluation must also consider probability error, calibration, dataset similarity, subgroup support and the conditions under which uncertainty was estimated. Classical statistical and machine-learning models remain essential comparators because they distinguish gains associated with a foundation model from gains attributable to the evaluation protocol, model capacity or additional training records.

The central research question is therefore:

> *How reliably can diagnostic and prognostic analyses for liver-metastatic pNET be reproduced from the available SEER data, and what is gained or lost when the real clinical variables are replaced or supplemented by TabPFN-derived representations, synthetic records, MLP models and other tabular foundation models?*

1.3 Aim and objectives

The overall aim is to evaluate tabular foundation-model methods as predictors, representations and data generators within a traceable clinical-registry workflow. The work is organised around nine objectives:

O1. SEER data preparation. Document how the data were downloaded using SEER*Stat and how the cohort eligibility criteria, outcome definitions and preprocessing steps were based on the reference study and adapted where required to construct the diagnostic and prognostic datasets.

O2. Reference-study reproduction. Reconstruct the diagnostic and prognostic workflows of Li et al. (2023) using the November 2023 SEER extract after preprocessing it according to the reference-study methodology. Record all necessary methodological adaptations and the limits of comparison because this extract is not the November 2019 SEER dataset used by the reference study, which was no longer available through the project's SEER*Stat access.

O3. External validation of the reference models. Apply, without refitting or recalibration, the diagnostic and prognostic models trained during the SEER-based reference-study reproduction to the independently supplied UK–Erlangen hospital cohort, and quantify discrimination, probability error, calibration and survival separation.

O4. TabPFN-derived datasets. Use TabPFN to create dense representations of the observed patients and to learn the joint distribution of the clinical variables and outcomes, from which complete synthetic patient records can be sampled.

O5. Reproduction on transformed data. Apply the diagnostic and prognostic analysis pipelines to the embedding and complete synthetic datasets and compare these results with the real-data reproduction under their respective evaluation designs.

O6. Similarity and synthetic-data fidelity. Quantify the extent to which the transformations preserve marginal distributions, relationships among variables, local geometry, clinically important categories, overall and subgroup survival patterns, and downstream model performance.

O7. Multi-model benchmarking. Compare tabular foundation models, a survival-pretrained foundation model and classical baselines on matched real-patient evaluation units using task-appropriate discrimination, probability-error and calibration measures.

O8. Nonlinear prediction with MLPs. Assess whether multilayer-perceptron (MLP) models applied to the clinical variables improve upon the performance achieved by the initial

reconstructed reference-study models. Use matched linear, tree-based and foundation-model results as additional context, and explore MLP models applied to the saved TabPFN embeddings separately.

O9. Synthetic augmentation. For each evaluation split, use only the real training patients to generate a synthetic pool. Add 1,000, 5,000, 10,000 or 15,000 of these synthetic records to the same real training set, fit each model on the resulting combined dataset and evaluate it only on the held-out real patients. Compare every augmented condition with the same model trained on the real training set alone.

1.4 Analytical structure

Table 1 summarises how the experimental layers answer the central question. Each layer examines a different aspect of the project, so its findings must be interpreted according to its own purpose and evaluation design. Transformation fidelity, predictive accuracy and augmentation utility are therefore assessed separately.

Table 1: Analytical structure of the project. "Real test" denotes evaluation on observed patients who were excluded from fitting and, for augmentation, excluded from synthetic-data generation.

Analytical layer	Principal comparison	Purpose
Data processing	Raw SEER extract to task-specific diagnostic and prognostic cohorts	Establishes the eligible populations, variables, outcomes and attrition.
Reference reproduction	Reconstructed nomogram workflows on real clinical variables	Provides interpretable clinical baselines and documents reproducibility limits.
External validation	SEER-trained reference models applied unchanged to independent UK–Erlangen patients	Tests whether the reference models retain predictive value in an independent cohort, while making the limitations of the small sample clear.
TabPFN transformations	Original clinical data compared with TabPFN embeddings and synthetic records containing clinical variables and outcomes	Tests whether predictive and structural information survives transformation.
Foundation-model benchmark	Tabular foundation models and classical baselines trained and tested on the same SEER data splits	Compares model families using the same patient groups and evaluation measures.
MLP models	MLPs compared with the initial reference models and with additional models trained and tested on the same SEER data splits	Tests whether additional nonlinear capacity improves the previously achieved performance.
Synthetic augmentation	Real-only training versus real plus 1K, 5K, 10K or 15K synthetic rows	Measures whether training-only generated records improve performance on held-out real patients.

1.5 Scope and principles of interpretation

This study evaluates how well models predict outcomes from previously recorded SEER data. It does not determine why an outcome occurred or whether a treatment caused benefit or harm. Treatment information in the registry may be incomplete, and some treatments may not have been known at the time when a prediction would need to be made. The results

should therefore be interpreted as an evaluation of predictive performance, not as evidence for treatment decisions or causal relationships.

Most comparative experiments use internal SEER validation. The reference study's hospital cohorts were unavailable, but an independently supplied UK–Erlangen cohort of 30 patients permits external validation of the conventional diagnostic and prognostic models trained during the SEER-based reproduction. Only 11 patients enter the prognostic analysis. This small evaluation supports a preliminary assessment of transportability, not precise subgroup inference, deployment readiness or patient benefit. The foundation- model, representation, MLP and augmentation experiments remain internally evaluated and are not externally validated by this cohort.

The uncertainty estimates are interpreted according to what was resampled. Intervals obtained by resampling predictions from fixed fitted models quantify test-patient sampling variability conditional on those models. They do not include the uncertainty that would arise from repeating model selection, training or synthetic-data generation. Repeated folds and overlapping test sets are likewise not treated as independent studies.

The project uses synthetic records in two ways. First, complete synthetic datasets containing both clinical variables and outcomes are compared with the original SEER data to examine which distributions and clinical relationships are preserved. These generated records are not independent patients and therefore cannot be used as external validation data. Second, the augmentation experiments generate synthetic records using only the real patients assigned to the training set in each evaluation split. The generated records are added only to that training set, and model performance is measured on the held-out real patients. Important limitations remain: grade 4 is absent from the synthetic data, and a small proportion of generated records contain a tumour-size category that does not agree with the generated tumour size in millimetres. Analyses based on the saved embeddings are considered exploratory because those embeddings were created from the full cohort before the later training and test splits were defined.

1.6 Organisation of the report

The report follows the analytical sequence in Table 1. It first documents SEER data acquisition, cohort construction and preprocessing, then presents the reference diagnostic and prognostic reproductions and their evaluation in the independent UK–Erlangen cohort. Subsequent sections describe the TabPFN transformations, reproduction on transformed data, dataset-similarity and synthetic-fidelity analyses, and a comparison of tabular foundation models trained and evaluated on the same SEER patient splits with the same performance measures. The final experimental sections assess nonlinear MLP models and synthetic augmentation in which generated records are added only to the training data and performance is evaluated on held-out real patients. An integrated discussion then compares the findings across experiments, distinguishes supported results from exploratory or unsupported ones, and closes with study limitations and conclusions. Readers who need further technical detail can consult the additional metric definitions, source records, code, tables and figures stored with the project files.

2 SEER Data Source, Cohort Construction and Preprocessing

2.1 Registry extract and relationship to the reference cohort

The source was a SEER*Stat Case Listing export from *Incidence—SEER Research Data, 17 Registries, Nov 2023 Sub (2000–2021)*, released in April 2024. The selected database is documented in the saved SEER*Stat session screenshots and in the NCI submission documentation (National Cancer Institute, 2024). The export used diagnosis years 2010–2016. These selection years must be distinguished from the full calendar-year coverage of the database.

The reference study used the November 2019 submission with 18 registries and the same diagnosis-year interval (Li et al., 2023). Consequently, the present extract is a reconstruction of the reference eligibility framework using a different registry release, not a recovery of the original patient-level dataset. Table 2 summarises the distinction. Differences in registry composition, record updates, follow-up and implementation of exclusions may affect the resulting cohort; their separate contributions have not been quantified here.

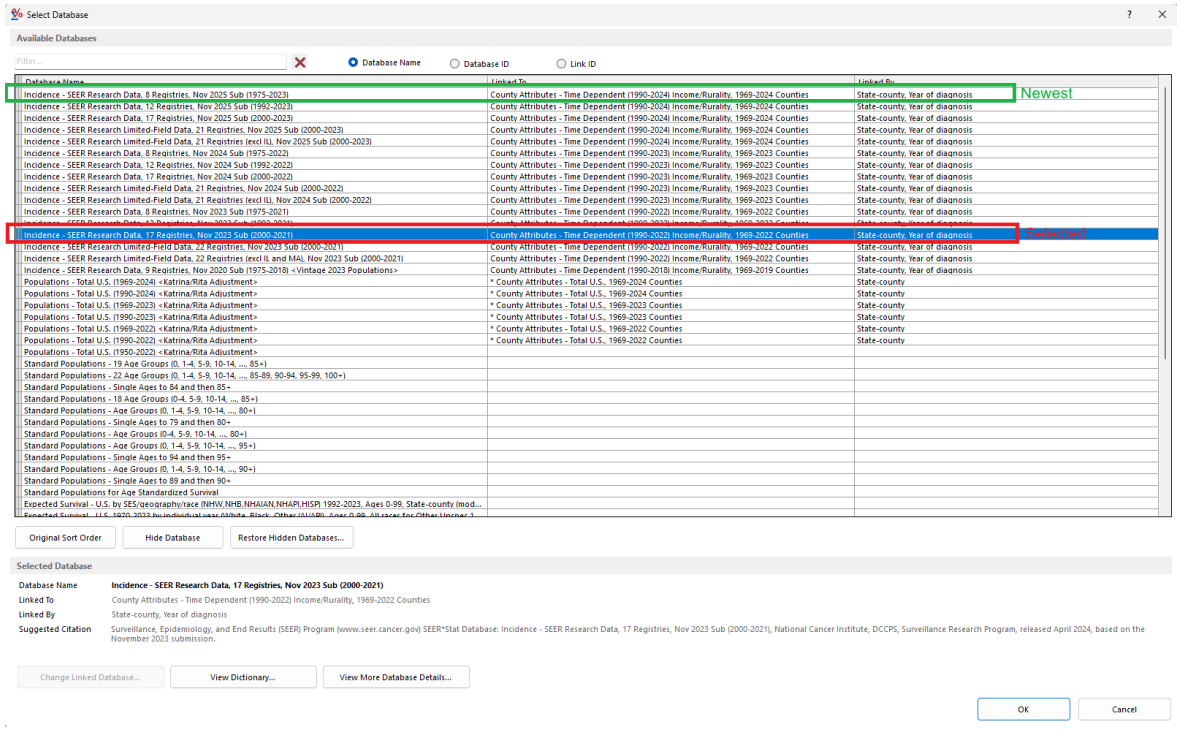
Table 2: Reference-study and project data sources. Reference counts are reported by Li et al. (2023); project counts are from the November 2023 SEER extract after applying the selection and preprocessing steps adapted from the reference-study methodology.

	Reference study	This project
SEER submission	November 2019	November 2023
Registry set	18 registries	17 registries
Database years	2000–2017	2000–2021
Selected diagnoses	2010–2016	2010–2016
Analytic cohort	1,998	2,672
Liver metastasis at diagnosis	343 (17.2%)	462 (17.3%)
External hospital data	Included in the reference study	UK–Erlangen cohort (30 patients)

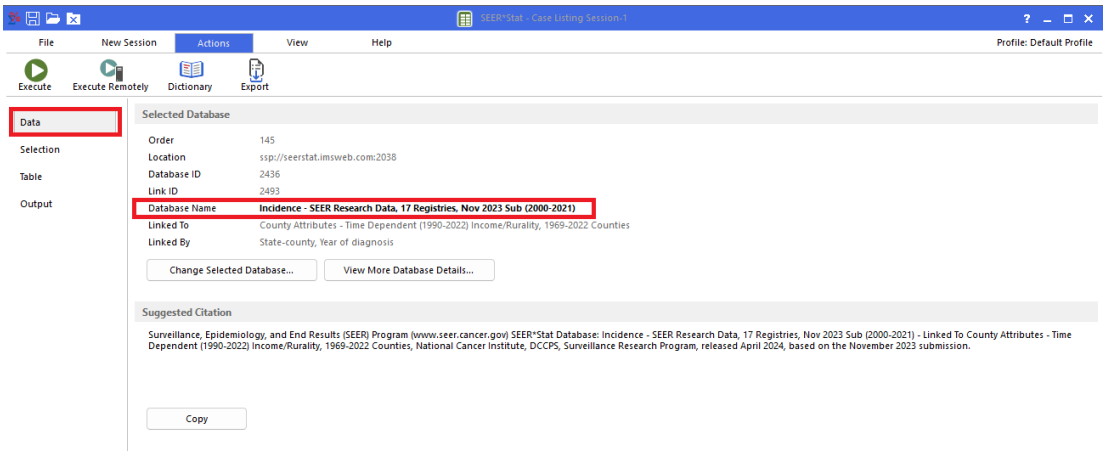
The similar metastasis proportions are descriptive, not a validation criterion: the pipeline was not adjusted to force agreement with the paper's sample size or prevalence. The larger retained sample also cannot, on its own, establish broader population coverage or improved representativeness.

2.2 SEER*Stat Case Listing session and raw export

The registry records were extracted with SEER*Stat software (https://seer.cancer.gov/seerstat/), version 9.0.43.0, using a Case Listing session. Figures 1 and 2 show the selected November 2023 database, the eligibility filters and the 30 fields included in the raw export. Together, these settings explain how the source dataset was obtained before Python preprocessing began. The database-selection screen also shows an optional link to county-level attributes, but none of those attributes was included among the exported fields or used in the analyses.

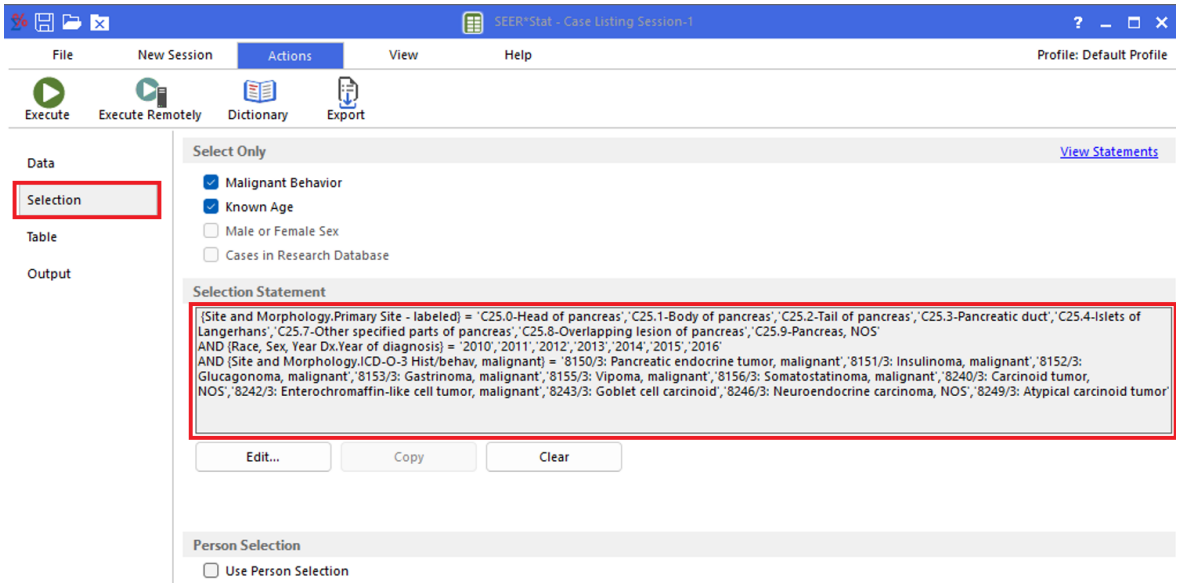


(a) Database-selection dialog. The blue row and red annotation identify the November 2023 SEER Research Data, 17 Registries submission used in this project. The green annotation identifies the newest release visible when the capture was made, not the database used for the analysis.

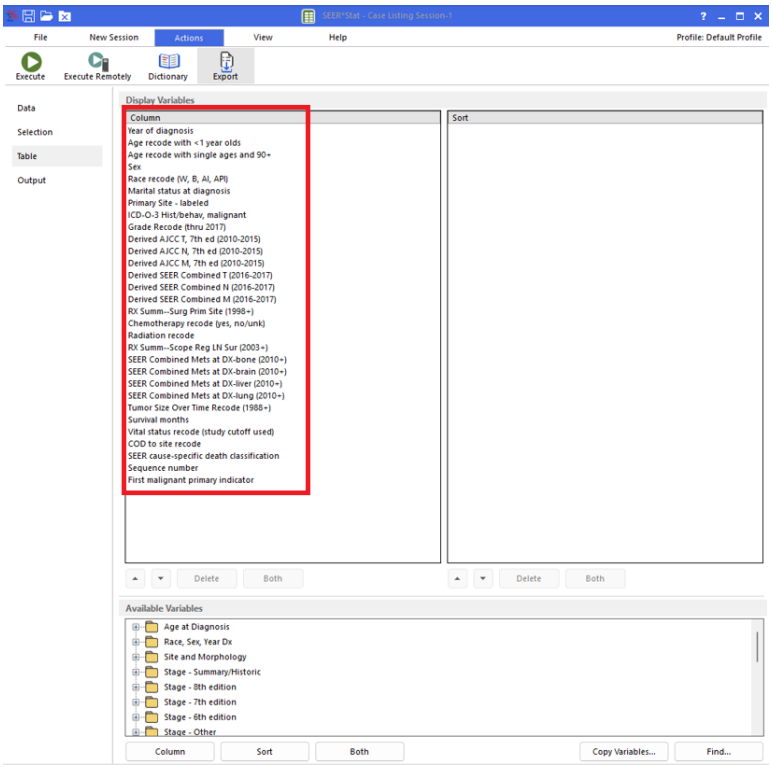


(b) Case Listing Data tab confirming the November 2023 database selected for the export.

Figure 1: SEER*Stat database selection for this project. The November 2023 submission was used for the present analysis; the reference study used the November 2019 submission.



(a) Selection tab showing malignant behaviour, known age, pancreatic primary-site codes, the 2010–2016 diagnosis interval and the accepted ICD-O-3 histology/behaviour codes.



(b) Table tab listing the 30 fields selected for the Case Listing export.

Figure 2: SEER*Stat Case Listing filters and exported fields used to create the raw dataset before Python preprocessing.

Executing the session produced `SEER_Dataset.csv`, a UTF-8 CSV with a byte-order mark. It contains 5,761 records and 30 fields spanning diagnosis year, demographics, primary site, histology, differentiation, TNM sources, treatment recodes, organ-specific metastases, tumour size, survival and eligibility indicators. The preprocessing loader explicitly uses `utf-8-sig` and disables pandas' default missing-value vocabulary so that the literal value "None" in the lymph-node-surgery field remains a valid source category rather than being converted to a missing value.

2.3 Extraction criteria and record-level eligibility

The saved SEER*Stat selection restricts the extract to malignant behaviour, known age, the specified pancreatic sites and histology codes, and the 2010–2016 diagnosis interval. Accepted topography codes are C25.0, C25.1, C25.2, C25.3, C25.4, C25.7, C25.8 and C25.9. The histology/behaviour selection comprises 8150/3, 8151/3, 8152/3, 8153/3, 8155/3, 8156/3, 8240/3, 8242/3, 8243/3, 8246/3 and 8249/3. These are the implemented registry categories; the pNET designation in this report should not be read as a new pathological reclassification of individual records.

Python preprocessing rechecks the year, site and histology criteria and applies the `First malignant primary indicator = Yes` rule. This is the configured rule, rather than an assumption that only people with a single lifetime primary tumour are eligible. Records with an excluded cause-of-death classification, invalid follow-up or unknown required analytic fields are then removed in the order shown below. There is no adult-only age filter; the retained recorded ages range from 4 to 90 years, with the upper value including the source category of 90 years or older.

2.4 Sequential attrition and resulting cohorts

Table 3 shows how the number of eligible records changes after each selection and preprocessing step. The year, pancreatic-site and histology checks remove no further records because the export already satisfies those criteria. The largest subsequent exclusion is the removal of 1,903 records with at least one unknown required analytic field: 41.6% of the 4,575 records reaching that step. Overall, 2,672 of the 5,761 exported records are retained (46.4%).

Table 3: Sequential preprocessing exclusions. Each row shows the number of records removed at that step and the number remaining.

Step	Criterion	Removed	Remaining
0	Raw SEER export	—	5,761
1	Diagnosis year 2010–2016	0	5,761
2	First malignant primary indicator	1,150	4,611
3	Accepted pancreatic primary site	0	4,611
4	Accepted histology/behaviour code	0	4,611
5	Exclude unclear cause-of-death classification	26	4,585
6	Exclude invalid or incomplete follow-up	10	4,575
7	Exclude unknown required analytic fields	1,903	2,672

The complete-case criterion covers 20 fields: age, sex, race, marital status, histology code,

grade, primary site, T stage, N stage, primary-site surgery, lymph-node surgery, chemotherapy, radiotherapy, four organ-metastasis indicators, tumour size, survival months and vital status. M stage is retained for reference in the full dataset but is not a required field in this run. The exclusion rule removes a record if at least one of these required fields is unknown, and one record may contain unknown values in several fields.

Figure 3 distinguishes the final analytic cohort from its liver-metastasis subset. The diagnostic and prognostic datasets are not independent samples: the latter is nested within the former.

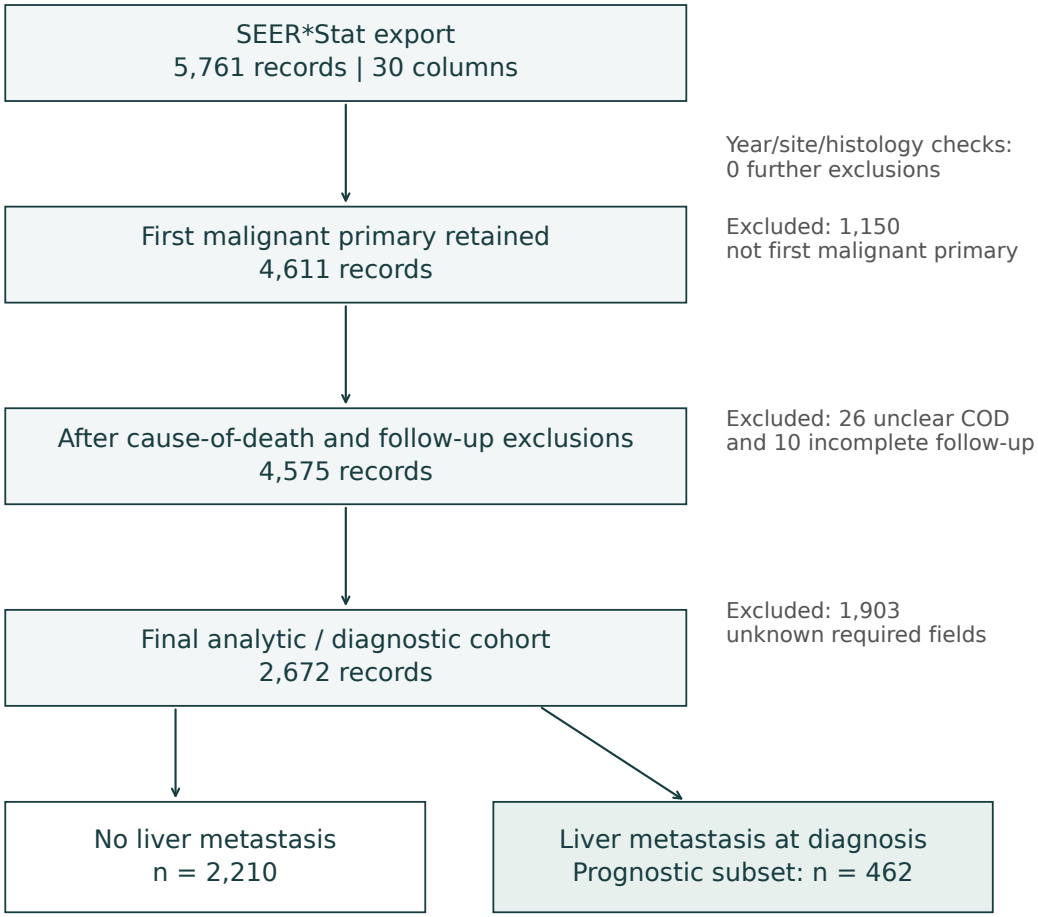


Figure 3: Cohort construction from the exported records. COD: cause of death. Both liver-metastasis groups contribute to the diagnostic task; only the 462 liver-positive records enter the prognostic task. Counts refer to this extract, not the reference paper.

2.5 Task definitions and analytic outputs

Three complementary files are produced: a full analytic table with 2,672 rows and 38 columns, a diagnostic table with 2,672 rows and 21 columns, and a prognostic table with 462 rows and 23 columns. Column totals include identifiers, targets and alternative encodings; they are not the number of independent predictors supplied to every model.

For diagnosis, the binary target is liver metastasis at diagnosis (`mets_liver`): 462 positive and 2,210 negative records. General M stage and its binary encoding are excluded from the

diagnostic file because they encode related metastatic status. Survival outcomes are also absent from this task-level file. Removing these fields prevents the diagnostic models from using information that directly reveals or closely reflects the target. Treatment variables remain available, but the registry does not establish whether each treatment was already known at the intended prediction time.

For prognosis, eligibility requires `mets_liver = 1`. Overall survival is represented by the recorded months from diagnosis and an event indicator equal to 1 for death from any cause and 0 for an alive/right-censored record. The prognostic dataset contains 311 patients who died and 151 whose last recorded status was alive, so their survival times are right-censored. Recorded follow-up ranges from 0 to 143 months, including 12 patients with zero recorded months. These values are kept as recorded. If a survival method requires a strictly positive time, the necessary adjustment is made and reported within that model's analysis. The observed range alone does not describe the typical follow-up time or median survival of the cohort.

Table 4: Selected characteristics of the diagnostic and prognostic datasets obtained after cohort selection and preprocessing. Percentages use the column-specific sample size. Q1–Q3 denotes the first and third age quartiles; the prognostic cohort is a subset of the diagnostic cohort.

Characteristic	Diagnostic	Prognostic
Patients	2,672	462
Age, median [Q1–Q3], years	60 [50–68]	59 [50–69]
Male sex	1,442 (54.0%)	261 (56.5%)
Legacy grade 4	60 (2.2%)	33 (7.1%)
Brain metastasis at diagnosis	6 (0.2%)	3 (0.6%)
Liver metastasis at diagnosis	462 (17.3%)	462 (100.0%)
Deaths (all causes)	Not a diagnostic outcome	311 (67.3%)
Alive / right-censored	Not a diagnostic outcome	151 (32.7%)
Recorded survival of zero months	Not a diagnostic outcome	12 (2.6%)

2.6 Clinical recoding and retained ambiguities

Table 5 summarises how the original SEER fields were converted into consistent variables for analysis. These rules include category grouping, binary coding, selection between available staging fields and survival coding, and they were applied in the same way to every eligible record. Steps that depend on a fitted model, such as pooling rare categories, scaling or feature selection, were performed later using only the relevant training data.

Table 5: Principal clinical recoding rules used to prepare the analysis datasets.

Domain	Implemented representation	Interpretation or limitation
Demographics	Single-year age; male indicator; four recorded race categories; married versus other known marital categories.	Age 90+ is stored as 90. Unknown race or marital status is excluded, not imputed.

Continued on next page

Table 5 continued

Domain	Implemented representation	Interpretation or limitation
Histology	Diagnostic labels retain the accepted histology categories. Prognosis groups codes 8240/3 and 8249/3 as "Carcinoid tumor", with other accepted codes grouped as "Others".	The binary prognostic grouping is an analysis convention, not equivalence of the underlying histologies.
Differentiation	Legacy grade recode mapped to integers 1–4; unknown grade excluded.	These are the registry differentiation categories, not a newly derived Ki-67/mitotic-count classification.
Anatomic site	Pancreatic head, body, tail and other accepted sites.	The "Others" category combines several source sites.
T and N stage	Prefer AJCC 7th-edition values for 2010–2015 and SEER Combined values for 2016; use a valid alternative source as fallback. Clinical/pathological prefixes are collapsed to T0–T4 and N0/N1.	Combining the available staging fields provides consistent model inputs across diagnosis years, but differences between clinical and pathological assessment are no longer retained. N1 is additionally encoded as `n_stage_positive`.
Primary surgery	Code 00 is 0; code 99 is missing; other parsed nonzero codes are 1.	Procedure detail and timing are lost in the binary representation.
Lymph-node surgery	"None" is 0; biopsy, removal and sentinel-node categories are 1. "Unknown or not applicable" is missing.	Not inferred from the absence of primary-site surgery; a positive value is not a lymph-node tumour finding.
Chemotherapy	"Yes" is 1; "No/Unknown" is 0.	Zero does not establish that chemotherapy was not received.
Radiotherapy	Active radiation categories are 1; "None/Unknown" and "Refused" are 0. "Recommended, unknown if administered" is missing.	The zero category combines different treatment-information states.
Tumour size	Store numerical millimetres and groups $0 \leq s \leq 26$, $26 < s \leq 36$, and $s > 36$. Unknown/unreasonable labels are missing.	The stored label "26–36 mm" excludes exactly 26 mm. Code 000 is retained as 0, not an unknown value.
Organ metastases	Bone, brain, liver and lung fields: Yes = 1, No = 0; unknown values excluded.	Liver status is the diagnostic outcome and prognostic eligibility criterion, not a prognostic predictor.
Survival	Numerical months; Dead = 1 and Alive = 0.	All-cause mortality, not cancer-specific survival; zero-month records remain in the input.

Treatment information deserves particular caution. NCI documentation states that these chemotherapy and radiotherapy variables concern first-course treatment and that non-recording cannot reliably distinguish no treatment from unknown treatment (National Cancer Institute, n.d.b). The report therefore retains the mixed meanings of the zero codes rather than describing them simply as untreated patients. The exported fields also do not establish that treatment

was known before the intended diagnostic or prognostic prediction time.

2.7 Data checks and implications for downstream evaluation

Before modelling, the processed data were checked to confirm that the exclusion steps lead to the final cohort of 2,672 records, that the diagnostic dataset contains all retained patients and that the prognostic dataset contains exactly the 462 patients with liver metastasis at diagnosis. The checks also confirmed that analysis identifiers are unique and that all required fields are complete under the implemented coding. Table 4 describes the main characteristics of these two datasets.

Each retained record receives a new sequential analysis identifier after filtering. This identifier is used only to keep the same record aligned across datasets and model outputs. It is not the original SEER patient identifier and cannot be used to link a patient across different SEER releases.

Four consequences constrain all later experiments. First, complete-case selection is substantial and need not be representative of the excluded records. Because both tasks descend from the same filtered cohort, the diagnostic dataset also inherits follow-up and cause-of-death exclusions, although those fields are not diagnostic predictors. Excluding an unclear cause of death follows the reproduction workflow; it is not intrinsically necessary to define all-cause mortality.

Second, no blank values in a task-level file does not imply complete clinical information. Treatment ambiguity persists within the stored codes, while stage-source harmonisation, coarse histology grouping and top-coded age reduce detail. There is no basis for treating the retained records as missingness-free measurements of every clinical construct.

Third, sparse strata are present even before modelling: the prognostic cohort contains only three records with brain metastasis and 33 with legacy grade 4. These counts provide context for unstable contrasts, bootstrap support and later synthetic-category loss. Failure to estimate a sparse subgroup is not evidence that its clinical association is absent.

Finally, every downstream finding is conditional on these eligibility and representation choices. Predictive models can identify patterns within the retained extract, but cannot recover excluded patients, establish treatment effects or validate transfer to a hospital population. The next section therefore begins with reproduction of the reference diagnostic and prognostic workflows on this fixed data foundation before any transformed-data or foundation-model comparison is considered.

3 Reference Diagnostic and Prognostic Model Reproduction

3.1 Purpose and evidence boundary

The reference workflow supplies the conventional statistical baseline for the project: logistic regression for liver metastasis at diagnosis and Cox regression for overall survival within the liver-positive cohort. The workflow of Li et al. (2023) is reconstructed using the cohorts in Section 2 so that subsequent TabPFN transformations, foundation-model comparisons, nonlinear heads and augmentation experiments can be interpreted against a transparent clinical reference.

This is a methodological reproduction rather than a patient-identical replication. The SEER release used by the article was no longer the available project source, so the preprocessing pipeline constructed a clinically similar cohort from the available SEER export. The article's external hospital cohort was also unavailable. Accordingly, agreement in selected variables or performance is informative, but a difference does not by itself indicate a reproduction failure or a model improvement.

The two tasks follow different evaluation designs from the reference study. For diagnosis, the complete SEER cohort is used to develop the models and to calculate ROC/AUC, calibration and decision curves. The reference article likewise calls its complete SEER diagnostic cohort the training cohort and evaluates it separately from its external hospital cohort. For prognosis, the liver-positive SEER cohort is divided into a 70% training subset and a 30% internal-validation subset. Diagnostic results therefore describe performance in the same cohort used for model development, whereas the prognostic results include evaluation in patients held out from model fitting. This distinction is stated whenever the two tasks are compared.

3.2 Candidate descriptors and task-specific designs

Both reproductions begin from the same 17 clinical descriptor families (Table 7), after the preprocessing and complete-case rules in Section 2. The task datasets differ in their target and eligible population: diagnosis uses the full cohort and predicts liver metastasis at diagnosis, whereas prognosis is restricted to patients already recorded with liver metastasis and predicts overall survival. Both histology definitions follow the reference study. The diagnostic analysis retains the individual ICD-O histological subtypes available in the cohort (10 categories), whereas the prognostic analysis applies the paper's coarser 2-group definition: carcinoid tumour NOS and atypical carcinoid tumour versus all other listed histological subtypes.

Table 6: Overview of the two reference-reproduction tasks. Evaluation procedures are described in detail in the task-specific subsections.

Task	Population and outcome	Development design	Principal evaluation
Diagnosis	2,672 patients; 462 liver-metastasis cases (17.3%)	Complete SEER cohort used for model development and within-cohort evaluation	ROC/AUC, Brier score, calibration and decision curves
Prognosis	462 liver-positive patients; overall survival and death indicator	323-record training subset and 139-record internal-validation subset	Harrell C-index, time-dependent AUC, calibration, decision curves and Kaplan–Meier (K–M) survival curves

Table 7: 17 candidate clinical descriptor families entering the reference feature-selection workflows.

Descriptor	Representation	Modelling description
Age at diagnosis	Continuous	Age in years at diagnosis
Race	Categorical	SEER race category after preprocessing harmonisation
Sex	Binary	Female or male indicator
Marital status	Binary	Married versus the harmonised non-married category
Histological subtype	Categorical	Both definitions follow the reference study: diagnosis retains the 10 subtype categories observed in this cohort; prognosis groups carcinoid and atypical carcinoid tumours together and places the other subtypes in an "Others" category
Tumour grade	Ordered categorical	Grade I–IV after the documented SEER mapping
Primary-site group	Categorical	Pancreatic head, body, tail or other site group
AJCC T stage	Ordered categorical	Harmonised T-stage category
AJCC N stage	Binary	N0 versus node-positive disease
Primary-site surgery	Binary	Recorded surgery of the primary tumour
Lymph-node surgery	Binary	Recorded regional lymph-node surgery
Radiotherapy	Binary	Recorded radiotherapy indicator
Chemotherapy	Binary	Recorded chemotherapy indicator
Tumour-size group	Ordered categorical	≤ 26 mm, 26–36 mm or > 36 mm
Bone metastasis	Binary	Bone metastasis recorded at diagnosis
Brain metastasis	Binary	Brain metastasis recorded at diagnosis
Lung metastasis	Binary	Lung metastasis recorded at diagnosis

General M stage is not included as a diagnostic predictor because it incorporates distant metastatic disease and would be too closely related to the liver-metastasis target. Treatment variables are retained to reproduce the reference workflow, but their temporal availability at a true diagnosis-time prediction point is uncertain. They must therefore be interpreted as retrospective predictors rather than as prospectively established decision-time inputs.

3.3 Diagnostic workflow and selected variables

The diagnostic analysis uses all 2,672 retained records, including 462 with liver metastasis, and 17 candidate clinical predictors. In accordance with the reference study, tumour size is entered as 3 categories (≤ 26 mm, 26–36 mm and > 36 mm), and N stage is entered once as N0 versus N1. The corresponding continuous tumour-size field and alternative N-stage encoding are not added as extra predictors. General M stage is excluded because it directly incorporates distant metastatic disease and would reveal information closely related to the liver-metastasis outcome. Treatment variables are retained to reproduce the reference model, although the SEER records do not establish whether every treatment was known at the intended prediction time.

The screening sequence starts with 10-fold cross-validated binomial LASSO. The implemented primary rule uses the one-standard-error penalty to obtain a smaller set of predictors, while the minimum-deviance penalty is retained as a sensitivity analysis. Nonzero encoded columns are then mapped back to their clinical descriptors. This differs slightly from the reference article's LASSO figure, which identifies the minimum-error penalty (Li et al., 2023).

As in the reference study, a 1,000-tree random forest provides a second view of predictor importance. Mean decrease in accuracy measures how much predictive accuracy is lost when a variable is permuted, whereas mean decrease in Gini summarises how much that variable reduces node impurity across the forest. These measures rank predictors in different ways; they do not by themselves confirm a clinical effect or establish causality. The reference article reports a 7-variable random-forest shortlist whose variables were also selected by LASSO. In this reproduction, the forest ranking is reported alongside LASSO but is not used as a mandatory filter. The final data-driven logistic model retains variables from the LASSO shortlist whose global likelihood-ratio-test has $p < 0.05$ in the joint multivariable model. These screening p-values are nominal and are not adjusted for the preceding selection.

The fitted penalties are $\lambda_{\min} = 0.0034$ and $\lambda_{1\text{se}} = 0.0182$ at the displayed precision. As shown in Figure 4, the one-standard-error rule deliberately accepts a small increase in cross-validated error in exchange for a sparser model. Because categorical descriptors expand into multiple design columns, selection is first performed at the encoded-column level and then mapped back to the originating clinical descriptor.

The selection results contain 15 clinical variables at $\lambda_{\min}$ and 9 at $\lambda_{1\text{se}}$. A total of 8 survive the multivariable screen: primary-site surgery, chemotherapy, bone metastasis, T stage, N stage, tumour-size group, primary site and histology. Grade is included in the 9-variable fit but does not meet the final threshold ($p = 0.07306$). This is a result of this particular adjustment and selection procedure, not evidence that grade is generally uninformative.

Two fixed reference sets are also fitted. The 5-variable *nomogram-figure* set contains grade, N stage, surgery, tumour size and bone metastasis; the 6-variable *text-defined* set additionally contains chemotherapy. Keeping both makes the reference article's text/figure discrepancy explicit (Li et al., 2023). The data-driven 8-variable model is a separate reproduction variant, not a relabelling of the published nomogram. The comparator contains only T stage and N stage and is therefore described as T/N rather than as a complete TNM-stage model.

Figure 5 also cautions against a single feature-importance interpretation. Chemotherapy and primary surgery rank 1st and 2nd by mean decrease in accuracy. Age ranks 2nd by Gini

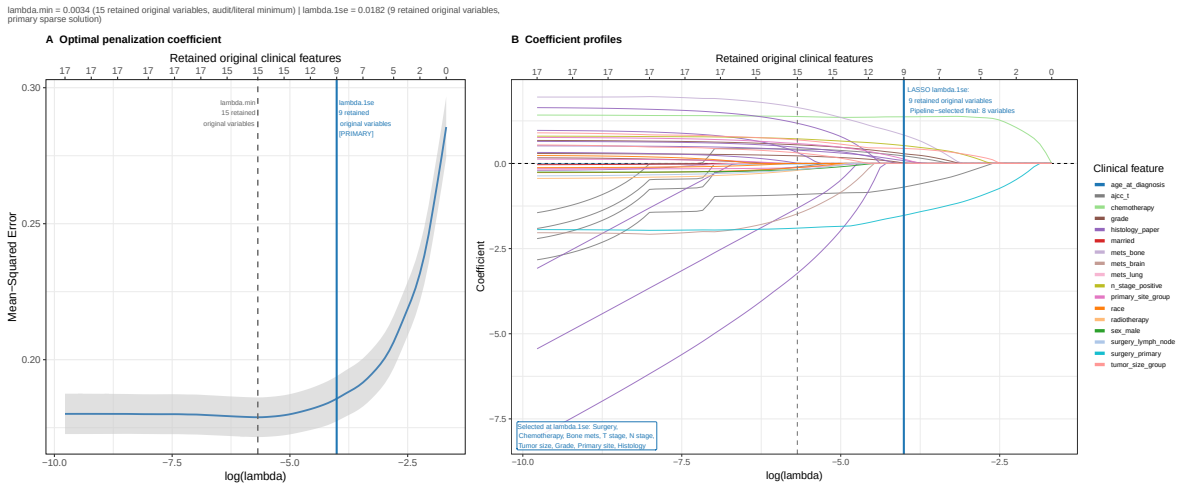


Figure 4: Diagnostic LASSO feature selection. A: ten-fold cross-validated binomial error with the minimum-error and one-standard-error penalties. B: coefficient shrinkage paths for the encoded predictor columns. The top axes count retained original clinical descriptor families after back-mapping; the primary $\lambda_{1\mathrm{se}}$ solution retains 9 families, whereas $\lambda_{\min}$ retains 15. The data-driven final model contains 8 variables after the subsequent multivariable screen.

decrease but 16th by permutation importance, with a negative latter value. Thus, the two importance definitions do not consistently corroborate the same ranking. This discrepancy supports treating importance as a model-dependent descriptive quantity; it does not by itself establish confounding or a causal mechanism.

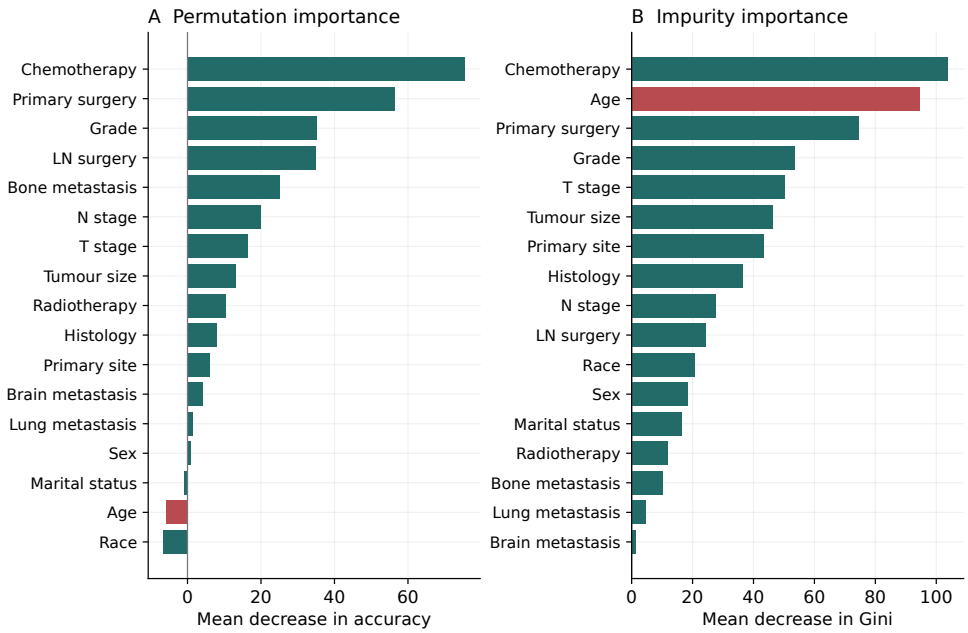


Figure 5: Diagnostic random-forest importance. Panels rank variables independently under their own criterion; the highlighted age row makes the disagreement visible. The horizontal scales measure different quantities and are not directly comparable. These are single-fit importance estimates, without ranking-stability intervals.

Table 8 separates the data-driven selection path from the fixed reference specifications. This distinction is important: the 5- and 6-variable models ask how the published variable sets behave in the available cohort, whereas the 8-variable model asks what this cohort's implemented screening pipeline selects.

Table 8: Diagnostic feature-selection path and evaluated specifications. The random forest provides an additional importance ranking but does not independently determine the final feature set.

Stage or model	Rule and role	Clinical variables retained or specified
LASSO, $\lambda_{\min}$	Minimum cross-validated deviance; sensitivity shortlist	15 of 17 candidates: all except marital status and lung metastasis
LASSO, λ_{1se}	Primary sparse shortlist	Primary surgery, chemotherapy, bone metastasis, T stage, N stage, tumour size, grade, primary site and histology
Pipeline-selected final	Global likelihood-ratio-test $p < 0.05$ within the 9-variable shortlist	Primary surgery, chemotherapy, bone metastasis, T stage, N stage, tumour size, primary site and histology; grade is excluded ($p = 0.0731$)
Paper-text model	Fixed 6-variable reference set	Grade, N stage, primary surgery, chemotherapy, tumour size and bone metastasis
Nomogram-figure model	Fixed 5-variable primary reproduction target	Grade, N stage, primary surgery, tumour size and bone metastasis
T/N baseline	Conventional comparator	T stage and N stage

The second-stage screen is shown variable by variable in Table 9. The reported likelihood-ratio-test p-values are global tests for each clinical descriptor in the joint 9-variable logistic model. They are used to reproduce the implemented selection rule and should not be interpreted as multiplicity-adjusted confirmatory inference.

Table 9: Diagnostic multivariable screen applied to the 9 λ_{1se} clinical descriptors.

Clinical descriptor	Global nominal p	Final status
Primary-site surgery	< 0.001	Retained
Chemotherapy	< 0.001	Retained
Bone metastasis	< 0.001	Retained
T stage	< 0.001	Retained
N stage	< 0.001	Retained
Tumour-size group	< 0.001	Retained
Grade	0.073	Not retained
Primary-site group	< 0.001	Retained
Histological subtype	0.004	Retained

3.4 Diagnostic model evaluation and results

Evaluation protocol. The reference article uses the complete SEER diagnostic cohort as its training cohort; it does not describe a separate internal hold-out set for the diagnostic model. Its 70/30 internal split applies to the prognostic analysis. The reproduction therefore follows the diagnostic design by fitting and evaluating all 4 logistic specifications in the complete 2,672-patient SEER cohort. Discrimination is summarised by ROC curves and AUC with a requested 1,000-resample interval. Calibration is assessed using the Brier score and calibration intercept/slope for the 5-variable primary model; the pipeline-selected model additionally has an uncorrected curve and a bootstrap-corrected calibration curve. Decision-curve analysis compares model net benefit with T/N, treat-all and treat-none strategies across threshold probabilities. Because the diagnostic models are evaluated in the same records used for fitting, these results may overestimate performance in new patients and are not equivalent to held-out or external validation.

Discrimination. Table 10 gives the AUC estimates in the complete 2,672-record diagnostic development cohort. The pipeline-selected model has the largest AUC, 0.9028, followed by 0.8947 for the paper-text model, 0.8795 for the nomogram-figure model and 0.7628 for T/N. The reference article reports a SEER training-cohort AUC of 0.877 and an external-hospital AUC of 0.893 (Li et al., 2023). The proximity of the published internal estimate to 0.8795 is descriptive agreement across different extracts, not a test of reproduction fidelity or model equivalence.

Table 10: Diagnostic discrimination in the complete SEER development cohort ($n = 2,672$), following the reference study's diagnostic training-cohort design. Each model is fitted and evaluated in these same records; brackets show 95% AUC intervals and not held-out performance intervals.

Model	Clinical variables	Within-cohort AUC [95% interval]
Nomogram-figure	5	0.880 [0.863–0.895]
Text-defined	6	0.895 [0.880–0.910]
Pipeline-selected	8	0.903 [0.888–0.917]
T/N baseline	2	0.763 [0.742–0.783]

The AUC routine requests 1,000 bootstrap resamples of the outcome–prediction pairs. It does not repeat feature selection or model fitting inside those resamples. If that calculation fails, the routine uses a DeLong interval, although the result table does not identify which method

supplied each interval. These intervals therefore quantify uncertainty around the reported AUCs but are not whole-pipeline bootstrap validation. The internal LASSO cross-validation selects the penalty; it does not turn the final full-cohort AUC into a held-out estimate.

For the 5-variable primary model, the within-cohort Brier score is 0.0969. The fitted calibration intercept and slope round to 0 and 1 on the training data. Such values are expected for an unpenalised logistic model evaluated on its own fitting sample and do not establish transportable calibration. The separate pipeline-selected calibration curve includes a bootstrap correction conditional on its already selected variable set (Figure 6B). That procedure does not account for reselecting the variables.

The diagnostic decision curves in Figure 6C compare the pipeline-selected model, T/N and the two default strategies. They describe net benefit in the same cohort used for model fitting. No clinical action or acceptable threshold range was prospectively defined here, and treatment-related inputs have not been shown to precede prediction. The curves therefore cannot establish a deployable diagnostic intervention or benefit in a new hospital.

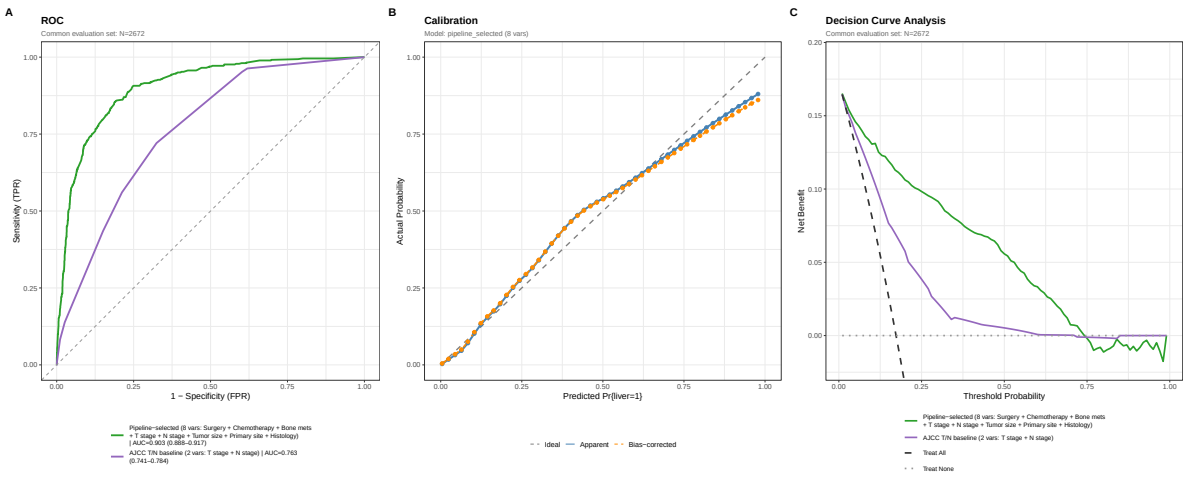


Figure 6: Diagnostic evaluation of the pipeline-selected 8-variable model in the complete SEER development cohort. A: ROC curve against the T/N baseline. B: uncorrected and bootstrap-corrected calibration. C: decision curves against T/N and the default strategies. The plotted evaluation is not external validation, and the Brier score discussed in the text belongs to the fixed 5-variable primary model rather than the plotted pipeline-selected model.

3.5 Prognostic design and model specifications

The prognostic analysis starts with 462 patients who had liver metastasis at diagnosis. Following the reference study, these patients are divided by simple random sampling into a 70% training subset and a 30% internal-validation subset. The resulting groups contain 323 training records (219 deaths and 104 censored) and 139 validation records (92 deaths and 47 censored). Because brain metastasis and some other categories are rare, the split was accepted only when every categorical predictor required by the primary model was represented in the training subset. The same patient assignments are used for all prognostic model development and internal-validation results in this section.

The 12 records with zero recorded survival months are assigned 0.5 months for model fitting: 11 fall in training and 1 in validation. This is the implementation's positive-time convention, not a

claim that all Cox estimators require it. The outcome remains death from any cause, with right censoring for records coded alive.

Table 11: Prognostic development split and overall-survival event counts.

Cohort	Records	Deaths	Censored
Training	323	219	104
Internal validation	139	92	47

The primary Cox specification contains age, grade, histology group, primary surgery and brain metastasis, following the reference prognostic nomogram (Li et al., 2023). The training subset contains 2 brain-metastasis records and the validation subset contains 1. Accordingly, the coefficient is estimable in these outputs, but very sparsely supported.

Additional models preserve the implementation's alternative selection paths. Backward Cox/AIC selection retains 9 variables, with bidirectional selection as a sensitivity procedure. The survival-forest analogue ranks candidates on the training cohort and evaluates top-k subsets in five folds, selecting 16 variables. The field labelled `rmse_cv` actually contains mean $1 - C$, where C is Harrell's concordance, not root mean squared error. Moreover, the ranking is formed before those internal folds; this is not nested validation of the complete ranking-and-selection process. It is used as a selection diagnostic, not as an unbiased performance estimate.

The union of the Cox/AIC and forest-selected sets, followed by a global multivariable $p < 0.05$ screen, produces the 5-variable data-driven set: age, histology, grade, N stage and primary surgery. A 6-variable text-screened sensitivity set and the 2-variable T/N comparator are also retained. These alternatives are not interchangeable with the primary model; the horizon-specific results below refer to the primary model and T/N only.

Table 12 makes the prognostic selection paths explicit. A total of 4 of the 5 data-driven variables agree with the published nomogram set. N stage replaces brain metastasis in the data-driven model; the fixed primary model nevertheless retains brain metastasis so that the paper's stated prognostic specification is evaluated directly.

Table 12: Prognostic feature-selection paths and evaluated specifications. All selection procedures use the 323-record training subset.

Model or path	Rule and role	Clinical variables retained or specified
Nomogram-figure primary	Fixed reproduction target from the published prognostic nomogram	Age, grade, histology, primary surgery and brain metastasis
Cox/AIC selection	Backward AIC reduction of the full Cox model	Age, sex, marital status, histology, grade, T stage, N stage, primary surgery and bone metastasis
Survival RF/RFE selection	Pre-ranked random survival forest variables; top-k sets assessed by five-fold mean $1 - C$; minimum at $k = 16$	All candidates except tumour-size group: grade, primary and lymph-node surgery, age, histology, chemotherapy, N stage, primary site, lung/bone/brain metastasis, race, sex, radiotherapy, T stage and marital status
Pipeline-selected final	Union of Cox/AIC and RF/RFE sets, then global multivariable $p < 0.05$ screen	Age, histology, grade, N stage and primary surgery
T/N baseline	Conventional comparator	T stage and N stage

Figure 7 shows the survival-forest top-k selection diagnostic using the quantity actually calculated by the implementation. The minimum mean five-fold error is 0.2629 at 16 variables, compared

with 0.2644 at 17 variables. The fold-level variation overlaps substantially, so the selected $k = 16$ should be understood as the numerical minimum of this run rather than evidence for a uniquely optimal subset.

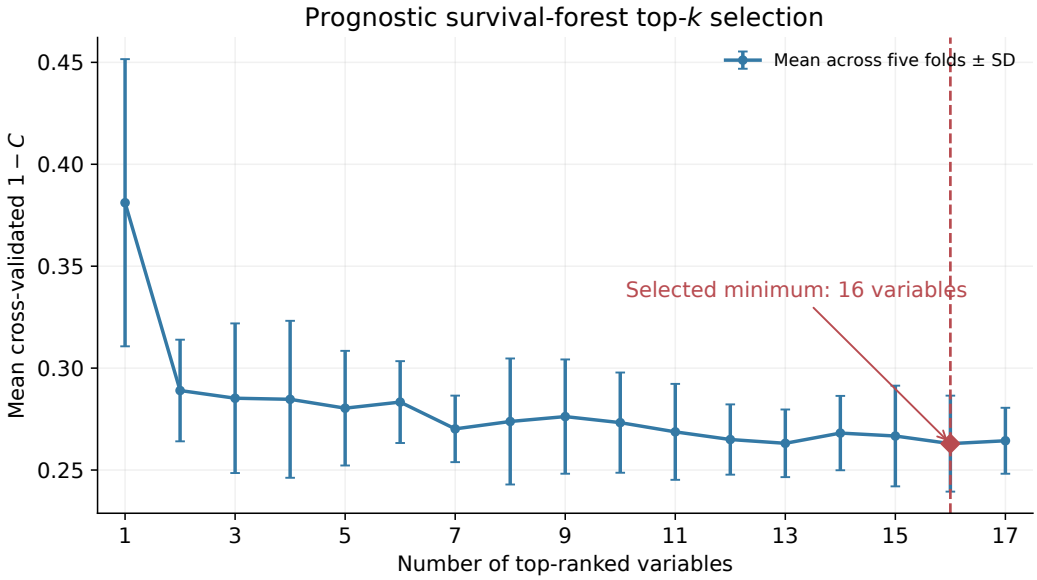


Figure 7: Prognostic survival-forest top-k selection. Points show mean five-fold $1 - C$, where C is Harrell concordance, and error bars show the fold standard deviation. The vertical line marks the minimum at 16 variables. The ranking was generated before the folds, so this is not nested validation of the complete ranking-and-selection procedure.

Evaluation protocol. All Cox specifications are fitted on the training subset. Training-set Harrell C-index is reported for the 6 specifications in Table 13; the fixed 5-variable primary Cox nomogram and T/N baseline are additionally compared by time-dependent ROC/AUC using marginal inverse probability of censoring weighting (IPCW) at 12, 24 and 36 months in training and validation. The primary model is also assessed through grouped survival calibration, horizon-specific decision curves and Kaplan–Meier curves for risk groups defined by the median training linear predictor. The same risk threshold is carried unchanged into validation.

3.6 Prognostic results: association, discrimination and stratification

Comparison of the evaluated Cox specifications. Six Cox specifications are retained because they answer different questions. The 5-variable primary model reproduces the prognostic nomogram displayed in the reference article. The 2-variable T/N model provides a simple clinical baseline using only tumour and nodal stage. The text-screened sensitivity model uses the variables named in the article's regression results. The Cox/AIC model applies backward selection using the Akaike information criterion (AIC), while the random forest with recursive feature elimination (RF/RFE) model uses a survival forest to rank predictors and compare progressively smaller subsets. Finally, the pipeline-selected model combines the variables identified by the Cox/AIC and RF/RFE procedures and retains those meeting the multivariable selection rule. Together, these models show how the choice of feature-selection method affects the result without replacing the prespecified primary model.

Table 13 reports training-set concordance for all 6 specifications. RF/RFE and Cox/AIC have the largest point estimates (0.758 and 0.757), followed by the 5-variable pipeline-selected model (0.753), the text-screened sensitivity model (0.743), the primary nomogram model (0.740) and T/N (0.573). Because model selection and concordance estimation use the same training subset, these values cannot establish that the more complex models will perform better in new patients. Horizon-specific internal validation was prespecified for the primary nomogram and T/N baseline, which form the main reference-study comparison; the other 4 models are retained to examine how alternative feature-selection procedures change the training results.

Table 13: Training-set performance of the 6 prognostic Cox specifications ($n = 323$, 219 deaths). The intervals are Harrell C-index intervals. AIC is comparable only among models fitted to this same outcome and sample; lower values indicate better relative fit after accounting for model complexity.

Specification	Variables	Harrell C [95% interval]	AIC
Primary nomogram model	5	0.740 [0.705–0.774]	2174.5
Text-screened sensitivity	6	0.743 [0.709–0.777]	2173.0
Cox/AIC-selected	9	0.757 [0.724–0.790]	2152.8
RF/RFE-selected	16	0.758 [0.726–0.791]	2169.1
Pipeline-selected	5	0.753 [0.719–0.786]	2155.9
T/N baseline	2	0.573 [0.533–0.613]	2288.1

Adjusted associations and model assumptions. Table 14 gives the primary training-model hazard ratios. Higher grade, older age and the other-histology group are associated with greater modelled hazard, while recorded surgery is associated with lower hazard. These are observational associations conditional on the model covariates. In particular, the surgery coefficient is not an estimate of a treatment benefit.

Table 14: Primary prognostic Cox model: nominal hazard ratios and Wald intervals from 323 training records and 219 deaths. Interpretations are limited by sparse brain-metastasis support and evidence against proportional hazards.

Contrast	HR [95% interval]	Nominal p
Brain metastasis: yes vs no	2.98 [0.72–12.40]	0.134
Age: per year	1.02 [1.01–1.03]	< 0.001
Grade II vs I	1.42 [1.01–1.99]	0.044
Grade III vs I	3.02 [2.12–4.30]	< 0.001
Grade IV vs I	3.44 [2.10–5.61]	< 0.001
Other histology vs carcinoid group	1.59 [1.16–2.17]	0.004
Primary surgery: yes vs no	0.37 [0.28–0.50]	< 0.001

Brain metastasis has HR 2.98 with a wide interval, 0.72–12.40. More importantly, the Schoenfeld-residual test rejects proportional hazards globally ($p = 2.61 \times 10^{-7}$), with evidence for grade ($p = 6.14 \times 10^{-7}$), histology ($p = 0.00156$) and surgery ($p = 0.00206$). This indicates that the associations of these predictors with mortality may change during follow-up rather than remain constant over time. The corresponding hazard ratios and survival predictions should therefore be interpreted cautiously, even when the model separates higher-risk from lower-risk patients well.

Discrimination. The primary prognostic model is the fixed 5-variable Cox nomogram containing age, grade, histology group, primary surgery and brain metastasis. Its training-set Harrell C-index is 0.7395 (95% interval 0.7047–0.7744), compared with 0.5728 for T/N. Harrell's C-index summarises how well each model orders patients by risk across the observed follow-up. Table 15 instead reports discrimination at 12, 24 and 36 months using marginal IPCW to account for censoring. The censoring weights are estimated separately within the training and validation subsets. The reported intervals quantify uncertainty in each AUC but do not repeat feature selection or model fitting.

Table 15: Time-dependent AUC [95% interval] for the fixed 5-variable primary prognostic Cox model (age, grade, histology group, primary surgery and brain metastasis) and the 2-variable T/N baseline. Training $n = 323$; internal validation $n = 139$. Marginal IPCW and influence-function intervals are used.

Cohort	Months	Primary Cox model	T/N baseline
Training	12	0.889 [0.842–0.936]	0.613 [0.539–0.687]
Training	24	0.826 [0.776–0.876]	0.625 [0.560–0.690]
Training	36	0.822 [0.775–0.869]	0.598 [0.536–0.660]
Internal validation	12	0.886 [0.826–0.946]	0.391 [0.284–0.498]
Internal validation	24	0.873 [0.817–0.930]	0.443 [0.346–0.540]
Internal validation	36	0.844 [0.779–0.910]	0.505 [0.410–0.599]

The primary model has higher point estimates than T/N at all three horizons in both subsets. However, T/N's validation AUCs of 0.391, 0.443 and 0.505 indicate poor transfer within this one split, not that clinical staging is intrinsically uninformative. No paired difference test or repeated-split comparison is supplied by this table. AUC also does not assess the accuracy of absolute survival probabilities.

Figure 8 complements the point estimates with the underlying horizon-specific ROC curves. Panels A–C use the training subset and Panels D–F use the fixed internal-validation subset; the comparison is restricted to the fixed 5-variable primary model and the T/N baseline, the 2 specifications evaluated at each survival horizon.

Calibration and decision curves. Survival calibration uses 1,000 bootstrap repetitions in the training subset and compares grouped predicted survival probabilities with Kaplan–Meier estimates in the validation subset. The reported slopes and intercepts are calculated by regressing the observed group-level survival estimates on the corresponding mean predictions. They are therefore descriptive group-level summaries rather than standard individual-level calibration measures, and they are not used to claim that the model is well calibrated in new patients.

The survival decision curves convert predicted survival into mortality risk at each time horizon and exclude patients censored before that horizon. Because this calculation does not use IPCW, it provides an approximate view of net benefit and is not treated as definitive evidence of clinical utility.

Risk-group separation. The median training linear predictor, 0.2328 at the reported precision, defines low- and high-risk groups. The same threshold is applied to validation without retuning. Median observed survival differs substantially between the groups (Table 16), with

log-rank $p < 0.001$ in both subsets. Values smaller than the reported precision are stated as $p < 0.001$ rather than as exact zero. The patient counts, event counts and median survival estimates use the same risk-group assignments in the table and figure.

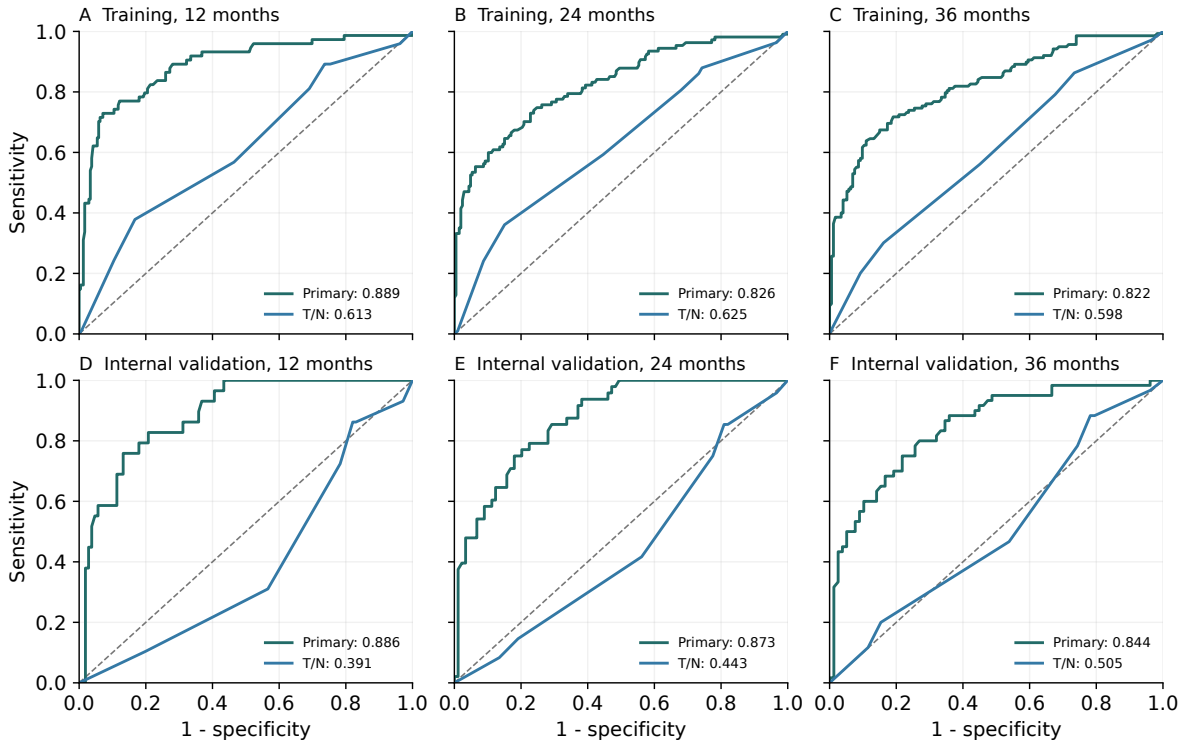


Figure 8: Time-dependent ROC curves for the fixed 5-variable primary prognostic Cox model (age, grade, histology group, primary surgery and brain metastasis) and the 2-variable T/N baseline at 12, 24 and 36 months. Panels A–C show training results and Panels D–F show internal-validation results. Marginal IPCW is used; the results are not estimates from repeated data splits or an external cohort.

Table 16: Risk groups from the fixed 5-variable primary prognostic Cox model. Median overall survival is estimated by Kaplan–Meier, not the median of recorded follow-up times.

Cohort	Group	Records	Deaths	Median OS (months)
Training	Low-risk	162	85	75
Training	High-risk	161	134	19
Internal validation	Low-risk	70	36	89
Internal validation	High-risk	69	56	17

Table 16 quantifies the size, number of deaths and median overall survival of each risk group, while Figure 9 shows the full survival trajectories. The curves remain separated through most of the observed follow-up, although the confidence bands widen at later times as the number of patients still under observation decreases.

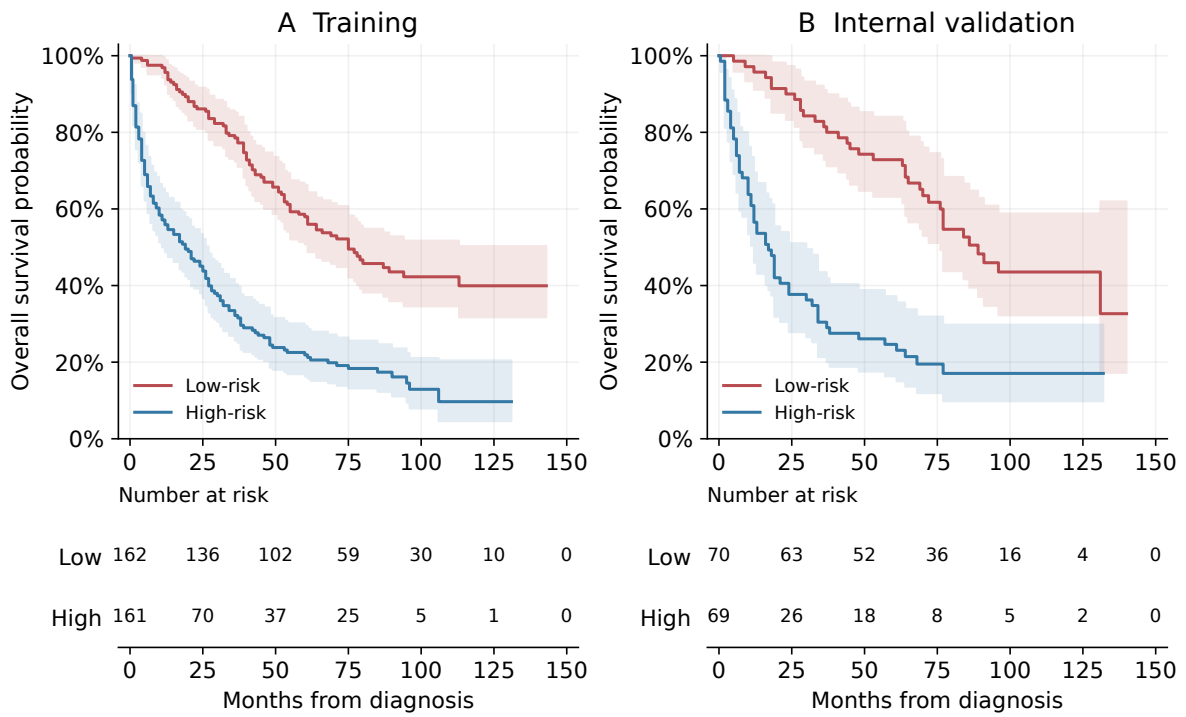


Figure 9: Overall-survival curves for risk groups defined by the fixed 5-variable primary prognostic Cox model. Shading denotes pointwise 95% log-type Kaplan–Meier intervals; rows below each panel give numbers at risk. The low/high threshold is learned in training and unchanged in validation. Sparse late follow-up widens uncertainty. These internally defined prognostic strata are not independently discovered patient archetypes or external validation cohorts.

3.7 Focused comparison with the published study

Table 17 summarises the comparisons that are most informative for this reproduction. For diagnosis, 5 of the paper's 6 stated risk factors are also in the data-driven model: N stage, primary surgery, chemotherapy, tumour size and bone metastasis. Grade is selected by LASSO but does not survive the adjusted screen, whereas T stage, primary site and histology are additional selected variables. For prognosis, the fixed primary model uses the same 5 variables as the published nomogram, while the data-driven model shares age, histology, grade and surgery and substitutes N stage for brain metastasis.

Table 17: Headline discrimination compared with Li et al. (2023). Published values are contextual benchmarks, not targets tested on the same patients. Prognostic AUC triplets are ordered as 1-, 2- and 3-year overall-survival AUC.

Task	Published study	Present reproduction on the available SEER cohort
Diagnosis	SEER training-cohort AUC 0.877; external-hospital AUC 0.893. The text identifies 6 variables, while the displayed nomogram omits chemotherapy.	AUC calculated in the complete SEER development cohort: 0.8795 for the fixed 5-variable nomogram, 0.8947 for the 6-variable text set, 0.9028 for the 8-variable data-driven model, and 0.7628 for T/N. An independent 30-patient external evaluation is reported in Section 4.
Prognosis	C-index 0.752/0.740/0.760 in training/internal/external cohorts. Training AUC 0.832/0.794/0.792 and internal AUC 0.832/0.857/0.818.	training Harrell C-index 0.7395. Primary-model AUC 0.889/0.826/0.822 in training and 0.886/0.873/0.844 in the single internal-validation subset.

The diagnostic AUCs are numerically close, and the prognostic discrimination is of the same general magnitude, but neither observation is evidence of equivalence. The available cohort is larger and comes from a different SEER release; category frequencies and coding availability changed; the exact feature-selection details could not always be recovered from the publication; and the paper's external hospital data were unavailable. Moreover, the diagnostic AUCs were calculated in the same cohort used to fit the models, so they may overestimate performance in new patients. The prognostic results come from one 70/30 internal split and may vary with a different split. These differences prevent a defensible claim that the present models outperform the published models.

3.8 Interpretation and transition

The reproduction provides conventional logistic- and Cox-regression reference models for the later experiments, but it does not establish clinical readiness. The diagnostic analysis shows that the selected predictors depend on the feature- selection rule, and that permutation and impurity importance can rank the same variables differently. Treatment variables must also be interpreted cautiously because their timing and the reasons for receiving treatment are not fully captured in SEER. In the prognostic analysis, the small number of brain-metastasis cases and the evidence against constant hazard ratios increase uncertainty. These findings make calibration, model assumptions and validation in new patients as important as the discrimination scores.

The primary reference models used in subsequent comparisons are the 5-variable diagnostic logistic nomogram (grade, N stage, primary surgery, tumour-size group and bone metastasis) and the 5-variable prognostic Cox nomogram (age, grade, histology group, primary surgery and brain metastasis). The text-defined, data-driven and T/N models are retained as comparators so that later analyses can distinguish a change in representation or model family from a change in the chosen clinical predictors.

The next section tests these SEER-trained models in the independent UK–Erlangen cohort. Here, *frozen* means that the fitted coefficients, categorical levels and decision thresholds are carried over without refitting, feature reselection, recalibration or threshold adjustment using the external patients. The diagnostic validation includes the 5-variable primary nomogram, the

6-variable paper-text model, the 8-variable data-driven model and the T/N baseline. The main prognostic comparison uses the 5-variable primary Cox model and the T/N baseline, while the remaining Cox variants are reported as supplementary comparisons. This external evaluation is performed before the report turns to the TabPFN-based extensions.

4 External Validation in the University Hospital Erlangen Cohort

4.1 Purpose and validation design

Section 3 reproduced the diagnostic and prognostic models of the reference study (Li et al., 2023) using data from the Surveillance, Epidemiology, and End Results (SEER) programme. This section evaluates whether those models retain predictive value in an independent hospital cohort from University Hospital Erlangen (*Universitätsklinikum Erlangen*, abbreviated as UK Erlangen). The cohort contains 30 patients and was not used to develop or select any of the models. This analysis concerns only the conventional reference-study models; it does not validate the foundation-model, embedding, multilayer-perceptron or synthetic-data experiments presented later in the report.

All model components were kept unchanged during external validation. In this report, an unchanged or "frozen" model means that its predictors, coefficients, categorical definitions and decision rules were taken directly from the SEER reproduction. No model was refitted, no variable was reselected, no predicted probability was recalibrated and no threshold was chosen using the Erlangen patients. The diagnostic analysis evaluates the 5-variable logistic-regression nomogram as the primary model. The 6-variable specification described in the reference-study text, the 8-variable data-driven reproduction and a comparator using tumour (T) and nodal (N) stage were also evaluated. The prognostic analysis evaluates the 5-variable Cox proportional-hazards nomogram as the primary model and compares it with a Cox model containing only T and N stage. This design assesses how the SEER-trained models perform in a different clinical population, without adapting them to the small external cohort.

4.2 Cohort composition

The external dataset contained 30 unique patients. All 30 entered the diagnostic cohort: 11 (36.7%) had liver metastasis at diagnosis and 19 did not. The same 11 liver-metastasis-positive patients formed the prognostic cohort; 7 died during follow-up and 4 were censored, meaning that follow-up ended before death was observed. Table 18 summarises the main cohort characteristics. Continuous variables are reported as median [first quartile, third quartile], and categorical variables as count (percentage). Exact tumour diameter was available for 26 diagnostic patients and 8 prognostic patients. One patient had missing N stage. Marital status was unknown for 11 diagnostic patients, including 2 patients in the prognostic cohort.

The external cohort differed from the SEER cohorts used for model development. Liver metastasis was present in 36.7% of the Erlangen diagnostic cohort compared with 17.3% of the SEER diagnostic cohort. Within the Erlangen prognostic cohort, 5 of 11 patients had T4 disease and only 1 underwent primary-tumour surgery. The different case mix provides a meaningful test of whether the model predictions transfer to another setting. However, the small number of patients limits the precision of every estimate and does not support reliable subgroup comparisons.

Table 18: Characteristics of the University Hospital Erlangen external cohort. The prognostic cohort is the liver-metastasis-positive subset of the diagnostic cohort. Percentages use the full cohort size as the denominator; consequently, the N1 percentages include the patient with missing N stage in the denominator. Tumour-size summaries include only patients with an exact measurement in millimetres.

Characteristic	Diagnostic cohort	Prognostic subset
Records	30	11
Age at diagnosis, years	54.0 [46.0, 64.8]	52.0 [43.5, 69.0]
Male sex	15 (50.0%)	7 (63.6%)
Grade III–IV	4 (13.3%)	3 (27.3%)
AJCC T4	5 (16.7%)	5 (45.5%)
AJCC N1	11 (36.7%)	6 (54.5%)
Primary-tumour surgery	20 (66.7%)	1 (9.1%)
Tumour size, mm	33.5 [22.8, 53.5]	42.5 [26.5, 65.0]
Liver metastasis at diagnosis	11 (36.7%)	11 (100%)
Overall-survival follow-up, months	–	24.0 [10.0, 37.0]
Deaths	–	7 (63.6%)

4.3 Data harmonisation and model compatibility

The Erlangen variables were converted to the same definitions, units and categories used to train the SEER models. Patient identifiers were replaced with project-specific pseudonyms, and only aggregate results are presented in this report. Sex and explicit yes/no variables were converted to binary indicators. Histological grades I–IV were represented as values 1–4. Morphology codes from the International Classification of Diseases for Oncology, Third Edition (ICD-O-3), were mapped to the detailed diagnostic categories and to the broader carcinoid-versus-other grouping required by the prognostic model. Primary pancreatic site was grouped as head, body, tail or other. Tumour and nodal stages followed the categories of the American Joint Committee on Cancer (AJCC) and were retained as categorical variables.

Exact tumour diameters were grouped as 0–26 mm, 26–36 mm or greater than 36 mm, matching the SEER model definitions. When a tumour size was reported only as a bound, such as less than 20 mm or greater than 40 mm, the bound was used to assign the valid size category but was not replaced by an invented exact value. Unknown marital status and missing N stage were likewise retained as missing rather than assigned to an observed category.

After recoding, the data were checked for unique patients, valid categories, compatible units and correct restriction of the prognostic analysis to patients with liver metastasis. All observed categories were compatible with the fitted models. Because the analyses used complete cases for the predictors required by each model, the evaluable sample varied slightly. The 4 diagnostic models shared a common set of 29 patients because 1 patient lacked N stage. The primary prognostic nomogram could be applied to all 11 patients, whereas the comparison with the T/N-stage model used 10 patients. Supplementary prognostic models that also required marital status could be applied to only 8 patients. These differences arise from missing predictor values, not from errors in recoding.

4.4 Evaluation methods and uncertainty

Diagnostic discrimination was measured with the area under the receiver operating characteristic curve (AUC). An AUC of 0.5 represents chance-level ranking, whereas 1.0 represents perfect separation of patients with and without liver metastasis. Overall probability error was measured with the Brier score, for which lower values indicate more accurate probabilities. Calibration was assessed with calibration-in-the-large and calibration slope. Their ideal values are 0 and 1, respectively. Decision-curve analysis compared the models' net benefit across possible risk thresholds.

Prognostic discrimination was assessed with Harrell's concordance index (C-index) and with cumulative/dynamic AUC at 12, 24 and 36 months. The C-index measures whether patients assigned higher risk tend to experience death earlier. The time-specific AUC measures discrimination between patients who died by a given time and those who remained event-free beyond that time, while accounting for censoring. Prediction error at each time point was measured with a Brier score using inverse probability of censoring weighting (IPCW). Observed survival was estimated with the Kaplan–Meier method and compared with the mean survival predicted by the model.

Uncertainty was estimated with 5,000 patient-level bootstrap resamples. The same resampled patients were used across models within each task to make comparisons as consistent as possible. Diagnostic resampling preserved the numbers of patients with and without liver metastasis. Reported intervals are percentile 95% confidence intervals for performance of the already fitted models in this external sample. They do not include uncertainty from the earlier SEER variable selection or model-development process.

4.5 Diagnostic external validation

All 3 multivariable diagnostic models showed strong discrimination in the common 29-patient set (Table 19). The primary 5-variable nomogram achieved an AUC of 0.934 (95% confidence interval 0.809–1.000), and the 8-variable data-driven model achieved 0.942 (0.837–1.000). The T/N-stage comparator achieved 0.739 (0.521–0.927). The wide, overlapping intervals do not support a precise ranking of the 3 multivariable models. Brier scores were approximately 0.10 for those models and 0.189 for the T/N-stage comparator.

Table 19: Diagnostic performance in the 29 patients with all predictors required by the 4 models. Values in parentheses are 95% patient-bootstrap confidence intervals. Calibration-in-the-large (CITL) compares the overall observed event frequency with the model predictions; ideal CITL is 0 and ideal calibration slope is 1.

Model	n	AUC	Brier score	CITL	Calibration slope
Nomogram figure (5 variables)	29	0.934 (0.809–1.000)	0.098 (0.045–0.162)	0.515 (0.009–1.009)	1.845 (1.077–4.616)
Paper-text specification (6 variables)	29	0.916 (0.772–1.000)	0.099 (0.037–0.176)	0.112 (−0.545–0.876)	1.132 (0.667–1.908)
Pipeline-selected model (8 variables)	29	0.942 (0.837–1.000)	0.097 (0.036–0.174)	0.435 (−0.255–1.293)	1.116 (0.716–7.701)
AJCC T/N baseline	29	0.739 (0.521–0.927)	0.189 (0.122–0.258)	0.719 (0.231–1.244)	0.667 (0.038–2.540)

For the primary nomogram, the positive CITL indicates that the observed frequency of liver metastasis was higher than the model predicted on average. Its calibration slope above 1 indicates

that the predicted risks varied less across patients than the observed outcomes supported. The 6-variable reference-text model was closer to the ideal calibration values by point estimate, but its confidence intervals were also broad. The grouped calibration plot and decision curves in Figure 10 are therefore descriptive. With only 10 patients with liver metastasis in the common analysis set, they cannot identify a stable risk threshold or establish clinical benefit.

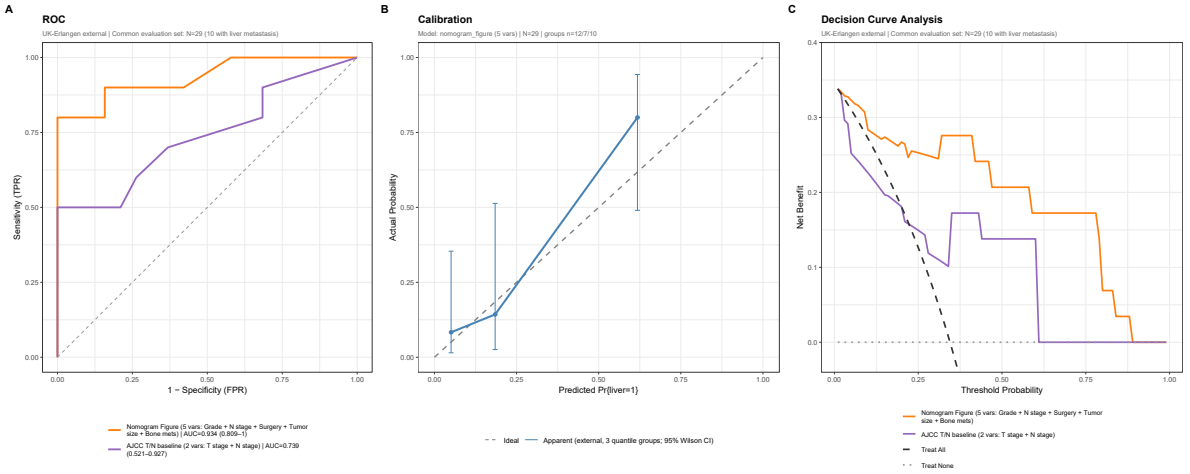


Figure 10: Diagnostic external validation in the common 29-patient set: receiver operating characteristic curves (A), calibration across 3 predicted-risk groups (B) and decision-curve analysis (C). The models were trained on the reproduced SEER cohort and applied without modification. The curves are imprecise because the common set contains only 10 patients with liver metastasis.

4.6 Prognostic external validation

The primary 5-variable prognostic nomogram—containing brain metastasis, age, grade, histological group and primary-tumour surgery—could be evaluated in all 11 liver-metastasis-positive patients. Its C-index was 0.833 (95% confidence interval 0.625–1.000). In the common 10-patient set, the nomogram's C-index was 0.824 compared with 0.676 for the T/N-stage model (Table 20). The nomogram's AUC estimates were 0.954 at 12 months, 0.936 at 24 months and 0.833 at 36 months when all 11 eligible patients were used. All confidence intervals were wide. The T/N-stage model's AUC also changed considerably across time points, reflecting the very small numbers of deaths and surviving patients available at each time rather than reliable evidence of superior performance at 36 months.

Table 20: External prognostic discrimination for the SEER-trained 5-variable nomogram and the T/N-stage comparator. The first row uses all 11 patients evaluable for the nomogram; the remaining rows use the same 10 patients for both models. Values in parentheses are 95% patient-bootstrap confidence intervals.

Model and evaluation set	n	Deaths	C-index	AUC_{12}	AUC_{24}	AUC_{36}
Nomogram, all eligible	11	7	0.833 (0.625–1.000)	0.954 (0.750–1.000)	0.936 (0.638–1.000)	0.833 (0.407–1.000)
Nomogram, common comparison	10	7	0.824 (0.611–1.000)	0.952 (0.750–1.000)	0.933 (0.625–1.000)	0.839 (0.424–1.000)
AJCC T/N, common comparison	10	7	0.676 (0.400–0.934)	0.667 (0.320–1.000)	0.533 (0.111–1.000)	0.845 (0.446–1.000)

For the primary nomogram, the IPCW Brier scores were 0.116, 0.123 and 0.153 at 12, 24 and 36

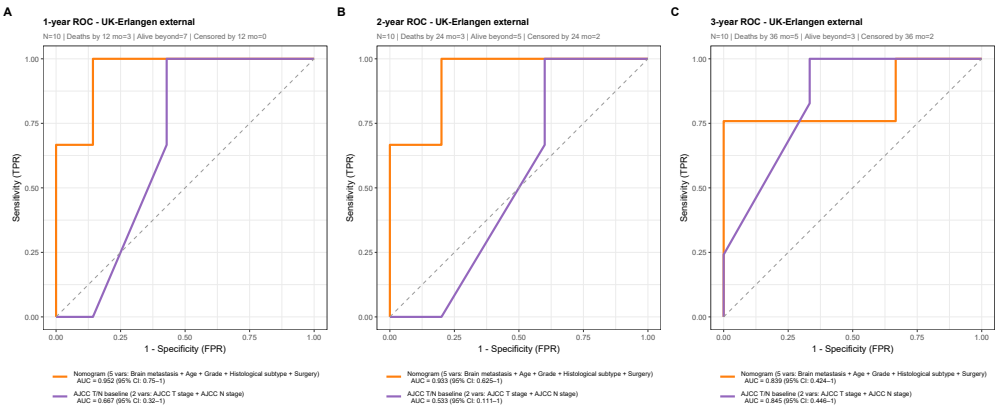


Figure 11: Time-specific receiver operating characteristic curves at 12, 24 and 36 months for the prognostic nomogram and T/N-stage model in their common 10-patient set. Only 3 deaths occurred by 12 and 24 months and 5 by 36 months, so the estimates are highly discrete and imprecise.

months, respectively (Table 21). The difference between Kaplan–Meier observed survival and mean predicted survival was 0.013 at 12 months, 0.139 at 24 months and −0.030 at 36 months. Every confidence interval included 0 and was too wide to demonstrate close agreement between observed and predicted survival. Figure 12 also displays calibration within 3 predicted-risk groups. Each group contains only 3 or 4 patients, so these group-level estimates are exploratory. Prognostic decision curves are not shown because the cohort did not meet the prespecified minimum of 20 evaluable patients and 5 deaths at a given time point.

Table 21: Prediction error and calibration of the primary 5-variable prognostic nomogram in all 11 eligible patients. The calibration gap is Kaplan–Meier observed survival minus mean predicted survival; 0 indicates agreement. Values in parentheses are 95% patient-bootstrap confidence intervals.

Months	n	IPCW Brier	Kaplan–Meier survival	Mean predicted survival	Calibration gap
12	11	0.116 (0.052–0.201)	0.716 (0.488–1.000)	0.703	0.013 (−0.208–0.204)
24	11	0.123 (0.046–0.222)	0.716 (0.488–1.000)	0.577	0.139 (−0.070–0.326)
36	11	0.153 (0.046–0.278)	0.430 (0.191–0.967)	0.459	−0.030 (−0.360–0.288)

The prognostic nomogram's risk cutpoint of 0.232769 was derived from the SEER training cohort and applied unchanged. It assigned 4 Erlangen patients to the low-risk group and 7 to the high-risk group. Median overall survival was 44 months in the low-risk group and 29 months in the high-risk group; the log-rank test gave $p = 0.191$. In Figure 13, the high-risk curve reaches 0 when the last patient at risk dies at 47 months. The low-risk curve remains flat at later times because 1 censored patient was followed for 199 months. These are valid Kaplan–Meier estimates for the observed patients, but the 2 small groups do not support stable estimates of population survival.

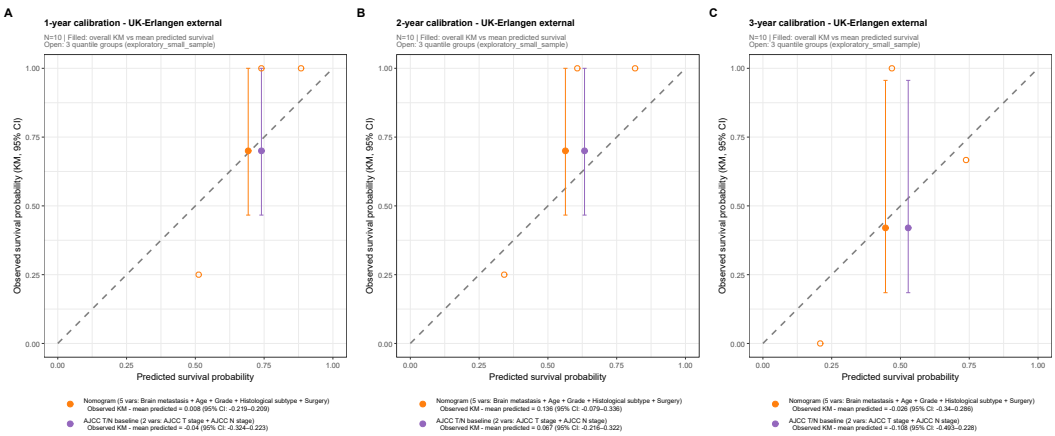


Figure 12: Calibration of the primary 5-variable prognostic nomogram at 12, 24 and 36 months in all 11 eligible patients. Filled points compare overall Kaplan–Meier survival with mean predicted survival; open points show the exploratory estimates for 3 predicted-risk groups. Wide confidence intervals and estimates at 0 or 1 reflect the very small group sizes.

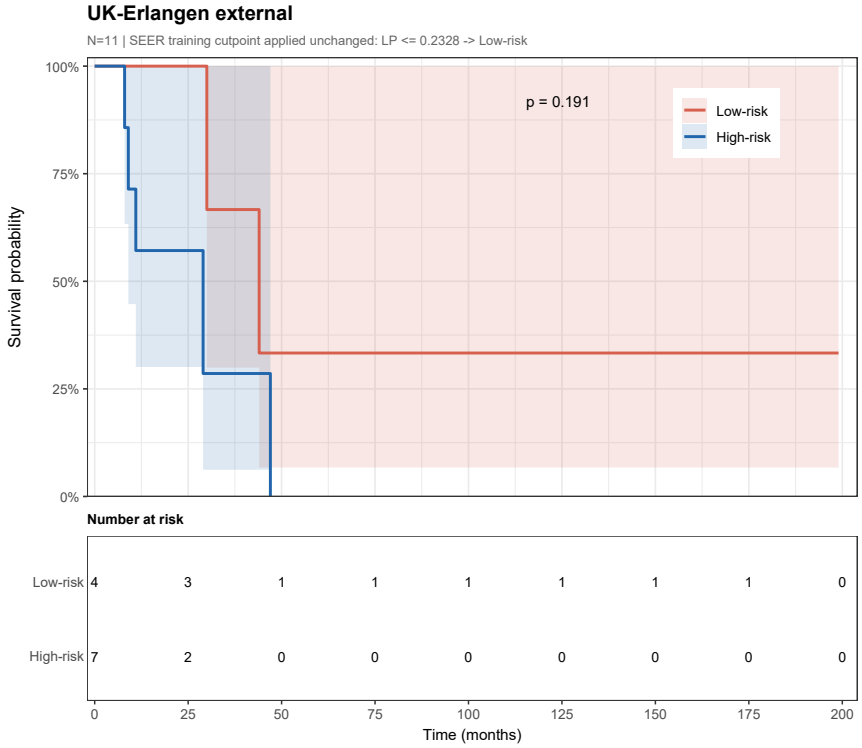


Figure 13: Kaplan–Meier survival after applying the unchanged SEER-derived risk cutpoint to the primary 5-variable prognostic nomogram. Shading shows pointwise 95% confidence intervals, and the lower panel gives the number of patients remaining at risk. The groups contain only 4 and 7 patients; therefore, the non-significant log-rank result is inconclusive and does not demonstrate equal survival.

4.7 Interpretation and transition

The independent Erlangen cohort shows that the SEER-trained diagnostic models retained strong ability to rank patients by the presence of liver metastasis. The primary prognostic nomogram also ranked survival outcomes better than the T/N-stage comparator by point estimate. These findings are encouraging, but the wide confidence intervals show that the exact performance values remain uncertain.

The calibration and decision-curve results require greater caution. The cohort contains only 30 patients for diagnosis and 11 for prognosis, and several analyses use even fewer complete cases. It is therefore not possible to make reliable claims about clinical benefit, performance in patient subgroups or readiness for clinical deployment. The results support the technical applicability of the unchanged SEER models to the available Erlangen data, while also showing the need for validation in a substantially larger independent cohort.

This section validates only the diagnostic and prognostic models reproduced from the reference study. The next section returns to the SEER data and introduces the Tabular Prior-Data Fitted Network (TabPFN) representations and the generation of synthetic cohorts containing both predictors and outcomes for the subsequent experiments.

5 Tabular Prior-Data Fitted Network Transformations and Dataset Construction

5.1 Purpose and scope

This section describes 2 ways in which the processed Surveillance, Epidemiology, and End Results (SEER) cohorts were transformed using a Tabular Prior-Data Fitted Network (TabPFN) (Hollmann et al., 2025). The first transformation creates a numerical embedding for every real patient. An embedding replaces the original clinical variables with a vector of numbers intended to summarise the information that TabPFN learned from those variables and the task outcome. The second transformation creates entirely new synthetic records by learning a distribution over the clinical predictors and outcomes and then sampling from it.

The 2 transformations serve different purposes. Embeddings retain the original patients and outcomes but change how their clinical information is represented. Synthetic generation retains the original clinical column structure but creates new combinations of predictors and outcomes. The resulting datasets are used in later sections to examine predictive performance and the extent to which the synthetic records resemble the real cohorts.

Both transformations in this section were created once from the complete diagnostic or prognostic cohort. This is important for interpretation. They are suitable for studying the fixed transformed datasets, but they do not provide an independent evaluation of a transformation learned using training data alone. The later augmentation experiment therefore generates synthetic records separately within each training portion and evaluates the resulting models only on real patients that were not used by the generator.

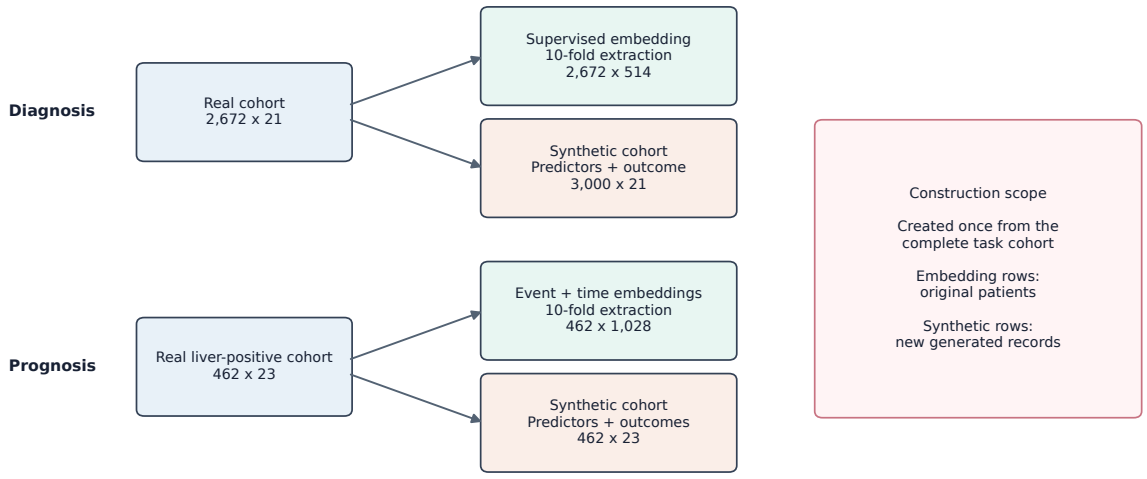


Figure 14: Transformation routes and resulting dataset dimensions. Embedding rows represent the original patients and retain their outcomes. Synthetic rows are newly generated records containing clinical predictors and outcomes. Both transformations shown here were constructed once from the complete cohort for the corresponding task.

5.2 Input variables and resulting datasets

Both transformations start from the diagnostic and prognostic cohorts described in Section 2. Each task contains 19 clinical predictor columns. This input is wider than the 17 clinical descriptor families considered during the reference-model variable-selection stage because 2 clinical concepts appear in 2 forms. Nodal stage is represented both as the American Joint Committee on Cancer N category and as a binary indicator of nodal involvement. Tumour size is represented both as an exact measurement in millimetres and as a grouped category. Comparisons with the reference models therefore reflect both a new numerical representation and a broader input specification.

Patient identifiers were retained only to match transformed rows to their corresponding patients and were never used as model inputs. For diagnosis, liver metastasis at diagnosis is the outcome. For prognosis, death status and observed follow-up time in months form the outcome. Liver-metastasis status is retained in the prognostic dataset for consistency, but it is not used as a predictor because all 462 patients in that cohort have liver metastasis.

Table 22: Inputs and resulting data for each transformation. Predictor counts exclude patient identifiers and outcomes.

Route	Task	Input to transformation	Outcome information	Resulting data
Embedding	Diagnostic	19 clinical predictors	Liver-metastasis label	patient identifier + 512 coordinates + label
Synthetic cohort	Diagnostic	19 predictors + label	Generated predictors and outcome	Original 21-column clinical schema
Embedding	Prognostic	19 clinical predictors	Event and log-follow-up blocks	patient identifier + 1,024 coordinates + 3 retained outcomes
Synthetic cohort	Prognostic	19 predictors + event + log-follow-up	Generated predictors and outcomes	Original 23-column clinical schema

Table 23 gives the dimensions of the 4 resulting datasets. The diagnostic embedding dataset contains 512 numerical coordinates for each of the 2,672 original patients. The prognostic embedding combines 2 blocks of 512 coordinates and therefore contains 1,024 coordinates for each of the 462 original patients. Synthetic generation produced 3,000 diagnostic records and 462 prognostic records. These generated records are not additional observed patients and do not increase the amount of independent clinical evidence.

The transformations used TabPFN 8.0.8, the TabPFN extensions package 0.4.2 and PyTorch 2.5.1. Both routes used a random seed of 42 and graphics processing unit (GPU) acceleration. Table 24 summarises the settings that differ between embedding extraction and synthetic generation.

Table 23: Dimensions and record identity of the 4 transformed datasets. Feature counts exclude patient identifiers and outcomes.

Dataset	Rows	Numerical or clinical features	Total columns	Record identity
Diagnostic embeddings	2672	512	514	Original patients
Diagnostic synthetic cohort	3000	19	21	New generated records
Prognostic embeddings	462	1024	1028	Original patients
Prognostic synthetic cohort	462	19	23	New generated records

Table 24: Configuration used to construct the embedding and synthetic datasets.

Route	Data used for fitting	Principal settings	Reproducibility settings
Embeddings	Complete task cohort; ten-fold row-wise extraction	4-member ensemble; mean aggregation	seed 42; GPU acceleration
Synthetic generation	One model on the complete task cohort	temperature 1.0; 24 feature-order permutations; 4 estimators	seed 42; GPU acceleration

5.3 Supervised embedding construction

The embeddings were created with 10-fold out-of-fold extraction. The patients were divided into 10 groups. For each group, TabPFN used the other 9 groups as labelled context and then produced embeddings for the patients in the group that had been left out. Consequently, a patient's own outcome was not used when constructing that patient's embedding. Six categorical predictors were passed directly to TabPFN as categorical variables; the alternative ordinal encoding was not needed.

The construction can be described compactly as follows. Let $\mathcal{I}_1, \ldots, \mathcal{I}_{10}$ denote the 10 patient groups, and let $\mathcal{D}_{-k}$ contain the predictors and outcomes of all patients outside group k. For patient $i \in \mathcal{I}_k$, ensemble member m produced the hidden representation

$$\mathbf{e}_i^{(m)} = h_m(\mathbf{x}_i; \mathcal{D}_{-k}), \qquad i \in \mathcal{I}_k, \tag{1}$$

where $\mathbf{x}_i$ is the patient's clinical predictor vector and h_m is the target-column representation from the final TabPFN layer, obtained using the other 9 groups as labelled context. The 4 ensemble representations were then averaged coordinate by coordinate:

$$\mathbf{e}_i = \frac{1}{4} \sum_{m=1}^{4} \mathbf{e}_i^{(m)} \in \mathbb{R}^{512}. \tag{2}$$

Thus, each output row still describes the same patient, but the original 19 predictors are replaced by a dense numerical summary learned from relationships between predictors and the task outcome.

For diagnosis, a TabPFN classifier used the 19 clinical predictors and liver-metastasis outcome to produce a 512-coordinate representation for each patient. Four ensemble members produced

separate representations, which were averaged coordinate by coordinate. The final diagnostic dataset contains the patient identifier, 512 embedding coordinates and the unchanged outcome, for a total of 514 columns and 2,672 rows.

For prognosis, 2 separate 512-coordinate representations were created. A classifier learned a representation associated with whether death was observed, while a regressor learned a representation associated with the logarithm of 1 plus observed follow-up time. The 2 blocks were combined into 1,024 coordinates. If t_i is follow-up time in months and δ_i indicates whether death was observed, this step is

$$u_i = \log(1 + t_i), \qquad \mathbf{e}_i^{\text{prog}} = \mathbf{e}_i^{\text{event}} \,\|\, \mathbf{e}_i^{\text{time}} \in \mathbb{R}^{1024}, \tag{3}$$

where the event block uses δ_i as its target, the time block uses u_i, and $\|$ denotes concatenation. This gives the representation information about both observed event status and follow-up time without treating either block as a complete survival model. The final prognostic dataset contains the patient identifier, the 1,024 coordinates, liver-metastasis status, follow-up time and death status, for a total of 1,028 columns and 462 rows. In both embedding datasets, patient membership and outcomes remained unchanged, and every embedding value was finite.

The 10-fold procedure prevents a patient's own outcome from directly entering that patient's representation, but it is not a complete outer validation design. The embeddings were created once for the entire cohort before the later predictive-model splits were defined. A patient assigned to a later evaluation set could therefore have contributed to the TabPFN models used to construct embeddings for other patients. Results based on these fixed embeddings are accordingly treated as exploratory. A stricter evaluation would repeat the full embedding procedure using only the training patients within every outer split.

The follow-up-time block also does not model censoring directly. It treats each patient's observed follow-up time as a regression target, including patients for whom death was not observed. A later Cox proportional-hazards model can account for censoring when it is fitted, but it cannot change how this upstream embedding was constructed.

5.4 Synthetic cohort generation

A separate TabPFN generator was fitted to the complete real cohort for each task. Rather than fitting one generator per outcome group, each generator learned the clinical predictors and outcomes together. Synthetic records were then sampled without imposing a fixed number of patients with liver metastasis or a fixed number of deaths.

In simple terms, the generator first represented each complete real record as one vector. For diagnosis this vector contained the clinical predictors and liver-metastasis outcome; for prognosis it contained the predictors, observed death status and transformed follow-up time:

$$\mathbf{z}_i^{\text{diag}} = (\mathbf{x}_i, y_i), \qquad \mathbf{z}_i^{\text{prog}} = (\mathbf{x}_i, \delta_i, \log(1 + t_i)). \tag{4}$$

Here y_i indicates liver metastasis at diagnosis. Categorical values were temporarily replaced by deterministic numerical codes so that all components of $\mathbf{z}_i$ could be passed to the generator.

TabPFN then formed an approximate sampling distribution from the complete set of real

records. The generator started with an empty row and filled its q columns sequentially. The resulting distribution can be written as

$$\widehat{p}(\mathbf{z}) = \prod_{\ell=1}^{q} \widehat{p}(z_{\pi_\ell} \mid z_{\pi_1}, \ldots, z_{\pi_{\ell-1}}) \,. \tag{5}$$

Here π is the fixed order in which columns appear in the generation matrix. At each step, the already generated columns provide the context for predicting the next one. For that conditional prediction, the generator averaged 24 permutations of the available context columns. This reduces sensitivity to how the context is presented to TabPFN; it does not mean that 24 independent synthetic cohorts were produced. New complete records were then sampled as

$$\widetilde{\mathbf{z}}_r \sim \widehat{p}(\mathbf{z} \mid \mathcal{D}_{\text{real}}), \qquad r = 1, \ldots, n_{\text{syn}}, \tag{6}$$

with temperature 1.0 and 4 ensemble estimators. The notation means that the new rows were sampled from patterns learned in the real cohort; it does not imply that the learned distribution is identical to the true patient population.

For diagnosis, the generator used 19 clinical predictors together with the liver-metastasis outcome and produced 3,000 synthetic records. For prognosis, it used the 19 clinical predictors, death status and the logarithm of 1 plus follow-up time, producing 462 records. Liver-metastasis status was excluded from the prognostic generator because it is constant in the real prognostic cohort; after generation, it was restored as present for every synthetic record. Both tasks followed the same generation procedure described above.

After sampling, the numerical codes were converted back to their clinical categories and continuous variables were restricted to the ranges observed in the corresponding real cohort. For the prognostic records, generated log-time $\widetilde{u}_r$ was returned to months by

$$\widetilde{t}_r = \text{round}[\text{clip}(\exp(\widetilde{u}_r) - 1, t_{\min}, t_{\max})] \,. \tag{7}$$

This final decoding step produced datasets with the same clinical columns and units as the real data, but with newly generated records and identifiers.

The chosen numbers of synthetic records are generation settings, not new effective sample sizes. Every synthetic record is derived from a model fitted to the corresponding finite real cohort. The prognostic generator is also not a specialised survival model: it generates death status and observed follow-up time together but does not explicitly represent the censoring process.

The reconstruction rules for each variable type are summarised in Table 25.

Every generated record received a new task-specific identifier, and no real patient identifier was reused. The diagnostic and prognostic synthetic cohorts retain the same column order and variable definitions as their corresponding real cohorts, which allows the same modelling procedures to be applied to both.

Table 25: Representation and reconstruction rules used during synthetic generation. These rules keep each individual variable within its allowed range but do not guarantee that related variables remain mutually consistent.

Variable type	Representation during generation	Reconstruction rule
Nominal category	Deterministic numerical codes	Round and restrict to a valid code, then restore the corresponding category
Binary variable	Numeric 0 or 1	Round and restrict to 0 or 1
Histological grade	Ordered numerical values	Restrict to the observed range and select the nearest observed grade
Continuous variable	Numeric value	Restrict to the real-data range and round variables recorded as integers
Follow-up time	Logarithm of 1 plus months	Reverse the logarithmic transformation, restrict to the observed range and round to complete months
Constant prognostic liver status	Excluded from generation	Restore liver metastasis as present

5.5 Quality checks and observed distortions

The 4 transformed datasets had the expected dimensions. The embedding datasets preserved all patient identifiers and outcomes, and all numerical coordinates were finite. The synthetic cohorts contained no missing values, duplicated clinical records or complete clinical records that exactly matched a real patient. These checks confirm basic data integrity, but they do not establish a formal privacy guarantee or rule out partial memorisation of real-data patterns.

The diagnostic synthetic cohort contained liver metastasis in 471 of 3,000 records (15.7%), compared with 462 of 2,672 real patients (17.3%). Because the outcome frequency was not fixed during generation, the difference is an observed result rather than an error. The prognostic synthetic cohort contained 311 deaths among 462 records (67.3%), exactly matching the real count by chance. This match does not demonstrate preservation of survival times, censoring patterns or relationships between predictors and outcome.

Table 26: Outcome and grade-IV representation in the real and synthetic cohorts. The diagnostic outcome is liver metastasis at diagnosis; the prognostic outcome is observed death.

Task	Data	n	Outcome present	Grade IV
Diagnostic	Real	2672	462 (17.3%)	60 (2.2%)
Diagnostic	Synthetic	3000	471 (15.7%)	0 (0.0%)
Prognostic	Real	462	311 (67.3%)	33 (7.1%)
Prognostic	Synthetic	462	311 (67.3%)	0 (0.0%)

A more important distortion concerns histological grade. Grade IV was present in 60 diagnostic and 33 prognostic real patients but was absent from both synthetic cohorts. As Figure 15 shows, the generated cohorts also placed a larger proportion of records in grade I. The absence of grade IV means that effects involving this rare category cannot be evaluated in the synthetic data; it must not be interpreted as evidence that grade IV has no clinical effect.

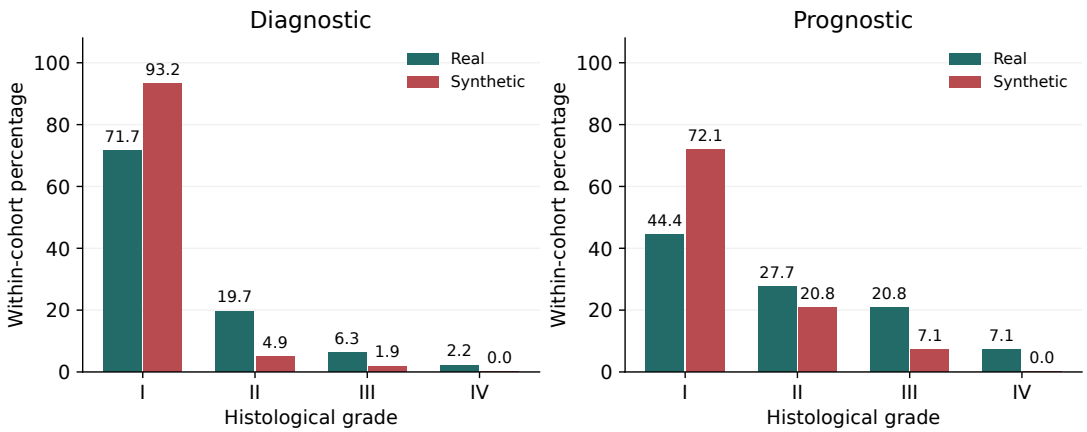


Figure 15: Distribution of histological grade in the real and synthetic cohorts. Bars show the percentage within each cohort. The synthetic bars describe one generated dataset per task and do not measure variation across repeated generator fits.

The consistency checks also compared pairs of variables that describe the same clinical concept. Nodal stage agreed with its binary indicator in every synthetic record. Tumour-size fields were less consistent: 46 of 3,000 diagnostic records and 11 of 462 prognostic records had a size group that did not match their generated measurement in millimetres. No such disagreement was present in the real cohorts. Neither synthetic cohort contained duplicate records or an exact match to a complete real record when the identifier was excluded. The tumour-size disagreements arose because the continuous measurement and grouped category were generated separately rather than deriving the group from the generated millimetre value. The records were retained as produced so that later analyses describe the generated datasets without post-generation correction. Section 7 examines these and other aspects of synthetic-data fidelity in greater detail.

5.6 Appropriate interpretation and transition

The software versions, random seed and transformation settings were recorded, but the exact pretrained TabPFN model-weight file was not. The existing datasets can therefore be analysed exactly as generated, but rerunning the transformations with a different downloaded weight version may not reproduce identical numerical values. This is a reproducibility limitation of both transformation routes.

Table 27: Appropriate and inappropriate interpretations of the transformed datasets.

Dataset	Appropriate use in this study	Interpretation not supported
Full-cohort embeddings	Exploratory analyses of models using the fixed TabPFN representation	Strict independent generalisation, survival-specific representation learning or a clinical meaning for individual coordinates
Full-cohort synthetic cohorts	Reproduction of the modelling workflow and descriptive assessment of similarity, support and internal consistency	External validation, additional observed patients, a privacy guarantee or an unbiased test of synthetic augmentation
Training-only synthetic cohorts used later	Model augmentation when generation is restricted to the training patients within each evaluation split	A replacement for evaluation on untouched real patients or for an independent external cohort

In summary, the embedding datasets provide new numerical representations of the original patients, whereas the synthetic datasets provide newly generated records in the original clinical format. Neither transformation creates new independent clinical evidence. The following section evaluates diagnostic and prognostic models using these transformed datasets, with the limitations of full-cohort construction, approximate censoring representation and incomplete rare-category support kept explicit.

6 Reproduction Using TabPFN-Derived Datasets

6.1 Purpose and evaluation designs

Section 5 introduced two distinct transformations of the clinical data. The embedding transformation represents each original patient by dense numerical coordinates learned by a Tabular Prior-Data Fitted Network (TabPFN). Synthetic generation instead produces new records containing clinical predictors and outcomes in the same format as the original cohort. This section examines whether the diagnostic and prognostic modelling procedures can be applied to both transformed datasets and how much predictive structure they retain.

These analyses do not constitute external validation. The embedding models are evaluated on the original patients, and the synthetic models are evaluated only on generated records. They also do not test synthetic augmentation, because no synthetic records are added to a real-data training set in this section.

Table 28 summarises the four evaluation designs. Out-of-fold (OOF) predictions are predictions for patients who were excluded from the downstream model fit that produced their prediction. For the embedding analyses, however, the TabPFN representations had already been constructed using the full cohort. The downstream OOF or training-validation procedures therefore reduce overfitting in the prediction model but do not provide a fully separated evaluation of the complete transformation-and-modelling process.

Table 28: Evaluation designs used for the TabPFN-derived datasets.

Dataset	Task	Model and evaluation	Interpretation
Real-patient embeddings	Diagnosis	Ridge logistic regression on 512 coordinates; 10-fold OOF downstream predictions for all 2,672 patients	Exploratory internal evaluation because the embeddings were created before the downstream folds
Real-patient embeddings	Prognosis	Ridge Cox model on 1,024 coordinates; trained on 323 patients and evaluated on 139 patients	Exploratory internal validation because the embeddings were created before the 323/139 split
Synthetic cohort	Diagnosis	Clinical logistic models fitted and evaluated on all 3,000 generated records	Fitted-cohort performance; no independent test set
Synthetic cohort	Prognosis	Clinical Cox models trained on 323 generated records and evaluated on 139 generated records	Internal validation within the generated cohort; no evaluation on real patients

6.2 Performance summary across data sources

Tables 29 and 30 bring together the main results from the published study, the SEER reproduction, the independent UK Erlangen evaluation and the TabPFN-derived datasets. They provide a compact overview, but they are not a single model ranking. The published values come from a different patient cohort; the diagnostic SEER and complete-synthetic results were calculated in the same records used for model fitting; the Erlangen results are external; and the embedding results use held-out downstream predictions even though the representations were created from the complete cohort. Direct comparisons are therefore most reliable within rows that share the same data and evaluation procedure.

Table 29: Compact diagnostic performance summary. ROC AUC is the area under the receiver operating characteristic curve; brackets give the 95% confidence interval. Published values are from Li et al. (2023). The data-type column distinguishes evaluations using clinical records, TabPFN embeddings or fully synthetic records.

Data and evaluation	Data type	Model	n	ROC AUC [95% CI]	Meaning of the estimate
Published study: SEER cohort	Real	Published diagnostic model	1,998	0.877 [0.873–0.908]	Model-development cohort
Published study: external hospitals	Real	Published diagnostic model	–	0.893 [0.855–0.935]	Independent hospital data
Present SEER reproduction	Real	Five-variable nomogram	2,672	0.880 [0.863–0.895]	Same cohort used for fitting and evaluation
Present SEER reproduction	Real	Six-variable text specification	2,672	0.895 [0.880–0.910]	Same cohort used for fitting and evaluation
Present SEER reproduction	Real	Eight-variable selected model	2,672	0.903 [0.888–0.917]	Same cohort used for fitting and evaluation
Present SEER reproduction	Real	T/N-stage comparator	2,672	0.763 [0.742–0.783]	Same cohort used for fitting and evaluation
UK Erlangen	Real	Five-variable nomogram	29	0.934 [0.809–1.000]	Independent external evaluation
UK Erlangen	Real	Six-variable text specification	29	0.916 [0.772–1.000]	Independent external evaluation
UK Erlangen	Real	Eight-variable selected model	29	0.942 [0.837–1.000]	Independent external evaluation
UK Erlangen	Real	T/N-stage comparator	29	0.739 [0.521–0.927]	Independent external evaluation
Real-patient TabPFN embeddings	Embedding	Ridge logistic regression	2,672	0.900 [0.885–0.915]	Ten-fold downstream held-out predictions; embeddings created from the complete cohort
Complete synthetic cohort	Synthetic	Five-variable nomogram	3,000	0.902 [0.889–0.915]	Same generated cohort used for fitting and evaluation
Complete synthetic cohort	Synthetic	Six-variable text specification	3,000	0.925 [0.914–0.937]	Same generated cohort used for fitting and evaluation
Complete synthetic cohort	Synthetic	Five-variable selected model	3,000	0.926 [0.915–0.938]	Same generated cohort used for fitting and evaluation
Complete synthetic cohort	Synthetic	T/N-stage comparator	3,000	0.770 [0.749–0.792]	Same generated cohort used for fitting and evaluation

Table 30: Compact prognostic performance summary. Harrell C is Harrell's concordance index, and the remaining columns give time-dependent AUC at 12, 24 and 36 months. The data-type column distinguishes evaluations using clinical records, TabPFN embeddings or fully synthetic records. Published values are from Li et al. (2023). For the synthetic rows, concordance is calculated in the 323-record training subset and AUC in the 139-record generated validation subset. AIC denotes Akaike information criterion; RF/RFE denotes random-forest recursive feature elimination. A dash indicates that the corresponding value was not reported.

Data and evaluation	Data type	Model	Harrell C	AUC 12 m	AUC 24 m	AUC 36 m
Published study: training	Real	Published primary model	0.752	0.832	0.794	0.792
Published study: internal validation	Real	Published primary model	0.740	0.832	0.857	0.818
Published study: external validation	Real	Published primary model	0.760	–	–	–
Present SEER training	Real	Five-variable primary model	0.740	0.889	0.826	0.822
Present SEER training	Real	Six-variable text-screened model	0.743	–	–	–
Present SEER training	Real	Cox/AIC-selected model	0.757	–	–	–
Present SEER training	Real	RF/RFE-selected model	0.758	–	–	–
Present SEER training	Real	Five-variable selected model	0.753	–	–	–
Present SEER training	Real	T/N-stage comparator	0.573	0.613	0.625	0.598
Present SEER internal validation	Real	Five-variable primary model	–	0.886	0.873	0.844
Present SEER internal validation	Real	T/N-stage comparator	–	0.391	0.443	0.505
UK Erlangen: all eligible	Real	Five-variable primary model	0.833	0.954	0.936	0.833
UK Erlangen: common set	Real	Five-variable primary model	0.824	0.952	0.933	0.839
UK Erlangen: common set	Real	T/N-stage comparator	0.676	0.667	0.533	0.845
Real-patient embedding validation	Embedding	Ridge Cox model	–	0.861	0.858	0.843
Synthetic training/validation	Synthetic	Five-variable primary model	0.706	0.831	0.848	0.830
Synthetic training/validation	Synthetic	T/N-stage comparator	0.538	0.585	0.544	0.567

6.3 Origin and interpretation of the clinical T/N comparators

The American Joint Committee on Cancer (AJCC) tumour stage (T) and nodal stage (N) were not included among the embedding coordinates. The T/N curves shown in this section come from separate models fitted with these two clinical variables. For the real-patient embedding analyses, the T and N values belong to the same SEER patients represented by the embeddings. All 2,672 diagnostic patients and all 462 prognostic patients matched one-to-one, with no missing or duplicated matches. For the synthetic analyses, the comparator uses the T and N values generated as part of each synthetic record.

The diagnostic comparison is not symmetric. The embedding ridge model is assessed with 10-fold OOF downstream predictions, whereas the clinical pipeline and the two-variable T/N model are fitted and evaluated on the complete diagnostic cohort. Their areas under the receiver operating characteristic curve (AUCs) should therefore be interpreted as descriptive results under different evaluation conditions, not as a formal head-to-head model comparison.

For prognosis, the embedding ridge model, the clinical reference model and the T/N comparator are trained on the same 323 patients and evaluated on the same 139 patients. This aligns the downstream evaluation sets. Nevertheless, the embeddings were constructed before this split, so the validation cohort was not independent of the representation-learning step.

6.4 Diagnostic reproduction

6.4.1 Real-patient embedding model

The diagnostic embedding model is a ridge-penalised logistic regression using all 512 embedding coordinates. Ridge regularisation is suitable for this dense, correlated representation and avoids selecting individual anonymous coordinates as if they were clinical variables. Ten-fold OOF downstream predictions produced an AUC of 0.900 and a Brier score of 0.0872. The Brier score is the mean squared difference between the predicted probabilities and the observed binary outcomes; lower values indicate smaller probability errors.

Table 31 places this result beside the eight-variable clinical pipeline and the two-variable clinical T/N comparator. Their AUCs were 0.903 and 0.763, respectively. Because the two clinical models were fitted and evaluated on the full cohort, the numerical similarity between the clinical pipeline and embedding model must not be interpreted as evidence of equivalent performance.

Table 31: Diagnostic discrimination in the 2,672 real SEER patients. Confidence intervals condition on the fitted models and the available predictions; they do not include uncertainty from repeating embedding construction or clinical variable selection.

Model and evaluation	Predictors	AUC [95% CI]	Predictor domain
Clinical pipeline, full-cohort fit/evaluation	8	0.903 [0.888–0.917]	Real clinical
Embedding ridge, 10-fold OOF downstream predictions	512	0.900 [0.885–0.915]	Real embedding
Clinical T/N comparator, full-cohort fit/evaluation	2	0.763 [0.741–0.784]	Real clinical

Calibration was summarised by grouping predictions into ten risk groups and regressing the

observed group frequencies on the mean predicted risks. This gave an intercept of 0.002 and a slope of 0.986. These grouped linear-regression values are not the conventional individual-level logistic calibration intercept and slope, and therefore do not establish near-perfect calibration. Calibration was not calculated for the clinical T/N comparator.

6.4.2 Synthetic diagnostic cohort

The clinical feature-selection procedure was repeated in the 3,000-record synthetic diagnostic cohort. The least absolute shrinkage and selection operator (LASSO) retained 15 clinical variables at the minimum-error penalty and five at the more conservative one-standard-error penalty. The five variables retained by the final pipeline were chemotherapy, primary-site surgery, nodal status, tumour stage and tumour-size group. This differs from the eight-variable model selected from the real cohort and indicates that the generated data contain a different predictor-outcome structure.

Table 32: Diagnostic discrimination within the synthetic cohort. Every model was fitted and evaluated on the same 3,000 generated records, so the values describe fitted-cohort performance rather than validation performance.

Model	Predictors	AUC [95% CI]	Evaluation
Text-defined	6	0.925 [0.914–0.937]	Full synthetic cohort
Nomogram-figure	5	0.902 [0.889–0.915]	Full synthetic cohort
Pipeline-selected	5	0.926 [0.915–0.938]	Full synthetic cohort
Synthetic T/N comparator	2	0.770 [0.749–0.792]	Full synthetic cohort

The pipeline-selected model had the largest AUC, 0.926. This value is optimistic because variable selection, model fitting and evaluation all used the same synthetic cohort. Re-estimating calibration on the same fitting data produced an intercept of zero and slope of one, as expected for an unpenalised logistic model evaluated on its training data; these values provide no evidence of performance on new records.

6.5 Prognostic reproduction

6.5.1 Real-patient embedding model

The prognostic embedding model is a ridge-penalised Cox proportional-hazards model using 1,024 coordinates: 512 from the death-status embedding and 512 from the follow-up-time embedding. The model was fitted on 323 patients and evaluated on 139 patients using the same split as the clinical reference model. The validation cohort contained 92 deaths during total follow-up, including 33 by 12 months, 50 by 24 months and 61 by 36 months.

The embedding model achieved AUCs of 0.861, 0.858 and 0.843 at 12, 24 and 36 months. The corresponding clinical-reference values were 0.886, 0.873 and 0.844. The confidence intervals overlap substantially, but the design does not support claims of equivalence or superiority

Table 33: Time-dependent AUC in the common real-patient validation cohort ($n = 139$). Marginal censoring weights were estimated within the validation cohort for each calculation.

Horizon (months)	Clinical reference	Embedding ridge	Clinical T/N comparator	Deaths by horizon
12	0.886 [0.826–0.946]	0.861 [0.789–0.933]	0.391 [0.284–0.498]	33
24	0.873 [0.817–0.930]	0.858 [0.797–0.920]	0.443 [0.346–0.540]	50
36	0.844 [0.779–0.910]	0.843 [0.779–0.907]	0.505 [0.410–0.599]	61

because the embeddings were created before the training-validation split. The clinical T/N comparator performed poorly, with AUCs of 0.391, 0.443 and 0.505.

The embedding model did not undergo a prognostic calibration analysis. Its results therefore support statements about discrimination and risk-group separation only, not about the accuracy of its predicted survival probabilities. The follow-up-time embedding also treats observed follow-up as a regression target without explicitly modelling censoring, which limits its interpretation as a survival-specific representation.

6.5.2 Synthetic prognostic cohort

The 462 generated prognostic records were divided into 323 training records and 139 validation records. Several Cox model specifications were considered because they answer different questions. The primary five-variable model reproduces the reference study's nomogram specification and contains brain metastasis, age, histological grade, histological group and primary-site surgery. Data-driven alternatives used backward selection by the Akaike information criterion (AIC), random-forest recursive feature elimination (RF-RFE), or the filtered union of both selections. A separate two-variable T/N model provided a simple stage-based comparator. The main validation results below use the primary five-variable Cox model, not the data-driven alternatives.

Table 34: Time-dependent AUC in the synthetic validation cohort ($n = 139$). Both predictors and outcomes are generated, so this evaluates consistency within the synthetic cohort rather than transfer to real patients.

Horizon (months)	Primary five-variable Cox model	Synthetic T/N comparator	Deaths by horizon
12	0.831 [0.753–0.909]	0.585 [0.451–0.719]	28
24	0.848 [0.784–0.911]	0.544 [0.432–0.657]	44
36	0.830 [0.762–0.898]	0.567 [0.470–0.665]	55

The primary model achieved validation AUCs of 0.831, 0.848 and 0.830 at 12, 24 and 36 months. Its concordance in the synthetic training cohort was 0.706 [0.672–0.741], compared with 0.538 [0.500–0.576] for the synthetic T/N model. The global proportional-hazards test for the primary model was significant ($p = 0.0027$); the surgery and histological-group terms also showed individual violations. Its hazard ratios should therefore not be assumed to remain constant throughout follow-up. Grade IV was absent from the entire synthetic cohort, so its association could not be estimated.

Calibration was assessed by grouped observed-versus-predicted regressions in the synthetic validation cohort. The slopes were 1.487, 1.323 and 1.226 at 12, 24 and 36 months. These are descriptive grouped summaries, not conventional individual-level survival-calibration slopes, and they apply only to this one generated cohort.

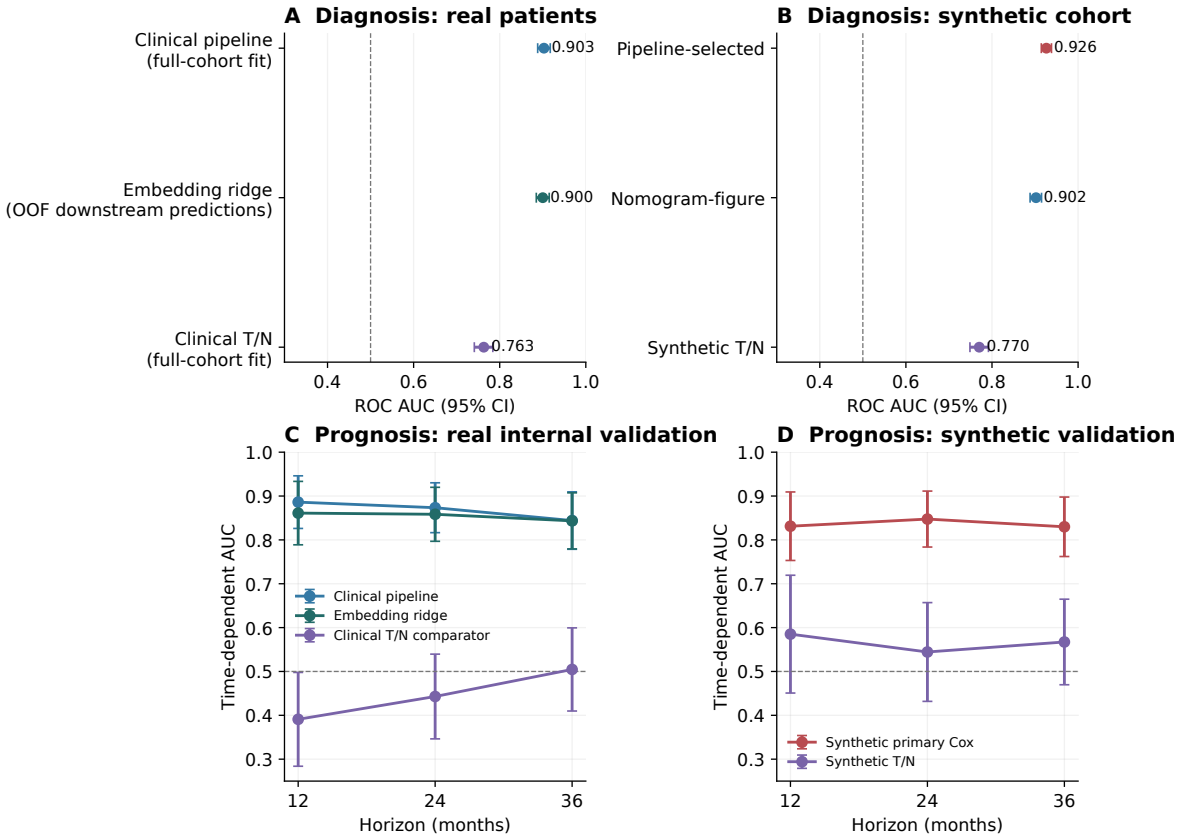


Figure 16: Discrimination results separated by data domain and evaluation design. Panels A and C use real SEER patients; Panels B and D use generated records. In Panel A, the embedding model uses downstream out-of-fold predictions while the clinical models use full-cohort fitted predictions. In Panel C, the downstream models share the same validation patients, but embedding construction preceded the split. Values in different panels are therefore not directly comparable.

For risk stratification, each prognostic model used the median linear predictor in its training cohort as the threshold between low- and high-risk groups. The same threshold was then applied unchanged to the corresponding validation cohort. Table 35 reports the ridge embedding model for the real patients and the primary five-variable Cox model for the synthetic patients.

Among the real validation patients, median overall survival (OS) was 84 months in the low-risk group and 18.5 months in the high-risk group. In the synthetic validation cohort, the corresponding medians were 94 and 24 months. These separations show that each model orders patients or generated records by risk within its own evaluation domain; they do not show that the synthetic model transfers to real patients.

Table 35: Validation-cohort risk groups for the real-patient embedding ridge model and the primary five-variable synthetic Cox model. The two cohorts contain different records and must not be pooled or compared as if they were one sample.

Data and model	Risk group	Patients	Deaths	Median OS (months)
Real-patient embedding ridge	Low-risk	73	40	84
Real-patient embedding ridge	High-risk	66	52	18.5
Primary synthetic Cox model	Low-risk	70	35	94
Primary synthetic Cox model	High-risk	69	59	24

6.6 Decision-curve analysis

Decision-curve analysis (DCA) expresses the clinical consequences of predictions as net benefit across a range of decision thresholds. For diagnosis, net benefit was calculated for all 2,672 real patients. The embedding curve uses OOF downstream predictions, whereas the clinical T/N curve uses probabilities from a model fitted on the complete cohort. This asymmetry makes their visual comparison descriptive rather than confirmatory.

For prognosis, Figure 17 shows the 24-month results in the 139 real validation patients. Patients censored before a given horizon would be excluded because their binary outcome at that time is unknown; all 139 patients were eligible at 24 months. This complete-case horizon approach does not use inverse-probability-of-censoring weights and may be biased when censoring is related to risk.

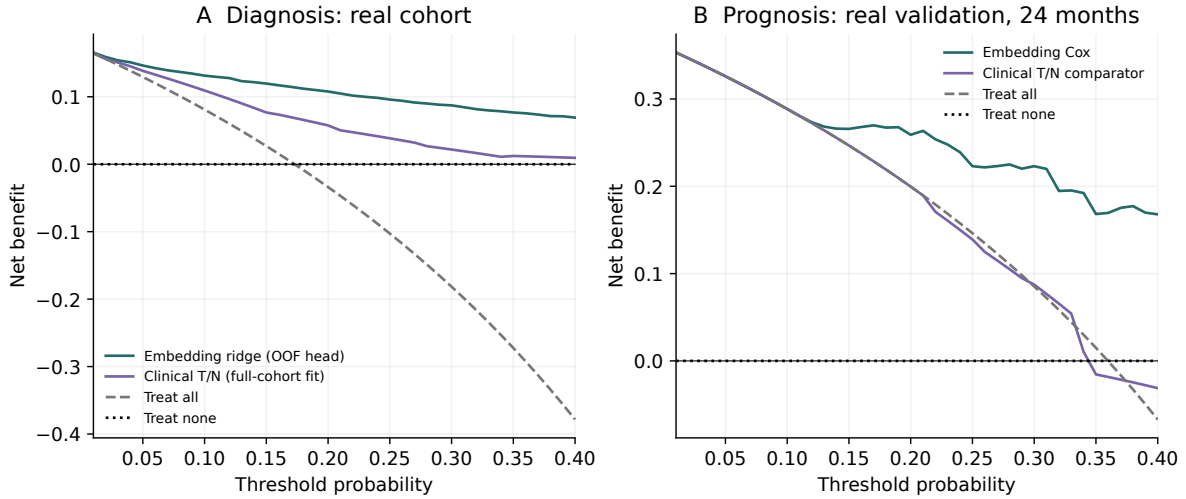


Figure 17: Decision curves for the real-patient embedding models and separate clinical T/N comparators. The diagnostic curves use different fitting and evaluation conditions. The prognostic curves use the common 139-patient validation cohort at 24 months and the complete-case horizon approximation.

No uncertainty intervals for net benefit were calculated, and no clinically preferred threshold range was specified in advance. The curves therefore show how the available predictions behave under the stated net-benefit calculation; they do not establish clinical usefulness.

6.7 Interpretation and transition

The embedding analyses show that regularised prediction models can recover strong diagnostic and prognostic ordering from the TabPFN coordinates. The resulting AUCs are close to those of the clinical reference models, but this is exploratory evidence. Full-cohort representation construction, different diagnostic evaluation procedures and incomplete calibration prevent a confirmatory comparison.

The synthetic analyses show that the clinical modelling procedures can be executed on generated cohorts and can recover internally discriminative models. They do not demonstrate clinical generalisation. Diagnostic models were fitted and evaluated on the same generated records, prognostic validation remained entirely synthetic, and the absence of grade IV patients limits the clinical support of both generated cohorts. The changed feature-selection results and the proportional-hazards violations further show that the synthetic data do not fully reproduce the structure of the real cohorts.

A stricter embedding comparison would construct each representation using only the training patients in an outer split, fit the clinical and embedding models within that split, and compare paired predictions on untouched real test patients. The following section complements the predictive results presented here by directly evaluating representation similarity and synthetic-data fidelity.

7 Dataset Similarity and Synthetic-Data Fidelity

7.1 Purpose and analysis framework

The two TabPFN transformations require different forms of evaluation. An embedding contains the original patients represented by new numerical coordinates, so the relevant question is whether relationships among those patients are retained. A synthetic cohort contains newly generated records in the original clinical format, so it can be compared with the real cohort in terms of distributions, clinical subgroups, survival patterns and fitted-model behaviour. No single similarity score can answer all of these questions.

Table 36 summarises the four complementary analyses. All synthetic-data results describe one generated diagnostic cohort and one generated prognostic cohort, each produced by a generator trained on the corresponding complete real cohort. The results do not include variation from retraining the generator or sampling additional synthetic cohorts, and they do not constitute external validation.

Table 36: Questions addressed by the similarity and fidelity analyses.

Analysis	Data compared	Main measures	Interpretation
Embedding structure	Clinical and embedding representations of the same patients	Trustworthiness, continuity and Mantel correlation	Preservation of local neighbours and overall distance ordering
Synthetic distribution	Real and generated records containing predictors and outcomes	Marginal differences, discriminator tests, correlation differences and nearest-record distance	Statistical differences detectable under the specified preprocessing and models
Clinical structure	Real and generated categories, subgroups, outcomes and survival curves	Category support, subgroup outcomes, survival differences and adjusted relationships	Clinical patterns that appear preserved, distorted or unavailable in the generated cohorts
Model portability	Identically specified models trained on real or generated records and scored on the real source cohort	Discrimination differences and coefficient-sign agreement	Compatibility within the source cohort, not independent generalisation

7.2 Preservation of embedding structure

Trustworthiness measures whether patients who are close in an embedding were also close in the clinical representation; it penalises new neighbours created by the transformation. Continuity measures whether clinically similar patients remain neighbours after transformation. Both metrics range from zero to one, with one indicating perfect neighbourhood preservation (Venna and Kaski, 2001). Results were calculated for neighbourhoods of 10 and 30 patients.

The Mantel analysis compares the ordering of all pairwise patient distances in the two representations using a Spearman correlation (Mantel, 1967). Significance was assessed with 499 permutations of patient labels. For every embedding block, none of the permuted correlations was as large as the observed value; the resulting value of 0.002 is the smallest permutation probability available with this number of permutations.

The diagnostic embedding preserved local structure most strongly, with trustworthiness of 0.953 and continuity of 0.952 for 10-neighbour sets (Table 37). Its Mantel correlation was 0.558, indicating moderate preservation of overall distance ordering. The prognostic event and follow-up-time blocks showed weaker local and global preservation. Combining the two prognostic blocks improved continuity and the Mantel correlation relative to either block alone, but remained below the diagnostic embedding.

Table 37: Structural preservation between the clinical representation and each TabPFN embedding. Every row uses all matched patients.

Embedding block	n	Trust $k = 10$	Continuity $k = 10$	Trust $k = 30$	Continuity $k = 30$	Mantel r	p
Diagnostic (512-D)	2672	0.953	0.952	0.929	0.925	0.558	0.0020
Prognostic event (512-D)	462	0.873	0.855	0.828	0.817	0.400	0.0020
Prognostic time (512-D)	462	0.842	0.838	0.805	0.809	0.401	0.0020
Prognostic combined (1,024-D)	462	0.873	0.884	0.845	0.856	0.495	0.0020

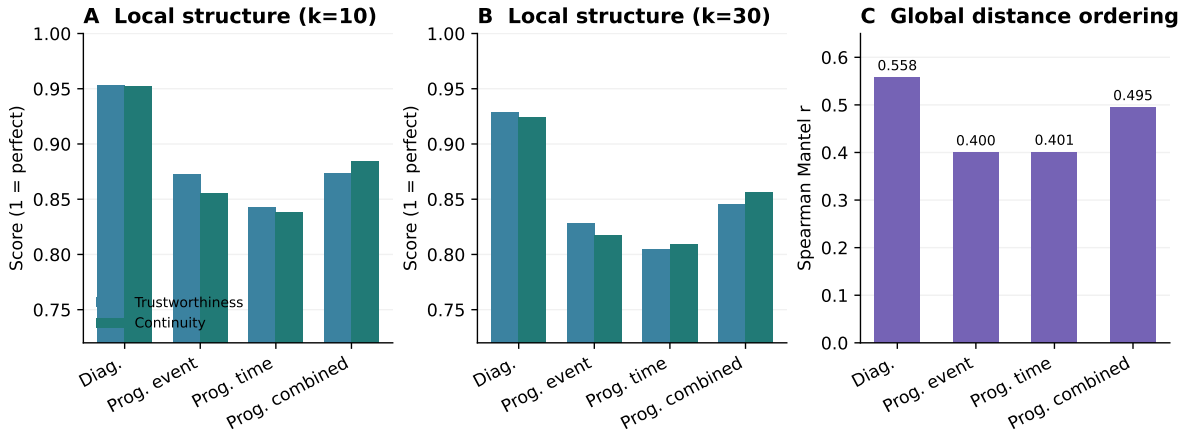


Figure 18: Local-neighbourhood preservation and overall distance-order preservation for the diagnostic and prognostic embeddings. These metrics describe representation structure, not predictive performance.

Two design choices are important when interpreting these results. First, the clinical representation used for the distance calculations includes the liver-metastasis outcome for diagnosis and both death status and follow-up time for prognosis. Because the embeddings were also supervised using these outcomes, the results describe preservation of outcome-inclusive clinical structure rather than predictor-only structure. Second, nominal categories were converted to numerical codes before standardised Euclidean distances were calculated. This imposes an artificial ordering on categories such as race and histology. The scores are therefore useful within this specified encoding but are not encoding-independent properties of the data.

Figure 19 provides a complementary qualitative view using t-distributed stochastic neighbour embedding (t-SNE). Each panel was fitted independently in its own feature space: Gower distances (Gower, 1971) were used for the real and synthetic clinical data, while standardised

Euclidean distances were used for the dense TabPFN coordinates. The axes and cluster locations therefore cannot be compared directly across panels.

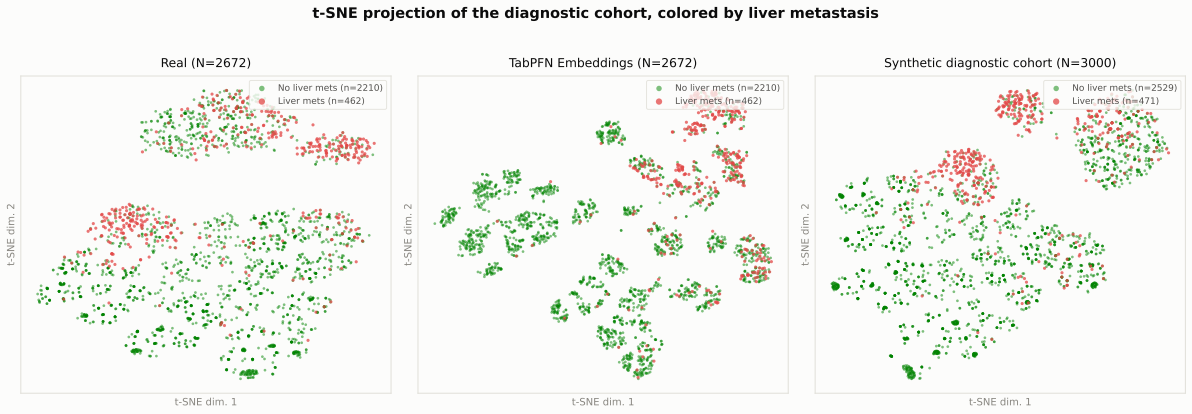


Figure 19: Independent t-SNE projections of the real diagnostic clinical space, the TabPFN embedding and the complete synthetic diagnostic cohort. Points are coloured by liver-metastasis status. The panels show local organisation within each representation; their coordinates do not form a shared space.

All three projections contain outcome-related local grouping together with substantial overlap between patients with and without liver metastasis. The embedding forms more compact local clusters, which is consistent with its high neighbourhood-preservation scores. The synthetic projection recreates broad outcome-associated grouping, but its cluster shapes and densities differ from the real clinical projection. Because t-SNE emphasises local neighbourhoods and can alter global geometry, the figure supports only a qualitative description; it does not establish distributional fidelity or performance in new patients.

7.3 Overall similarity of the synthetic cohorts

Two discriminator-based tests assessed whether real and synthetic records could be distinguished when predictors and outcomes were considered together. Nominal variables were represented by separate indicator columns, continuous variables were standardised and binary or ordinal variables were entered numerically.

The propensity-score mean squared error (pMSE) ratio compares the separation achieved by an in-sample logistic discriminator with the separation expected if real and synthetic records came from the same distribution (Snoke et al., 2018). A ratio near one indicates little detectable separation under this model. The classifier two-sample test (C2ST) uses five-fold out-of-fold logistic predictions; its area under the receiver operating characteristic curve (AUC) is expected to be 0.5 when the datasets cannot be distinguished (Lopez-Paz and Oquab, 2017).

Both synthetic cohorts were distinguishable from their real counterparts (Table 38). The C2ST AUC was 0.630 for diagnosis and 0.664 for prognosis, while the pMSE ratios were 31.748 and 9.500. The pMSE ratios should not be ranked across tasks because their reference values depend on sample size and the number of discriminator variables.

Average pairwise correlations were similar, with mean absolute differences of 0.018 for diagnosis and 0.045 for prognosis. The largest correlation differences were 0.084 and 0.154, showing that average agreement can conceal larger discrepancies for individual variable pairs.

Table 38: Overall real-versus-synthetic distribution checks. Correlation differences are absolute differences between Pearson-correlation matrices over numeric, binary and ordinal fields. Nearest-record distances use min–max scaled variables and numerical category codes, so their magnitudes are not comparable across tasks.

Task	pMSE ratio	C2ST AUC	Mean correlation difference	Maximum correlation difference	Median nearest-real distance	Exact real-row matches
Diagnosis	31.748	0.630	0.018	0.084	0.180	0
Prognosis	9.500	0.664	0.045	0.154	0.706	0

Most continuous-variable distributions were also close. The Wasserstein distance for age was 0.024 diagnostic interquartile ranges (IQRs) in the diagnostic cohort and 0.049 IQRs in the prognostic cohort. For tumour size, the corresponding values were 0.026 and 0.084 IQRs. The diagnostic tumour-size Kolmogorov–Smirnov test gave $p = 0.044$, but this was one of many unadjusted marginal tests and is not interpreted as an isolated finding.

No synthetic record exactly matched a complete real record. This is a useful basic check, but it does not demonstrate privacy. The nearest-record distance depends on cohort size, dimensionality and the numerical coding of nominal categories, and neither membership-inference risk nor a formal differential-privacy guarantee was evaluated.

7.4 Grade and subgroup preservation

Histological grade was the clearest marginal distortion (Table 39). In the diagnostic cohort, grade I increased from 71.7% of real patients to 93.2% of synthetic records, while grades II and III decreased from 19.7% and 6.3% to 4.9% and 1.9%. All 60 real grade-IV patients were absent from the synthetic cohort. In the prognostic cohort, grade I increased from 44.4% to 72.1%, grade III decreased from 20.8% to 7.1%, and all 33 grade-IV patients were absent. The generator also failed to reproduce the single diagnostic patient recorded with Goblet-cell carcinoid histology.

Table 39: Histological-grade representation in the real and synthetic cohorts. The distortion categories use the predefined relative-share thresholds: moderate for changes of at least 25% and high for changes of at least 50%. Lost indicates that a real category is absent from the synthetic cohort.

Task	Grade	Real (n)	Synthetic (n)	Real %	Synthetic %	Distortion category
Diagnostic	I	1917	2795	71.7	93.2	Moderate
Diagnostic	II	527	147	19.7	4.9	High
Diagnostic	III	168	58	6.3	1.9	High
Diagnostic	IV	60	0	2.2	0.0	Lost
Prognostic	I	205	333	44.4	72.1	High
Prognostic	II	128	96	27.7	20.8	Moderate
Prognostic	III	96	33	20.8	7.1	High
Prognostic	IV	33	0	7.1	0.0	Lost

The relationships between grade and outcome also changed. Among diagnostic grade-II records, liver-metastasis prevalence increased from 0.243 in the real cohort to 0.592 in the synthetic cohort. For grade III, it increased from 0.571 to 0.724. In the prognostic grade-II subgroup, mortality increased from 0.672 to 0.875 and median observed follow-up or death time decreased from 48 to 21.5 months. These are descriptive differences between cohorts, not causal effects of

grade. Because grade IV is completely absent, its outcome relationships cannot be evaluated in either synthetic cohort.

7.5 Survival-curve fidelity

Kaplan–Meier survival estimates were compared between the 462 real and 462 synthetic prognostic records. Each cohort contained 311 observed deaths. Synthetic survival was higher by 0.057 at 12 months, 0.030 at 24 months and 0.053 at 36 months (Table 40). Restricted mean survival time (RMST) through 36 months was 1.55 months longer in the synthetic cohort.

Table 40: Overall Kaplan–Meier survival and 36-month restricted mean survival time in the real and synthetic prognostic cohorts. Differences are synthetic minus real.

Horizon or summary	Real	Synthetic	Difference
12	0.759	0.816	0.057
24	0.651	0.681	0.030
36	0.556	0.609	0.053
RMST to 36	26.02	27.57	1.55

The overall two-sample log-rank test gave $p = 0.315$. This failure to detect a difference is not evidence that the curves are equivalent. The generated cohort was learned from the real cohort, and the analysis does not include uncertainty from repeating generation.

Subgroup comparisons revealed larger differences than the overall curve. The largest survival-probability difference across 12, 24 and 36 months was 0.310 for grade II, 0.233 for patients with bone metastasis and 0.200 for patients with lung metastasis. A difference of 0.667 for T0 disease was based on only three real and one synthetic record and is therefore a sparse-data warning, not a stable estimate. Grade IV had no synthetic survival curve. These subgroup comparisons did not include pointwise confidence intervals or adjustment for the number of comparisons.

7.6 Conditional relationships and interactions

To summarise a diverse set of discrepancies, each comparison was converted to a heuristic distortion score from zero to one. Scores of at least 0.2 were classified as moderate and scores of at least 0.4 as high. These are screening thresholds chosen for this analysis; they are not probabilities, confidence levels or limits of clinical acceptability.

Diagnostic adjusted relationships were mostly stable, although the estimated directions for lymph-node surgery and radiotherapy changed after treatment adjustment. Prognostic relationships were less stable: 8 of 21 conditional comparisons were classified as highly distorted, and chemotherapy changed direction under the treatment-adjusted specification. Ratio-based attenuation measures become unstable when the original association is close to zero, so comparisons below the predefined minimum-signal threshold were reported as unsupported rather than assigned a distortion score.

The interaction analysis used the same predefined coefficient contrast in the real and synthetic cohorts and repeated the comparison in 200 paired bootstrap samples. In diagnosis, 28 of 56 planned terms were estimable in both cohorts, and 2 had synthetic-minus-real percentile

Table 41: Distortion screening by analysis domain. Each cell gives the mean normalised score and the number of high-distortion comparisons among all scored comparisons. Means from different domains should not be compared as common effect sizes because each domain uses a different normalisation.

Domain	Diagnostic mean	Diagnostic high/total	Prognostic mean	Prognostic high/total
Category support	0.221	5/30	0.286	5/22
Conditional dependence	0.074	2/30	0.302	8/21
Model behaviour	0.062	1/4	0.179	2/4
Subgroup outcome	0.219	3/21	0.171	2/21
Subgroup size	0.159	3/22	0.208	4/22
Survival curves	–	–	0.460	9/28

intervals excluding zero. In prognosis, 21 of 32 terms were estimable, and 4 intervals excluded zero. The remaining 39 terms were unsupported, mainly because grade IV was absent or rare histological combinations could not be estimated. The six supported differences are listed in Table 42. These results are exploratory because no correction for multiple comparisons or clinically meaningful equivalence range was specified.

Table 42: Synthetic-minus-real interaction differences with 2.5th–97.5th percentile intervals excluding zero. Diagnostic differences are on the log-odds scale and prognostic differences on the log-hazard scale.

Task	Interaction contrast	Median difference	Percentile interval
Diagnostic	grade III x primary surgery	2.110	[0.017, 3.701]
Diagnostic	carcinoid NOS x grade III	-2.108	[-4.889, -0.015]
Prognostic	grade II x T2	0.481	[0.044, 1.048]
Prognostic	grade II x T3	0.317	[0.055, 0.653]
Prognostic	grade III x T3	1.053	[0.098, 2.002]
Prognostic	other histology x grade II	0.463	[0.089, 0.850]

7.7 Model portability within the source cohort

Identically specified models were trained separately on the real and synthetic cohorts and then scored on the complete real source cohort. For diagnosis, the paper-text logistic model differed by only −0.0001 AUC when trained on synthetic instead of real data, while the nomogram specification differed by −0.0070. For prognosis, the synthetic-trained text-screened Cox model was 0.026 lower in Harrell's concordance statistic, whereas the T/N Cox model was 0.0036 lower (Table 43).

Similar discrimination did not guarantee similar coefficient interpretation. Only 60% of coefficients in the text-screened prognostic model retained their sign, and the grade-II coefficient changed direction materially. The grade-IV coefficient could not be fitted because that category is absent from the synthetic cohort.

Table 43: Models trained on real or synthetic data and scored on the complete real source cohort. Differences are synthetic-data-trained minus real-data-trained scores. Neither score is based on an independent test cohort.

Model specification	Metric	Real-data trained	Synthetic-data trained	Difference	Sign agreement (%)	Material sign change
Diagnosis: paper-text logistic	AUC	0.894	0.894	-0.0001	90	None
Diagnosis: nomogram logistic	AUC	0.879	0.872	-0.0070	89	Grade II
Prognosis: text-screened Cox	Harrell's C	0.737	0.711	-0.0260	60	Grade II
Prognosis: T/N Cox	Harrell's C	0.576	0.572	-0.0036	83	Pooled rare T-stage indicator

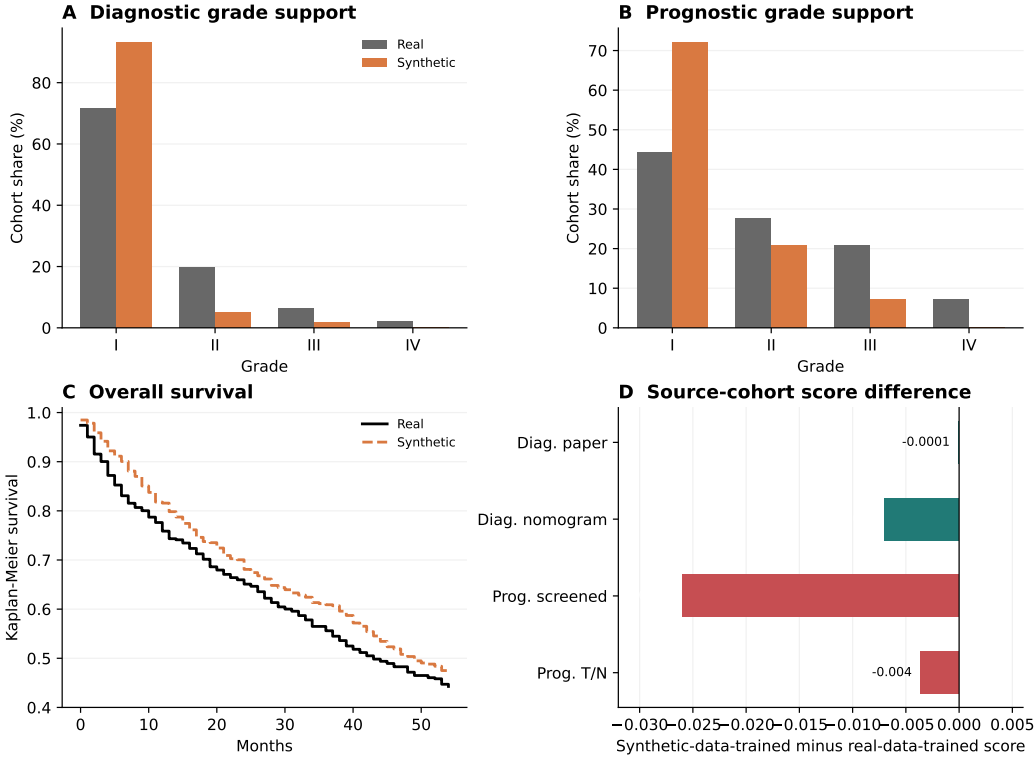


Figure 20: Summary of selected fidelity results. Panels A and B show the histological-grade distributions, Panel C compares overall Kaplan–Meier survival and Panel D shows the change in source-cohort discrimination when each model is trained on synthetic rather than real data.

These model comparisons are optimistic because the generator was trained on the same real cohort used for scoring. The real-data-trained reference models were also fitted and scored on that complete cohort. Small differences therefore show that the modelling procedures can produce similar rankings within the source data; they do not demonstrate performance at another institution or preservation of clinical effects.

7.8 Integrated interpretation and transition

The results show partial rather than complete preservation. The TabPFN embeddings retain substantial local structure and moderate overall distance ordering, although those findings depend on outcome-inclusive clinical distances and numerical category coding. The synthetic

cohorts reproduce many continuous-variable distributions, average correlations and overall survival features, and some synthetic-trained models retain similar discrimination on the real source cohort.

Important discrepancies remain. Statistical discriminators separate real and synthetic records, grade is strongly distorted, grade IV disappears, several subgroup survival curves and adjusted relationships change, and 39 of 88 planned interaction terms cannot be estimated in both cohorts. The synthetic cohorts are therefore suitable for testing whether modelling procedures can run and for carefully qualified exploratory analyses. They should not replace the real cohort when interpreting rare categories, regression coefficients, interactions or clinical survival effects. The next section broadens the predictive comparison to additional tabular foundation models while retaining these evaluation limits.

8 Multi-Foundation-Model Benchmarking

8.1 Purpose and comparison set

This section compares pretrained tabular foundation models (TFMs) with conventional prediction models using the same 17 clinical variables and the same held-out real patients. No synthetic records or precomputed embeddings are used. The diagnostic comparison includes logistic regression, XGBoost, TabPFN-3, TabICLv2 and TabDPT. The prognostic comparison includes elastic-net Cox regression, a random survival forest, a TabPFN landmark adaptation and SurvivalPFN.

The diagnostic question is whether the three classification foundation models improve prediction of liver metastasis relative to logistic regression and XGBoost. The prognostic question is whether a classification foundation model adapted to fixed survival horizons, or a model pretrained specifically for right-censored survival data, improves prediction relative to Cox regression and a random survival forest. These are internal comparisons within the SEER cohort; they do not replace the independent hospital validation of the reference models presented earlier.

Data used in this section. All experiments use the 2 processed real SEER datasets constructed in Section 2: the 2,672-patient diagnostic cohort and the 462-patient prognostic cohort. Models receive the same 17 retained clinical predictors from these datasets. The Erlangen cohort, the full-cohort synthetic datasets and the precomputed TabPFN embeddings are not used in this benchmark. The test patients are real SEER patients held out under the evaluation design described below.

All model specifications were fixed before evaluating the held-out patients. The benchmark comprises 125 diagnostic fits: five models evaluated in five repeats of five-fold cross-validation. It also comprises 20 prognostic fits: four models evaluated on five common 70/30 training–test partitions. Every fit completed and contributed to the reported results.

8.2 Evaluation design

For diagnosis, each cross-validation repeat gives every one of the 2,672 patients one prediction from a model that was not trained on that patient. Performance was calculated separately in each repeat and then averaged across the five repeats. For prognosis, every partition contained 323 training and 139 test patients from the 462-patient cohort. Performance was calculated in each test set and averaged equally across the five partitions.

Preprocessing was fitted separately within each training partition. This included pooling categories represented by fewer than 15 training patients, defining the category vocabulary, scaling continuous variables and removing constant encoded columns. The fitted transformation was then applied without modification to the corresponding test patients. Consequently, neither test predictors nor test outcomes influenced model fitting or preprocessing.

Uncertainty was estimated from 2,000 patient-level bootstrap samples of the held-out predictions. The same resampled patients were used for all models, which preserves the pairing needed for direct model comparisons. Diagnostic resampling retained the observed number of patients

with and without liver metastasis; prognostic resampling used the original patients and retained the same patient weights wherever a patient appeared in a test partition. The reported 95% percentile intervals therefore describe variation in the evaluation patients while keeping the fitted models and data partitions unchanged. They do not include uncertainty from refitting, alternative model seeds, pretraining or a different clinical population. The intervals are pointwise and were not adjusted for the number of comparisons.

8.3 Models and implementation

Table 44 summarises the evaluated methods. Logistic and Cox regression used one-hot encoded categories, whereas the tree models and foundation models used the training-derived ordinal representation. All models received the same underlying 17 clinical variables.

TabPFN-3, TabICLv2 and TabDPT are pretrained tabular classification models that use the training table as context at prediction time rather than updating their pretrained weights (Hollmann et al., 2025; Prior Labs, 2026; Qu et al., 2026; Ma et al., 2025). Their exact checkpoint versions were fixed and verified before evaluation. The TabPFN landmark method fits separate TabPFN classifiers for death by 12, 24 and 36 months; it is therefore an adaptation of a classification model, not a survival-pretrained model. SurvivalPFN was pretrained for right-censored time-to-event prediction and returns a survival distribution directly (Qi et al., 2026).

Table 44: Models included in the matched benchmark. One-hot and ordinal encodings were learned separately within every training partition. The pretrained weights of the foundation models remained unchanged during evaluation.

Task	Model	Model type	Clinical encoding	Main specification
Diagnosis	Logistic regression	Linear classifier	One-hot	L2 penalty; regularisation strength C=1
Diagnosis	XGBoost	Tree ensemble	Ordinal	400 trees; depth 4; learning rate 0.05
Diagnosis	TabPFN-3	Classification TFM	Ordinal	Eight-estimator ensemble; in-context prediction
Diagnosis	TabICLv2	Classification TFM	Ordinal	Eight-estimator ensemble; in-context prediction
Diagnosis	TabDPT	Classification TFM	Ordinal	Complete training context; in-context prediction
Prognosis	Cox elastic net	Proportional-hazards model	One-hot	Penalty 0.1; L1 ratio 0.5
Prognosis	Random survival forest	Survival-tree ensemble	Ordinal	500 trees; minimum leaf size 5
Prognosis	TabPFN landmark	Classification TFM adapted to survival	Ordinal	Separate classifiers at 12, 24 and 36 months
Prognosis	SurvivalPFN	Survival-pretrained TFM	Ordinal	In-context right-censored survival distribution

All foundation models ran on the graphics processing unit (GPU). Logistic regression, XGBoost, Cox regression and the random survival forest ran on the central processing unit (CPU). Device choice was fixed for the experiment and no automatic CPU fallback was allowed for a foundation model.

8.4 Diagnostic results

Diagnostic performance was assessed with the area under the receiver operating characteristic curve (ROC AUC), average precision and the Brier score. ROC AUC measures ranking across patients with and without liver metastasis. Average precision summarises the precision–recall relationship from the ranked predictions (scikit-learn developers, n.d.). The Brier score is the mean squared error of the predicted probabilities and is better when lower.

The three foundation models formed a narrow leading group (Table 45). TabPFN-3 had the highest point ROC AUC (0.903) and average precision (0.665). TabICLv2 was nearly identical, with ROC AUC 0.902, average precision 0.664 and the same rounded Brier score of 0.086. TabDPT also reached ROC AUC 0.902. Logistic regression remained competitive at ROC AUC 0.895, whereas XGBoost reached 0.883.

Table 45: Diagnostic performance on held-out real patients. ROC AUC, average precision and Brier score are shown with pointwise 95% patient-bootstrap intervals. Calibration slope and calibration-in-the-large intercept have ideal values of 1 and 0, respectively.

Model	ROC AUC [95% interval]	Average precision [95% interval]	Brier score [95% interval]	Slope	Intercept
Logistic regression	0.895 [0.878, 0.909]	0.635 [0.591, 0.681]	0.090 [0.084, 0.097]	0.960	0.004
XGBoost	0.883 [0.866, 0.898]	0.615 [0.576, 0.658]	0.096 [0.089, 0.103]	0.731	0.039
TabPFN-3	0.903 [0.888, 0.918]	0.665 [0.625, 0.708]	0.086 [0.080, 0.092]	0.899	0.091
TabICLv2	0.902 [0.886, 0.917]	0.664 [0.624, 0.708]	0.086 [0.080, 0.092]	0.910	0.084
TabDPT	0.902 [0.887, 0.916]	0.652 [0.611, 0.696]	0.087 [0.081, 0.093]	1.001	0.008

The paired differences in Table 46 show the size and uncertainty of the changes relative to logistic regression. The ROC AUC improvements were 0.0088 for TabPFN-3, 0.0075 for TabICLv2 and 0.0074 for TabDPT; all three intervals remained above zero. Their Brier-score improvements were also positive. Average precision improved for TabPFN-3 and TabICLv2, whereas the TabDPT interval included zero. XGBoost was lower than logistic regression in ROC AUC and Brier score, while its average-precision interval included zero.

Table 46: Paired diagnostic changes relative to logistic regression. ROC AUC and average precision are candidate minus logistic regression; Brier improvement is logistic regression minus candidate. Positive values favour the listed model. Intervals are pointwise and are not adjusted for multiple comparisons.

Model	Δ ROC AUC [95% interval]	Δ average precision [95% interval]	Brier improvement [95% interval]
TabPFN-3	0.0088 [0.0034, 0.0145]	0.0303 [0.0049, 0.0539]	0.0043 [0.0019, 0.0069]
TabICLv2	0.0075 [0.0021, 0.0132]	0.0288 [0.0057, 0.0496]	0.0043 [0.0020, 0.0066]
TabDPT	0.0074 [0.0016, 0.0133]	0.0171 [-0.0076, 0.0410]	0.0033 [0.0010, 0.0057]
XGBoost	-0.0120 [-0.0188, -0.0049]	-0.0203 [-0.0486, 0.0068]	-0.0059 [-0.0091, -0.0024]

Calibration did not follow exactly the same ordering. TabDPT had the slope closest to one (1.001) and an intercept close to zero (0.008). Logistic regression also remained close to these targets (0.960 and 0.004). TabPFN-3 and TabICLv2 had shallower slopes of 0.899 and 0.910 and positive intercepts of 0.091 and 0.084; XGBoost had the shallowest slope, 0.731. The diagnostic result is therefore a small foundation-model advantage in discrimination and overall probability error, not a uniform improvement in calibration.

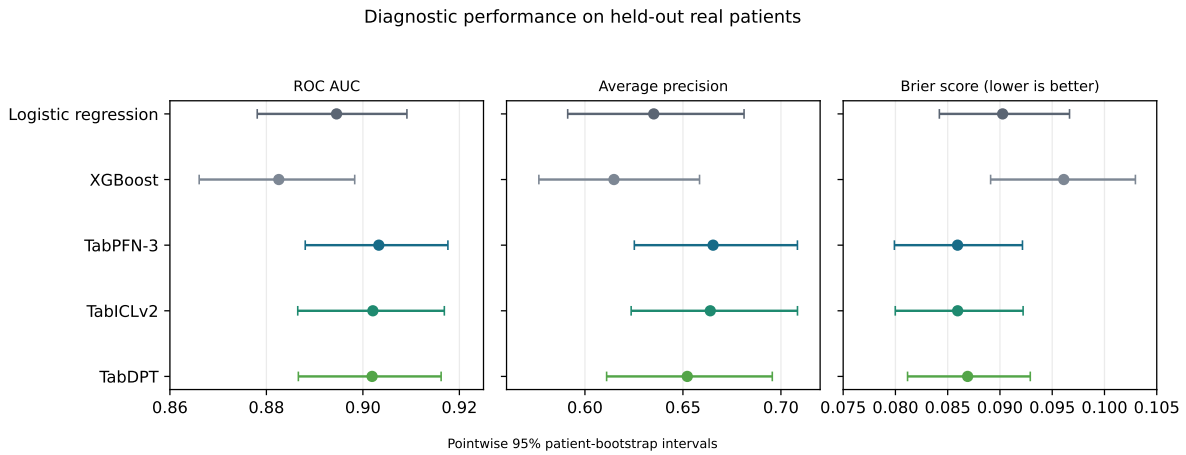


Figure 21: Diagnostic performance on held-out real patients under the common cross-validation design. Points are the primary estimates and bars are pointwise 95% patient-bootstrap intervals. Each panel has its own horizontal scale; only the Brier score is better when lower.

8.5 Prognostic results

Prognostic discrimination was assessed with Uno's concordance index, which uses inverse-probability-of-censoring weights to account for right censoring, and with time-dependent ROC AUC at 12, 24 and 36 months (scikit-survival developers, n.d.b). The same censoring distribution, estimated from the real training patients in each partition, was used for every model evaluated in that partition.

The four models had similar discrimination (Table 47). Mean Uno concordance ranged from 0.710 for SurvivalPFN to 0.717 for the TabPFN landmark method. Every paired concordance interval relative to Cox regression included zero. Time-dependent ROC AUC values ranged from 0.825 to 0.845 at 12 months and from 0.806 to 0.818 at 36 months. These results do not show a clear discrimination advantage for any foundation-model approach.

Table 47: Prognostic discrimination on held-out real patients. Uno concordance is shown with its pointwise 95% patient-bootstrap interval. The horizon-specific values are cumulative/dynamic ROC AUCs calculated from predicted death risk.

Model	Uno concordance [95% interval]	ROC AUC 12 m	ROC AUC 24 m	ROC AUC 36 m
Cox elastic net	0.711 [0.672, 0.749]	0.845	0.818	0.808
Random survival forest	0.712 [0.672, 0.751]	0.845	0.827	0.818
TabPFN landmark	0.717 [0.675, 0.755]	0.845	0.827	0.818
SurvivalPFN	0.710 [0.672, 0.746]	0.825	0.812	0.806

The Brier score evaluates survival-probability error and is better when lower. The integrated Brier score (IBS) summarises this error over the specified 12-, 24- and 36-month grid; it is not an integral over the complete follow-up period (scikit-survival developers, n.d.a). The random survival forest had the lowest IBS (0.153), followed by the TabPFN landmark method (0.155), Cox regression (0.160) and SurvivalPFN (0.164). Its paired IBS improvement over Cox regression was 0.0071 with a 95% interval from 0.0012 to 0.0128 (Tables 48 and 49). The corresponding intervals for both foundation-model approaches included zero.

For each model, the prognostic calibration summary divided test patients into three equally sized groups according to predicted risk and compared mean predicted survival with Kaplan–Meier (KM) survival at the three horizons. All 15 partition–horizon summaries, each based on three risk groups, were estimable for every model. The random survival forest had the smallest mean absolute gap (0.044), followed by SurvivalPFN (0.052), the TabPFN landmark method (0.054) and Cox regression (0.069). These are descriptive summaries without comparative confidence intervals, and each model defines its own risk groups.

Table 48: Prognostic probability error and grouped calibration. Brier scores and IBS are better when lower. The Kaplan–Meier gap is the mean absolute difference between predicted and observed survival across the 15 estimable partition–horizon summaries, each based on three risk groups.

Model	Brier 12 m	Brier 24 m	Brier 36 m	IBS [95% interval]	KM gap
Cox elastic net	0.130	0.164	0.182	0.160 [0.146, 0.175]	0.069
Random survival forest	0.122	0.157	0.176	0.153 [0.137, 0.171]	0.044
TabPFN landmark	0.125	0.160	0.177	0.155 [0.135, 0.176]	0.054
SurvivalPFN	0.135	0.170	0.183	0.164 [0.144, 0.186]	0.052

Table 49: Paired prognostic changes relative to elastic-net Cox regression. Uno concordance is candidate minus Cox; IBS improvement is Cox minus candidate. Positive values favour the listed model. The intervals are pointwise and are not adjusted for multiple comparisons.

Model	Δ Uno concordance [95% interval]	IBS improvement [95% interval]
TabPFN landmark	0.0056 [-0.0122, 0.0210]	0.0049 [-0.0034, 0.0130]
SurvivalPFN	-0.0009 [-0.0200, 0.0168]	-0.0042 [-0.0151, 0.0062]
Random survival forest	0.0003 [-0.0164, 0.0167]	0.0071 [0.0012, 0.0128]

Although the random survival forest had a positive pointwise IBS interval relative to Cox regression, the benchmark was not designed as a formal multiple-comparison or superiority study. The result supports a small probability-error improvement within this evaluation, not universal superiority of the random survival forest. Likewise, intervals that include zero do not establish that two methods are equivalent.

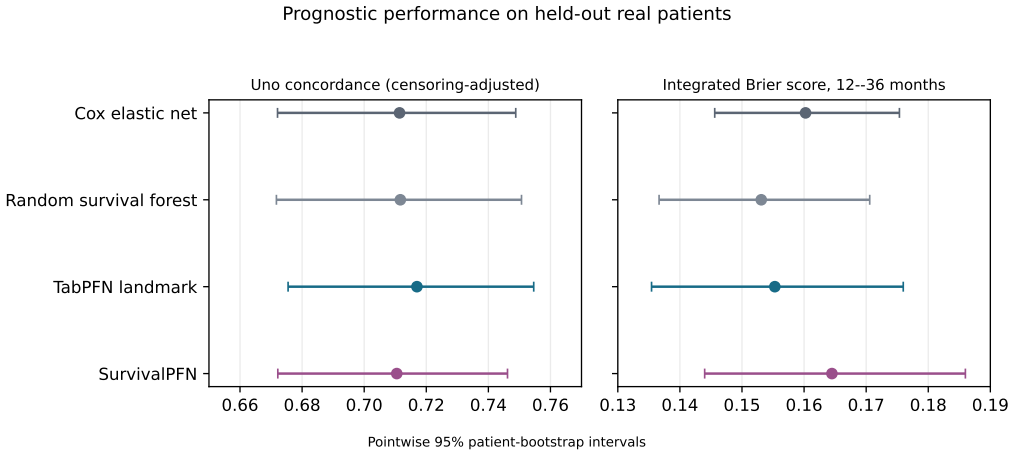


Figure 22: Prognostic discrimination and probability error on held-out real patients. Points are primary estimates and bars are pointwise 95% patient-bootstrap intervals. Higher Uno concordance and lower IBS are favourable.

8.6 Calibration and validity of survival predictions

Figure 23 presents two different calibration summaries. The diagnostic panel shows the calibration slope calculated from the out-of-fold probabilities, with an ideal value of one. The prognostic panel shows the mean absolute difference between predicted survival and Kaplan–Meier survival in model-specific risk groups, with an ideal value of zero. Their numerical values should not be compared directly because they measure different aspects of calibration.

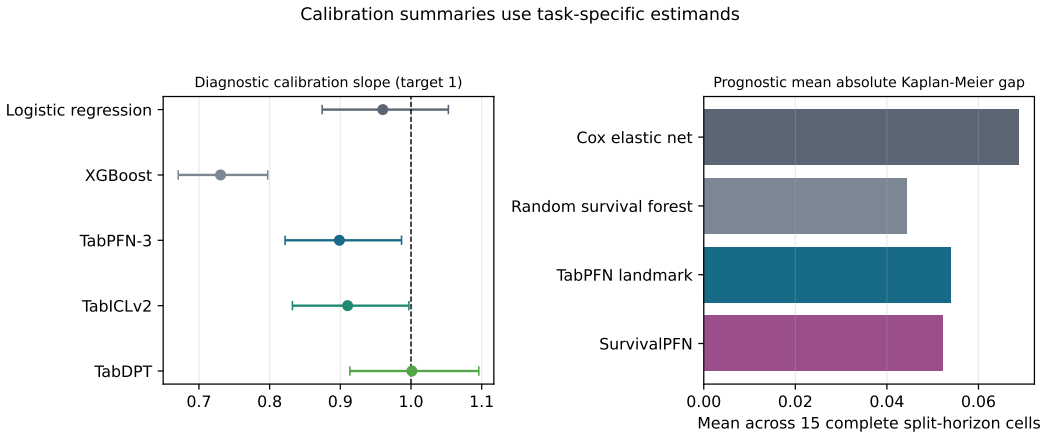


Figure 23: Task-specific calibration summaries. Diagnostic bars are pointwise 95% patient-bootstrap intervals around the calibration slope, and the dashed line marks the ideal value of one. Prognostic bars are descriptive mean absolute Kaplan–Meier gaps across 15 estimable partition–horizon comparisons and have no confidence intervals.

The three TabPFN landmark classifiers predict survival independently at 12, 24 and 36 months. They can therefore produce an invalid sequence in which predicted survival increases at a later time. This occurred in 41 of the 695 held-out prediction records and in every prognostic partition. The unmodified probabilities remain the primary results. As a sensitivity analysis, each patient's sequence was replaced by its cumulative minimum so that survival could only remain constant or decrease. This changed IBS only from 0.15529 to 0.15512.

SurvivalPFN returns a coherent survival distribution, but its predictions changed slightly when the same test patients were arranged into different query batches. The largest observed change was 0.00035 in its risk score and less than 10^{-6} in predicted survival. To keep the calculation reproducible, every test partition was scored once in a fixed patient order.

8.7 Computational profile

Table 50 reports the median combined fitting and prediction time for the real-data benchmark on the study computer. Classical models estimated new parameters in every training partition. The foundation models instead used each training table as context while keeping their pretrained weights fixed. The timings exclude environment setup, initial checkpoint loading, bootstrap calculations and report preparation. They describe this hardware and implementation only; they are not general measures of computational efficiency.

Median diagnostic times were 0.20 seconds for logistic regression, 0.28 for XGBoost, 1.33 for TabDPT, 1.42 for TabICLv2 and 2.82 for TabPFN-3. Median prognostic times were 0.58 seconds

Table 50: Computational profile on the study computer. Training/context rows are the median number of real training patients supplied to each model. Wall time is the median combined fitting and prediction time per evaluation unit on the RTX 3060 Laptop GPU and host CPU.

Task	Model	Units	Device	Training/context rows	Median wall time (s)	Training behaviour
Diagnosis	Logistic regression	25	CPU	2138	0.20	Model fitted in each partition
Diagnosis	XGBoost	25	CPU	2138	0.28	Model fitted in each partition
Diagnosis	TabPFN-3	25	CUDA	2138	2.82	In-context; weights fixed
Diagnosis	TabICLv2	25	CUDA	2138	1.42	In-context; weights fixed
Diagnosis	TabDPT	25	CUDA	2138	1.33	In-context; weights fixed
Prognosis	Cox elastic net	5	CPU	323	0.58	Model fitted in each partition
Prognosis	Random survival forest	5	CPU	323	0.73	Model fitted in each partition
Prognosis	TabPFN landmark	5	CUDA	323	5.81	In-context; weights fixed
Prognosis	SurvivalPFN	5	CUDA	323	6.80	In-context; weights fixed

for Cox regression, 0.73 for the random survival forest, 5.81 for the three TabPFN landmark classifiers together and 6.80 for SurvivalPFN. These results show that all methods were feasible on the available 6-GB GPU system, but they do not measure energy use, checkpoint loading latency or deployment throughput.

8.8 Interpretation and transition

The diagnostic results favour all three pretrained classification TFMs over logistic regression in ROC AUC and Brier score, but the ROC AUC gains are small (0.0074–0.0088). TabPFN-3 and TabICLv2 also improve average precision, while TabDPT has the calibration slope closest to one. No model is best on every criterion, and third-decimal differences should not be treated as a definitive ranking.

No corresponding prognostic TFM advantage was demonstrated. All four methods have similar discrimination, and SurvivalPFN has neither the lowest IBS nor the smallest grouped calibration gap. This result applies to the evaluated cohort, model specifications and five internal test partitions; it does not show that survival pretraining is ineffective in other settings.

The main limitations are internal validation in one SEER extract, bootstrap intervals that omit refitting and seed variability, and multiple pointwise comparisons without adjustment or a predefined clinically important margin. Checkpoint identities were verified, but the upstream development datasets were not independently reconstructed. The precomputed TabPFN embeddings were excluded because they were created from the complete cohort before the evaluation partitions and are reported separately as exploratory analyses.

Overall, the benchmark supports a modest diagnostic advantage for the three classification TFMs, but no prognostic advantage over Cox regression or the random survival forest. Section 9 tests whether a trainable nonlinear neural-network head changes either conclusion.

9 Nonlinear Multilayer Perceptron Heads

9.1 Purpose and comparison design

This section tests whether a compact nonlinear multilayer perceptron (MLP) improves prediction when it receives the same clinical variables as the models in Section 8. The diagnostic MLP is compared with logistic regression, XGBoost and 3 pretrained tabular foundation models (TFMs). The prognostic MLP is compared with Cox regression, a random survival forest (RSF), the TabPFN landmark approach and SurvivalPFN. These comparisons address the main question: does adding a trainable nonlinear head improve on the conventional and foundation-model results already obtained from the clinical data?

Data used in this section. The main MLP comparison uses the same processed real SEER cohorts, 17 clinical predictors and held-out evaluation partitions as Section 8: 2,672 patients for diagnosis and 462 for prognosis. A separate exploratory analysis uses the fixed TabPFN embedding datasets from Section 5, with 512 diagnostic coordinates and 1,024 prognostic coordinates for those same patients. No synthetic records or Erlangen patients are used. Section 8 compared conventional and pretrained model families; this section adds a trainable MLP and asks whether its nonlinear prediction head improves on those existing results.

Only observed patient records are used for training in the main clinical comparison. Diagnosis is evaluated on the same 5 repeats of 5-fold cross-validation among 2,672 patients, and prognosis on the same 5 predefined 70/30 splits among 462 patients. All preprocessing and model fitting are confined to the training portion of each fold or split, and performance is measured on the held-out real patients. The synthetic augmentation experiment is presented separately in Section 10.

A secondary analysis applies an MLP to previously computed TabPFN representations. It asks whether a nonlinear head performs better than a regularized logistic or Cox head when both receive the same representation. It does not compare clinical variables with representations. Because the TabPFN representations were produced from the complete cohort before the evaluation partitions were formed, this analysis cannot provide a fully isolated train–test comparison and is treated as exploratory.

Three predefined network initializations were evaluated for every MLP. This produced 75 diagnostic and 15 prognostic fits for each input type, or 180 MLP evaluation fits in total. A separate training-stage fit selected the number of epochs for each task, evaluation partition, input type and initialization.

9.2 Network architecture and task-specific outputs

Both MLPs are small feed-forward networks rather than additional foundation models. Each hidden layer uses a linear transformation, a rectified linear unit (ReLU) activation and 20% dropout. Batch normalization is not used. The networks are optimized with Adam with decoupled weight decay (AdamW), using a learning rate of 10^{-3} and weight decay of 10^{-4}. These settings were specified before the matched evaluation and were not chosen by comparing performance on the held-out patients.

The diagnostic network returns one logit, which is converted to a probability of liver metastasis, and is trained with binary cross-entropy. Class weighting is not used. The prognostic network follows the neural Cox formulation of Katzman et al. (2018): it returns one log-relative-risk value and minimizes the negative Cox partial log-likelihood. Tied event times use the Breslow approximation. After fitting, a Breslow baseline cumulative hazard is estimated from the same training patients so that survival probabilities can be obtained at 12, 24 and 36 months. Full-batch optimization is used for prognosis so that each Cox risk set contains all relevant training patients.

Table 51: MLP architecture used in the matched evaluation. Input widths are those obtained after training-only preprocessing. The prognostic clinical width varies between 26 and 27 because retained categories and constant columns can differ across training splits.

Task	Inputs after preprocessing	Hidden widths	Activation / dropout	Output	Training objective	Batch
Diagnosis	31 clinical / 512 embedding	64–32	ReLU; dropout 0.20	One logit; probability	Binary cross-entropy	128
Prognosis	26–27 clinical / 1,024 embedding	32–16	ReLU; dropout 0.20	One log-risk; Breslow survival	Cox partial likelihood	Full batch

For the clinical MLPs, rare categorical levels are pooled, categories are one-hot encoded, continuous variables are standardized and constant columns are removed. Every one of these steps is estimated anew from the corresponding training patients. This produces 31 diagnostic columns and 26 or 27 prognostic columns. The exploratory representation analysis uses 512 diagnostic and 1,024 prognostic inputs; their means and standard deviations are also estimated from the relevant training partition. Patient identifiers, outcomes, follow-up times and variables defining the evaluation partition are excluded from the predictors.

9.3 Training, model selection and uncertainty

The number of training epochs is selected without using any held-out test patient. Within each evaluation partition, 80% of the real training patients form an inner training set and 20% form an inner validation set. Diagnostic splits preserve the liver-metastasis proportion. Prognostic splits preserve event status together with broad follow-up-time groups when the available sample permits it. Preprocessing is fitted only on the inner training set.

Training can continue for at most 300 epochs. It stops after 20 consecutive epochs without an improvement of at least 10^{-4} in inner-validation loss. If epoch E gives the lowest eligible validation loss, the selection network is discarded. A newly initialized network is then trained for exactly E epochs on all patients in the outer training partition, with preprocessing refitted on those patients and no validation set. The outer test outcomes are used only after the final model has been fitted.

The main estimates use the first of the 3 predefined initializations; the other 2 show how much the result changes with network initialization. Confidence intervals are calculated from 2,000 resamples of the held-out patients. The same resampled patients are used for the MLP and each comparator, preserving the paired design. These intervals quantify variation in the evaluated patients while holding the fitted models, evaluation partitions and censoring estimates fixed.

They therefore do not include uncertainty from refitting the models or selecting a different initialization. Each 95% interval applies to one metric or comparison and is not adjusted for the number of results examined.

For diagnosis, discrimination is summarized by the area under the receiver operating characteristic curve (ROC AUC) and average precision; probability error is summarized by the Brier score. Calibration slope and intercept have ideal values of 1 and 0, respectively. For prognosis, discrimination is summarized by Uno's concordance and time-dependent ROC AUC calculated with inverse probability of censoring weighting (IPCW). Probability error is summarized by the integrated Brier score (IBS) over 12, 24 and 36 months. Calibration is described by the mean absolute difference between predicted survival and Kaplan–Meier survival across model-defined risk groups.

9.4 Diagnostic results using clinical variables

The clinical MLP achieved a ROC AUC of 0.897, average precision of 0.652 and Brier score of 0.0878 (Table 52). Its discrimination and probability error fall between the conventional models and the 3 diagnostic TFMs. The calibration slope of 0.893 suggests that its probabilities are somewhat more extreme than the observed outcomes support. The positive calibration intercept of 0.063 indicates a small overall tendency to underpredict risk on the log-odds scale. Logistic regression is better calibrated by these measures, with slope 0.960 and intercept 0.004.

Table 52: Diagnostic performance on held-out real patients. Brackets contain pointwise 95% patient-bootstrap intervals. Calibration slope and intercept have ideal values of 1 and 0.

Model	ROC AUC [95% interval]	Average precision [95% interval]	Brier score [95% interval]	Slope	Intercept
Logistic regression	0.895 [0.878, 0.909]	0.635 [0.591, 0.681]	0.090 [0.084, 0.097]	0.960	0.004
XGBoost	0.883 [0.866, 0.898]	0.615 [0.576, 0.658]	0.096 [0.089, 0.103]	0.731	0.039
TabPFN-3	0.903 [0.888, 0.918]	0.665 [0.625, 0.708]	0.086 [0.080, 0.092]	0.899	0.091
TabICLv2	0.902 [0.886, 0.917]	0.664 [0.624, 0.708]	0.086 [0.080, 0.092]	0.910	0.084
TabDPT	0.902 [0.887, 0.916]	0.652 [0.611, 0.696]	0.087 [0.081, 0.093]	1.001	0.008
Clinical MLP	0.897 [0.880, 0.912]	0.652 [0.612, 0.695]	0.088 [0.082, 0.094]	0.893	0.063

Paired comparisons show which differences are supported under this evaluation design (Table 53). Relative to logistic regression, the MLP improves ROC AUC by 0.0022 and average precision by 0.0166, but both intervals include zero. Its Brier score improves by 0.0024 (95% interval 0.0006–0.0044). The MLP performs better than XGBoost on all 3 reported metrics. In contrast, its ROC AUC is lower than those of TabPFN-3, TabICLv2 and TabDPT. Its Brier score is also higher than those of TabPFN-3 and TabICLv2. These results support individual contrasts only; they are not a multiple-comparison-adjusted model ranking.

Decision-curve net benefit was also calculated at risk thresholds from 0.05 to 0.40. The MLP, logistic regression and TabPFN-3 all perform better than the treat-all and treat-none strategies throughout this range. Their net benefits are close and no model is consistently best. Table 54 shows 3 representative thresholds. These values are descriptive because no paired interval was specified for this secondary analysis.

Table 53: Paired diagnostic improvement of the clinical MLP relative to each comparator. ROC AUC and average precision are MLP minus comparator; Brier improvement is comparator minus MLP. Positive values favour the MLP.

Comparator	Δ ROC AUC [95% interval]	Δ average precision [95% interval]	Brier improvement [95% interval]
Logistic regression	0.0022 [-0.0024, 0.0069]	0.0166 [-0.0024, 0.0341]	0.0024 [0.0006, 0.0044]
XGBoost	0.0142 [0.0086, 0.0196]	0.0369 [0.0155, 0.0566]	0.0083 [0.0055, 0.0111]
TabPFN-3	-0.0065 [-0.0105, -0.0025]	-0.0137 [-0.0295, 0.0015]	-0.0019 [-0.0036, -0.0002]
TabICLv2	-0.0053 [-0.0091, -0.0015]	-0.0122 [-0.0257, 0.0019]	-0.0019 [-0.0034, -0.0004]
TabDPT	-0.0051 [-0.0098, -0.0004]	-0.0005 [-0.0181, 0.0164]	-0.0009 [-0.0028, 0.0009]

Table 54: Mean diagnostic net benefit at selected risk thresholds. The values are averages over the 5 repeated sets of out-of-fold predictions and do not constitute external clinical-utility validation.

Model	Threshold 0.10	Threshold 0.20	Threshold 0.30
Logistic regression	0.1299	0.1044	0.0837
Clinical MLP	0.1308	0.1053	0.0870
TabPFN-3	0.1336	0.1056	0.0867

9.5 Prognostic results using clinical variables

The prognostic MLP reached Uno concordance 0.709 and an IBS of 0.162 (Table 55). Its time-dependent ROC AUC decreases from 0.827 at 12 months to 0.805 at 36 months. Across the 5 splits and 3 prediction horizons, the mean absolute Kaplan–Meier calibration gap is 0.065. This is close to the Cox-regression value of 0.069, but larger than the RSF value of 0.044.

Table 55: Prognostic performance on held-out real patients. Uno concordance and IBS include pointwise 95% patient-bootstrap intervals. Horizon-specific ROC AUC values use IPCW; lower IBS and Kaplan–Meier calibration gap are favourable.

Model	Uno concordance [95% interval]	AUC 12 m	AUC 24 m	AUC 36 m	IBS [95% interval]	Calibration gap
Cox elastic net	0.711 [0.672, 0.749]	0.845	0.818	0.808	0.160 [0.146, 0.175]	0.069
Random survival forest	0.712 [0.672, 0.751]	0.845	0.827	0.818	0.153 [0.137, 0.171]	0.044
TabPFN landmark	0.717 [0.675, 0.755]	0.845	0.827	0.818	0.155 [0.135, 0.176]	0.054
SurvivalPFN	0.710 [0.672, 0.746]	0.825	0.812	0.806	0.164 [0.144, 0.186]	0.052
Clinical MLP	0.709 [0.672, 0.746]	0.827	0.808	0.805	0.162 [0.146, 0.179]	0.065

None of the paired Uno-concordance intervals excludes zero (Table 56). The MLP's IBS does not differ clearly from Cox regression, TabPFN landmark or SurvivalPFN, but it is worse than the RSF: the RSF-relative IBS improvement is −0.0090 (95% interval −0.0166 to −0.0010). Thus, the nonlinear head does not establish a prognostic advantage over the existing models.

Table 56: Paired prognostic improvement of the clinical MLP relative to each comparator. Uno concordance is MLP minus comparator; IBS improvement is comparator minus MLP. Positive values favour the MLP.

Comparator	Δ Uno concordance [95% interval]	IBS improvement [95% interval]
Cox elastic net	-0.0021 [-0.0168, 0.0125]	-0.0018 [-0.0079, 0.0044]
Random survival forest	-0.0024 [-0.0183, 0.0148]	-0.0090 [-0.0166, -0.0010]
TabPFN landmark	-0.0077 [-0.0265, 0.0115]	-0.0068 [-0.0172, 0.0041]
SurvivalPFN	-0.0012 [-0.0199, 0.0171]	0.0024 [-0.0079, 0.0136]

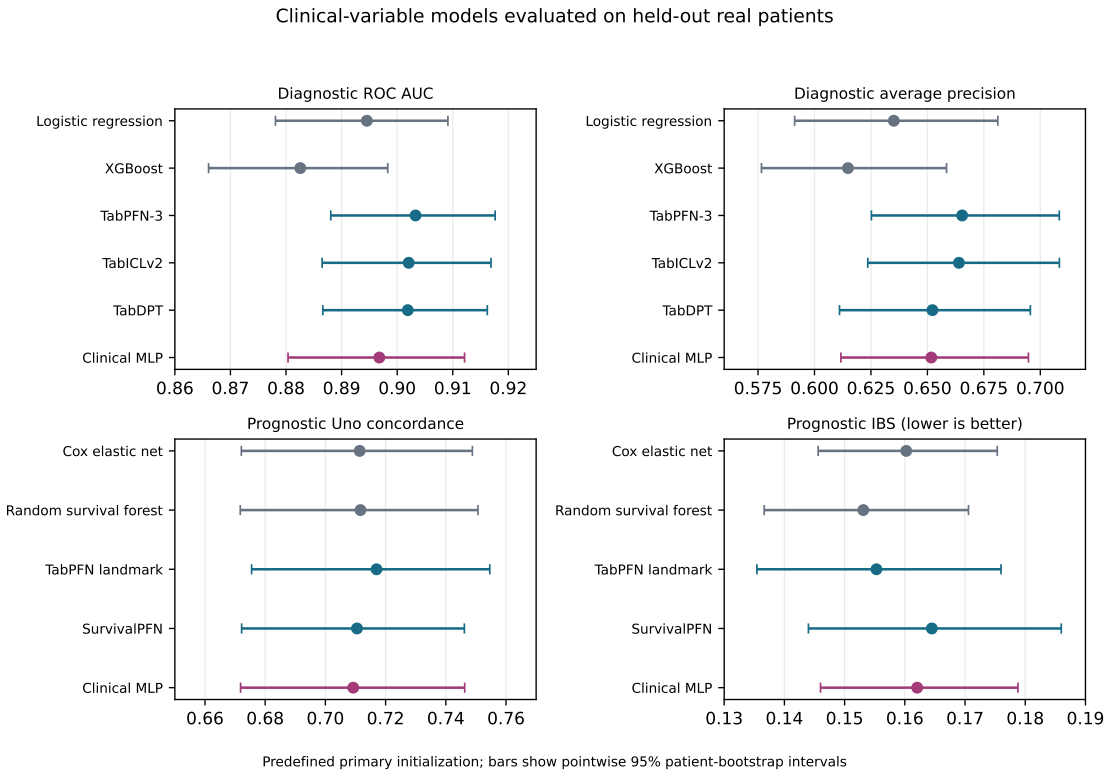


Figure 24: Clinical-variable performance on held-out real patients using the predefined primary network initialization. Points are estimates across the matched evaluation partitions and bars are pointwise 95% patient-bootstrap intervals. The clinical MLP is highlighted; lower IBS is favourable.

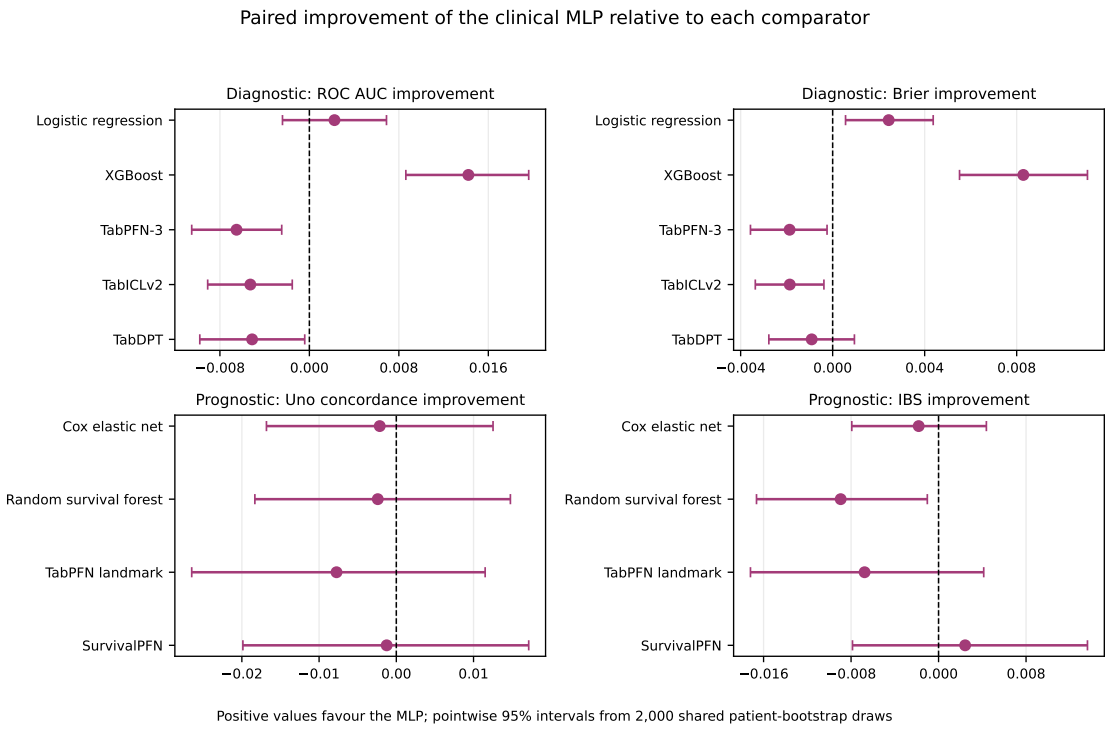


Figure 25: Paired improvement of the clinical MLP relative to each comparator. Positive values favour the MLP in every panel. The intervals use the same 2,000 patient resamples for both models and are not adjusted for the number of comparisons.

9.6 Exploratory analysis of TabPFN representations

For diagnosis, applying the MLP to the 512-dimensional TabPFN representation produces essentially the same ROC AUC as regularized logistic regression: both round to 0.897. The paired ROC AUC difference is approximately zero, and the intervals for average precision and Brier score also include zero (Table 57). For prognosis, the representation-based MLP has lower point concordance and higher point IBS than the representation-based Cox model, although both paired intervals narrowly include zero. There is therefore no consistent evidence that the nonlinear head extracts more useful information from these representations than the corresponding linear heads.

Table 57: Exploratory comparison of MLP and linear heads applied to the same TabPFN representations. Improvement is MLP minus linear head for discrimination and linear head minus MLP for Brier score or IBS, so positive values favour the MLP. The representations were computed before the evaluation partitions were formed.

Task	Metric	MLP head	Linear head	Paired improvement [95% interval]
Diagnosis	ROC AUC	0.897	0.897	0.0000 [-0.0063, 0.0063]
Diagnosis	Average precision	0.659	0.653	0.0069 [-0.0148, 0.0279]
Diagnosis	Brier score	0.088	0.089	0.0010 [-0.0016, 0.0037]
Prognosis	Uno concordance	0.705	0.727	-0.0226 [-0.0454, 0.0001]
Prognosis	Integrated Brier score	0.163	0.153	-0.0098 [-0.0201, 0.0003]

These findings cannot be used to rank the clinical variables against the TabPFN representations.

Although scaling is estimated within each training partition, the underlying representations were already created using the full cohort. The 2 input types also differ substantially in dimension: the representation heads receive 512 or 1,024 inputs, whereas the clinical heads receive 31 or 26–27 encoded inputs. The effects of representation, dimensionality and optimization therefore cannot be separated in a direct comparison.

9.7 Epoch selection, initialization variability and computation

The selected training duration differs by task and input type (Table 58). The clinical diagnostic MLP selects a median of 27 epochs, compared with 92 for clinical prognosis. The representation heads select medians of 7 and 13 epochs. These differences do not show that one input type learns a better predictor more quickly, because the input dimensions, loss functions and validation-loss scales differ. Every final network is refitted for exactly its selected number of epochs.

Table 58: Epoch selection, variation across 3 network initializations and observed computation time. Epochs are shown as minimum/median/maximum. Discrimination is ROC AUC for diagnosis and Uno concordance for prognosis; probability error is Brier score and IBS, respectively. Time is the median fit-and-predict duration per evaluation partition for the primary initialization.

Task	Input	Selections	Epochs	Discrimination range	Error range	Time (s)
Diagnosis	Clinical	75	11/27/50	0.8952–0.8968	0.0878–0.0880	1.44
Diagnosis	Representation	75	3/7/21	0.8957–0.8973	0.0879–0.0883	1.15
Prognosis	Clinical	15	69/92/154	0.7026–0.7092	0.1621–0.1656	0.64
Prognosis	Representation	15	2/13/29	0.7049–0.7115	0.1588–0.1661	0.48

Diagnostic estimates change little across the 3 initializations: clinical ROC AUC ranges from 0.8952 to 0.8968 and Brier score from 0.0878 to 0.0880. Prognostic variation is larger but remains modest, with Uno concordance from 0.7026 to 0.7092 and IBS from 0.1621 to 0.1656. The predefined primary initialization gives the best clinical point estimate among these 3 runs for the displayed metrics, but it was fixed before the results were examined. Three initializations are not sufficient to characterize the complete range of possible optimization outcomes.

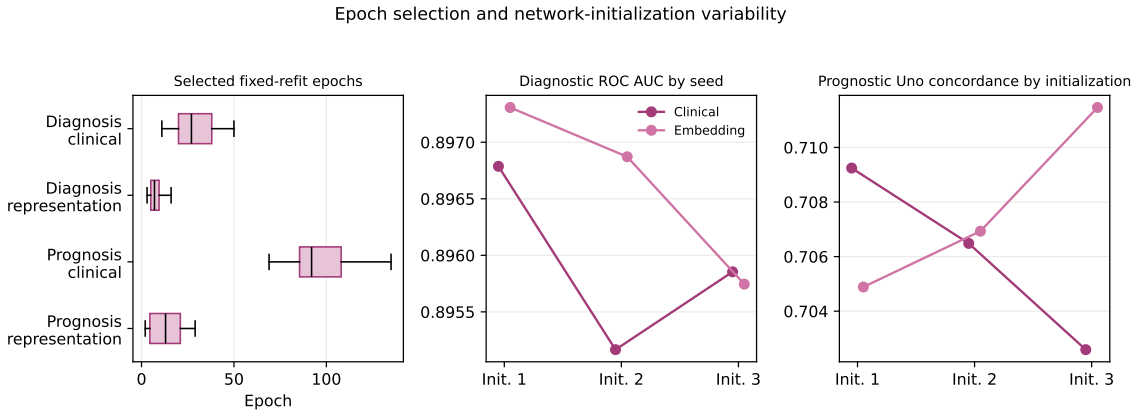


Figure 26: Selected epochs and variation across the 3 network initializations. The left panel summarizes the selected epoch counts across evaluation partitions. The other panels show separate point estimates for each initialization; connecting lines are visual guides and are not confidence intervals or time trends.

All networks were trained on the NVIDIA RTX 3060 Laptop graphics processing unit (GPU) using NVIDIA's Compute Unified Device Architecture (CUDA). Training used 32-bit floating-

point arithmetic and deterministic algorithms; the lower-precision TensorFloat-32 mode was disabled. The median fit-and-predict time per evaluation partition was 1.44 seconds for clinical diagnosis and 0.64 seconds for clinical prognosis. Maximum allocated GPU memory across the real-patient MLP fits was approximately 70 MB. These timings cover model fitting and prediction on the study computer and should not be interpreted as general hardware benchmarks.

9.8 Interpretation and limitations

The clinical MLP provides a direct test of whether a compact nonlinear model improves on the preceding clinical-data benchmarks. For diagnosis, it reduces Brier score relative to logistic regression and performs better than XGBoost, but it does not improve ROC AUC over logistic regression with clear statistical support and remains below all 3 diagnostic TFMs on ROC AUC. For prognosis, it does not improve discrimination over any comparator and has worse IBS than the RSF. The exploratory representation analysis likewise finds no consistent advantage over regularized logistic or Cox heads. Nonlinearity alone therefore does not explain the diagnostic advantage of the pretrained TFMs in this cohort.

The conclusion is limited to one architecture and optimizer configuration; no hyperparameter search was performed. The bootstrap intervals hold the fitted networks fixed, and 3 initializations provide only a partial view of optimization variability. Grade 4 is too rare for a reliable fold-level diagnostic conclusion and does not meet the prognostic subgroup requirements in any split. All primary results are internal to one SEER extract, and these MLPs were not tested in an independent clinical cohort. The representation analysis is further limited because the TabPFN transformation was computed before the evaluation partitions were formed. Overall, no general MLP advantage was demonstrated. Section 10 next evaluates partition-specific augmentation.

10 Synthetic Training-Data Augmentation

10.1 Research question and comparison

Section 9 asked whether a nonlinear MLP head improved prediction when models were trained on real patients only. This section asks a different question: whether adding generated records to each real training partition changes the performance of the same model families, including the MLP. It is therefore an augmentation experiment, not a second test of MLP architecture. Performance is always measured in held-out real patients who were not used for model training or synthetic-data generation. Four augmentation sizes were examined: 1,000, 5,000, 10,000 and 15,000 synthetic records. Throughout this section, these conditions are abbreviated as `+1K`, `+5K`, `+10K` and `+15K`, where K denotes one thousand records. Each augmented model was compared with the same model trained on real data alone (`+0`).

Data used in this section. The starting data are the same 2,672-patient diagnostic and 462-patient prognostic SEER cohorts used in Sections 8 and 9. Within every evaluation fold or split, a new synthetic pool is generated only from that fold's real training patients. Models are trained on either the real training partition alone or the combination of that real partition and a synthetic prefix. Evaluation uses only the corresponding held-out real SEER patients. The full-cohort synthetic datasets from Section 5, the precomputed embeddings and the Erlangen cohort are not used here.

The analysis included the six diagnostic models and five prognostic models evaluated in Sections 8 and 9. For diagnosis, these were logistic regression, XGBoost, TabPFN-3, TabICLv2, TabDPT and the clinical multilayer perceptron (MLP). For prognosis, they were Cox elastic net, random survival forest, TabPFN landmark, SurvivalPFN and the clinical MLP. The comparison therefore measures the effect of augmentation within each model; it is not a new comparison of unrelated algorithms.

The main diagnostic outcomes were the area under the receiver operating characteristic curve (ROC AUC), average precision (AP) and Brier score. The main prognostic outcomes were Uno's concordance index (Uno C), the 12-month time-dependent AUC and the integrated Brier score (IBS) from 12 to 36 months. Higher values are favourable for discrimination measures, whereas lower Brier scores and IBS values are favourable. Calibration and decision-curve net benefit were secondary outcomes.

The analysis comprised 875 primary model fits: 750 diagnostic fits from five repeats of five-fold cross-validation and 125 prognostic fits from five held-out splits. Of these, 175 used real data alone and 700 used an augmented training condition. A further 300 MLP fits assessed sensitivity to three initialization seeds. All planned fits completed. TabPFN representations that had previously been generated from the full cohort were not included because they did not preserve the training–test separation required for this experiment.

10.2 Training-partition-specific generation

Synthetic data were generated separately within every evaluation partition. For diagnosis, one generator was fitted to the real training patients in each of the 25 cross-validation folds. For prognosis, one generator was fitted to the real training patients in each of the five held-out splits.

Each generator learned the distribution of the clinical predictors together with the relevant outcome and produced a pool of 15,000 synthetic records. The held-out real patients were used only for evaluation and never entered the generator or an augmented training set.

The four augmentation sizes were nested. The `+1K` condition used the first 1,000 records from a partition's synthetic pool, `+5K` used the first 5,000, and so forth. Every condition retained all real training records. Using prefixes of the same pool makes the comparison across augmentation sizes more direct because it avoids generating an unrelated synthetic sample for each size. The model seed was also held constant across sizes for the primary comparison.

Table 59: Composition of the real-only and augmented training conditions. Real row counts and percentages are means across the 25 diagnostic or five prognostic evaluation partitions. Grade-4 prevalence decreases because no grade-4 record was generated.

	Diagnosis (25 partitions)			Prognosis (5 partitions)		
Condition	Real rows	Synthetic share	Grade 4	Real rows	Synthetic share	Grade 4
+0	2,138	0.0%	2.25%	323	0.0%	6.93%
+1K	2,138	31.9%	1.53%	323	75.6%	1.69%
+5K	2,138	70.1%	0.67%	323	93.9%	0.42%
+10K	2,138	82.4%	0.40%	323	96.9%	0.22%
+15K	2,138	87.5%	0.28%	323	97.9%	0.15%

The effect on training-set composition differed markedly between tasks (Table 59). Diagnostic partitions contained a mean of 2,138 real training records, so synthetic records represented 31.9% of the `+1K` condition and 87.5% of `+15K`. Prognostic partitions contained only 323 real training records; synthetic records therefore already represented 75.6% at `+1K` and 97.9% at `+15K`. In the larger prognostic conditions, model training was consequently dominated by the generated distribution.

Preprocessing was learned separately within each training condition, using its combined real and synthetic records and no held-out patients. Synthetic data could therefore affect which infrequent categorical levels were retained, the number of encoded variables and the scaling parameters. At `+1K`, a category was retained only because of its synthetic frequency in 120 of 150 diagnostic and 15 of 25 prognostic preprocessing fits. From `+5K` onward, this occurred in all 150 diagnostic and all 25 prognostic fits.

For the MLP, training duration was chosen from the real training data only. An inner real-data split selected the number of epochs, after which the model was reinitialized and trained for that fixed number of epochs on each complete training condition. The same epoch count was used across augmentation sizes within a partition and seed. This prevented synthetic validation records from determining training duration, although larger conditions still required more mini-batch updates per epoch.

10.3 Synthetic-data quality and distributional similarity

All 30 partition-specific pools contained the expected 15,000 records and no synthetic record exactly matched a real training record. One duplicate pair occurred within one diagnostic pool. Two extremely large generated values would not have been representable in the numerical format used for modelling; each was therefore replaced by the corresponding minimum or maximum observed in that partition's real training data. These two replacements affected two different

pools.

Two variables were generated both in their original form and as a derived clinical category. At `+15K`, the tumour-size category did not agree with the generated continuous tumour size in 1.47% of diagnostic and 2.62% of prognostic synthetic records. The binary indicator for positive nodal stage disagreed with the generated N stage in approximately 0.001% of records for each task. No such inconsistencies were present in the real training records, and the synthetic inconsistencies were retained rather than corrected before model fitting.

Table 60: Quality checks for the 30 partition-specific synthetic pools.

Check	Result	Scope or interpretation
Complete partition-specific pools	30/30	25 diagnostic and 5 prognostic pools
Exact synthetic–real matches	0	Across all partition-specific pools
Duplicate synthetic rows	1	One duplicated row in one diagnostic pool
Extreme values replaced by training bounds	2	One value in each of two pools
Grade-4 synthetic rows	0	Absent from all 30 pools
Tumour-size category mismatch	1.47% / 2.62%	Diagnostic / prognostic synthetic rows at +15K
N-stage indicator mismatch	0.001% / 0.001%	Diagnostic / prognostic synthetic rows at +15K
Fits retaining a category only through synthetic support	150/150 / 25/25	All diagnostic / prognostic fits from +5K

Table 61 compares each synthetic prefix with the real training records from which it was generated. Jensen–Shannon divergence (JSD) measures differences in categorical proportions and equals zero when the two distributions match exactly. The standardized Wasserstein distance measures differences in continuous variables relative to the standard deviation in the real training data. The reported correlation difference is the mean absolute difference between the Pearson correlation matrices of age, tumour size and grade.

Table 61: Differences between synthetic and real training distributions. JSD uses base 2, and Wasserstein distances are divided by the standard deviation in the corresponding real training partition. Outcome and survival-time shifts are calculated as synthetic minus real.

	Diagnosis		Prognosis	
Measure	+1K	+15K	+1K	+15K
Mean categorical JSD	0.004	0.004	0.006	0.006
Grade JSD	0.063	0.064	0.095	0.094
Mean standardized Wasserstein	0.177	0.165	0.214	0.202
Mean absolute correlation difference	0.034	0.033	0.057	0.055
Outcome-rate shift (percentage points)	-1.75	-2.04	0.96	1.01
Median survival-time shift (months)	–	–	6.6	7.0

Overall differences changed little as more synthetic records were included. Mean categorical JSD was 0.004 for diagnosis and 0.006 for prognosis at both displayed sizes. Grade was the main exception, with JSD values of 0.063–0.064 for diagnosis and 0.094–0.095 for prognosis because grade 4 was absent from every synthetic pool. Continuous-variable and correlation differences were slightly smaller at `+15K`, which is consistent with less random variation in the larger synthetic prefixes but does not correct the missing grade-4 support. At `+15K`, the synthetic liver-metastasis rate was 2.04 percentage points below the corresponding real diagnostic training

rate. In prognosis, the synthetic death-event rate was 1.01 percentage points higher and median observed survival was 7.0 months longer than in the real training data.

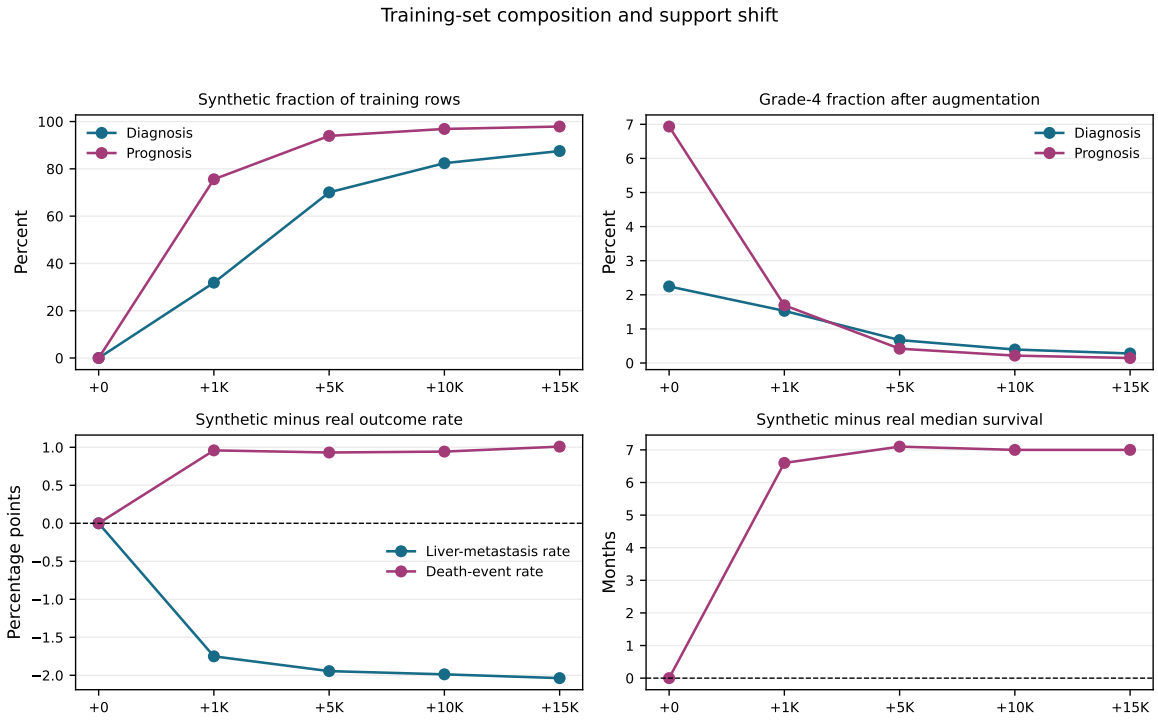


Figure 27: Changes in training-set composition across augmentation sizes. The synthetic share rises faster for prognosis because its real training partitions are smaller. Grade-4 prevalence approaches zero, while outcome prevalence and median observed survival remain shifted relative to the real training data.

10.4 Diagnostic performance

Diagnostic effects depended on the model rather than showing a uniform benefit (Table 62 and Figure 28). XGBoost showed the clearest improvement. From `+0` to `+15K`, its ROC AUC increased from 0.883 to 0.895, a paired change of 0.0126 with a 95% interval of 0.0074 to 0.0180. Average precision increased by 0.0249 (0.0077 to 0.0433), and Brier score decreased from 0.0961 to 0.0901. Expressed so that positive values favour augmentation, the Brier improvement was 0.0060 (0.0035 to 0.0086). The AUC and Brier gains were already evident at `+1K` and increased with augmentation size. Despite this improvement, XGBoost's `+15K` AUC remained below the approximately 0.900–0.901 values of the three tabular foundation models.

Table 62: Diagnostic results for real-only and maximum augmentation. Paired changes compare `+15K` with the same model at `+0`; positive values favour augmentation. Intervals are pointwise 95% patient-bootstrap intervals conditional on the fitted models.

Model	AUC +0	AUC +15K	Δ AUC [95% interval]	Δ AP [95% interval]	Brier improvement [95% interval]
Logistic regression	0.895	0.895	0.0009 [-0.0013, 0.0033]	-0.0003 [-0.0073, 0.0064]	-0.0003 [-0.0012, 0.0006]
XGBoost	0.883	0.895	0.0126 [0.0074, 0.0180]	0.0249 [0.0077, 0.0433]	0.0060 [0.0035, 0.0086]
TabPFN-3	0.903	0.901	-0.0027 [-0.0051, -0.0001]	-0.0078 [-0.0156, 0.0000]	-0.0016 [-0.0027, -0.0007]
TabICLv2	0.902	0.901	-0.0015 [-0.0043, 0.0017]	-0.0052 [-0.0138, 0.0044]	-0.0014 [-0.0025, -0.0003]
TabDPT	0.902	0.900	-0.0015 [-0.0037, 0.0008]	0.0057 [-0.0047, 0.0155]	-0.0005 [-0.0014, 0.0004]
Clinical MLP	0.897	0.898	0.0013 [-0.0023, 0.0051]	0.0027 [-0.0093, 0.0171]	0.0002 [-0.0013, 0.0018]

Logistic regression and the clinical MLP showed little evidence of change. At +15K, their AUC changes were 0.0009 (−0.0013 to 0.0033) and 0.0013 (−0.0023 to 0.0051), respectively, and the average-precision and Brier-score intervals also included zero. TabPFN-3 deteriorated slightly: its AUC change was −0.0027 (−0.0051 to −0.0001), and its Brier improvement was −0.0016 (−0.0027 to −0.0007). Its AUC, average precision and Brier score were already worse at +1K. TabICLv2 also had a worse Brier score at every augmented size, reaching a change of −0.0014 (−0.0025 to −0.0003) at +15K, while its AUC change remained inconclusive. TabDPT showed a small Brier deterioration at +5K, but its +15K intervals included zero.

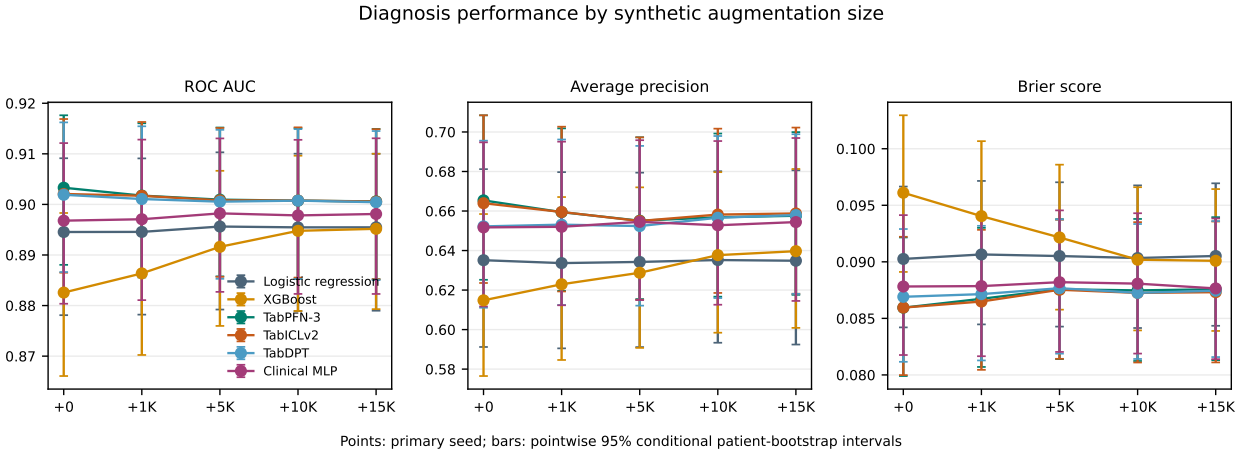


Figure 28: Diagnostic performance across nested augmentation sizes. Points are primary-seed estimates, and bars are pointwise 95% patient-bootstrap intervals conditional on the fitted models. The +0 condition is the matched real-only result for each model.

These results show that synthetic records were not equivalent to additional independent patients. They helped the initially weaker XGBoost classifier, had little effect on logistic regression and the clinical MLP, and slightly worsened the strongest diagnostic foundation models. Because the generator learned from the same finite real training sample, larger prefixes primarily repeated and smoothed patterns inferred from that sample; they could not add new clinical information beyond it.

10.5 Prognostic performance

No model showed a clear improvement across the main prognostic outcomes (Table 63 and Figure 29). Cox elastic net deteriorated from +5K onward. At +15K, Uno's concordance index fell from 0.711 to 0.686, a paired change of −0.0250 (95% interval −0.0477 to −0.0010). IBS increased from 0.160 to 0.182; when oriented so that positive values favour augmentation, the paired change was −0.0214 (−0.0335 to −0.0098). Its 12-month AUC also decreased by 0.0554 (−0.0991 to −0.0171).

For the clinical MLP, the concordance intervals included zero, but IBS was worse at every augmented size. The IBS-oriented improvement was −0.0095 (−0.0153 to −0.0038) at +1K and −0.0136 (−0.0216 to −0.0058) at +15K. Random survival forest had a small IBS deterioration at +1K; its larger-size concordance and IBS intervals included zero. TabPFN landmark was the most stable model. At +15K, its concordance change was −0.0006 (−0.0080 to 0.0080), IBS improvement was 0.0020 (−0.0020 to 0.0062), and 12-month AUC change was 0.0113 (−0.0011

Table 63: Prognostic results for real-only and maximum augmentation. Paired changes compare `+15K` with the same model at `+0`; positive values favour augmentation. IBS improvement is real-only minus augmented IBS.

Model	Uno C +0	Uno C +15K	Δ Uno C [95% interval]	IBS +0	IBS +15K	IBS improvement [95% interval]
Cox elastic net	0.711	0.686	-0.0250 [-0.0477, -0.0010]	0.160	0.182	-0.0214 [-0.0335, -0.0098]
Random survival forest	0.712	0.702	-0.0099 [-0.0284, 0.0116]	0.153	0.158	-0.0045 [-0.0099, 0.0011]
TabPFN landmark	0.717	0.716	-0.0006 [-0.0080, 0.0080]	0.155	0.153	0.0020 [-0.0020, 0.0062]
SurvivalPFN	0.710	0.705	-0.0056 [-0.0158, 0.0042]	0.164	0.167	-0.0021 [-0.0098, 0.0057]
Clinical MLP	0.709	0.701	-0.0080 [-0.0249, 0.0085]	0.162	0.176	-0.0136 [-0.0216, -0.0058]

to 0.0239). SurvivalPFN likewise showed no supported change at the maximum augmentation size.

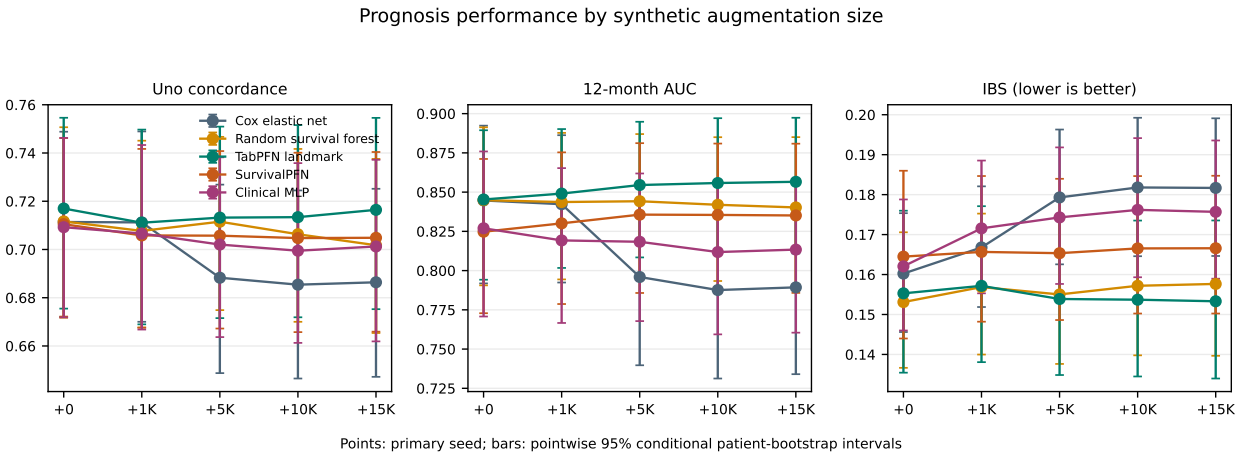


Figure 29: Prognostic performance across nested augmentation sizes. Points are primary-seed estimates, and bars are pointwise 95% patient-bootstrap intervals conditional on the fitted models. Lower IBS is favourable.

The more pronounced deterioration in prognosis occurred while synthetic records made up a much larger share of training than in diagnosis. This association does not identify a single cause. Outcome shifts, missing grade-4 support, inconsistencies between linked variables, preprocessing changes and repeated emphasis of patterns learned from the small real training sample all changed together. The results therefore describe the complete augmentation procedure rather than the isolated effect of any one difference.

10.6 Paired effects, calibration and decision-curve results

Paired intervals were calculated with 2,000 bootstrap resamples of the real evaluation patients. The same resample was used for an augmented fit and its real-only reference, preserving the patient-level pairing. These intervals quantify evaluation-sample variation while treating the trained models and synthetic pools as fixed. Figure 30 summarizes the main paired changes on a common scale where positive values favour augmentation. It shows the increasing diagnostic benefit for XGBoost, the worsening prognostic error for Cox elastic net and the MLP, and changes close to zero for most other comparisons.

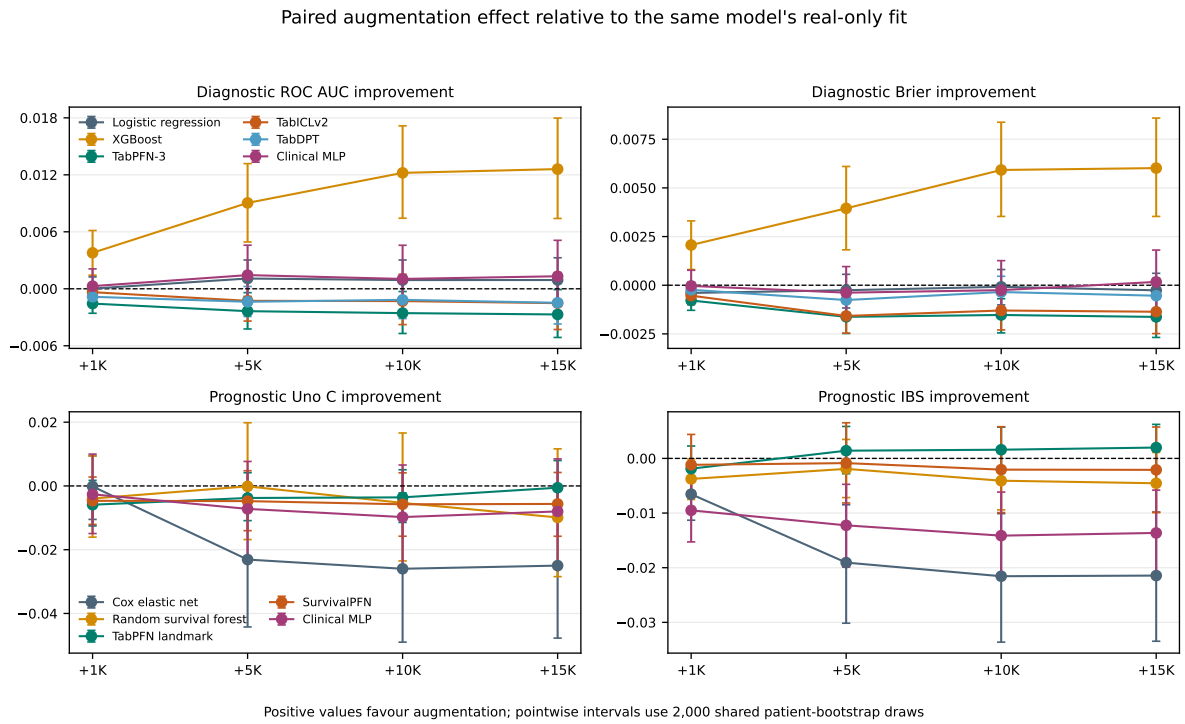


Figure 30: Paired effect of each augmentation size relative to the same model trained on real data alone. Positive values favour augmentation. Intervals are pointwise 95% patient-bootstrap intervals conditional on the fitted models; they are not adjusted for the number of models, outcomes or sizes.

Diagnostic calibration also varied by model (Table 64). A calibration slope of one is ideal. XGBoost's slope moved from 0.731 to 0.878, reducing its absolute distance from one by 0.1479 (0.1184 to 0.1801). In contrast, TabPFN-3 moved from 0.899 to 0.851 and TabICLv2 from 0.910 to 0.868; for both models, the change away from one had an interval excluding zero. Changes for logistic regression, TabDPT and the MLP were inconclusive at `+15K`.

For prognosis, calibration was summarized as the mean absolute difference between predicted survival and the Kaplan–Meier (KM) estimate across three risk groups and the 12-, 24- and 36-month horizons. The mean gap increased from 0.069 to 0.086 for Cox elastic net and from 0.065 to 0.086 for the MLP, remained 0.044 for random survival forest, decreased from 0.054 to 0.049 for TabPFN landmark, and changed from 0.052 to 0.054 for SurvivalPFN. These are descriptive point estimates. The planned bootstrap intervals were not reported because fewer than 1,900 of the 2,000 resamples retained at least five patients at risk in every required risk-group and time-horizon cell.

Diagnostic decision curves were summarized at risk thresholds of 0.10, 0.20 and 0.30 (Table 65). Net benefit balances true-positive detections against false-positive classifications at a chosen threshold. At `+15K`, XGBoost gained approximately 0.006–0.007 net-benefit units at the three thresholds, whereas every other model changed by no more than about 0.0022 in either direction. No bootstrap interval was planned for these summaries, so they provide descriptive evidence only and do not establish clinical utility in another population.

Table 64: Calibration for real-only and maximum augmentation. Diagnostic improvement is the reduction in the absolute distance of the slope from one. Prognostic KM gaps are descriptive because the required number of valid bootstrap resamples was not reached.

Task	Model	Metric	+0	+15K	Paired change [95% interval]
Diagnosis	Logistic regression	Calibration slope	0.960	0.923	-0.0369 [-0.0493, 0.0399]
Diagnosis	XGBoost	Calibration slope	0.731	0.878	0.1479 [0.1184, 0.1801]
Diagnosis	TabPFN-3	Calibration slope	0.899	0.851	-0.0475 [-0.0612, -0.0330]
Diagnosis	TabICLv2	Calibration slope	0.910	0.868	-0.0424 [-0.0584, -0.0229]
Diagnosis	TabDPT	Calibration slope	1.001	0.966	-0.0330 [-0.0463, 0.0464]
Diagnosis	Clinical MLP	Calibration slope	0.893	0.872	-0.0215 [-0.0433, 0.0030]
Prognosis	Cox elastic net	KM mean absolute gap	0.069	0.086	Point estimates only
Prognosis	Random survival forest	KM mean absolute gap	0.044	0.044	Point estimates only
Prognosis	TabPFN landmark	KM mean absolute gap	0.054	0.049	Point estimates only
Prognosis	SurvivalPFN	KM mean absolute gap	0.052	0.054	Point estimates only
Prognosis	Clinical MLP	KM mean absolute gap	0.065	0.086	Point estimates only

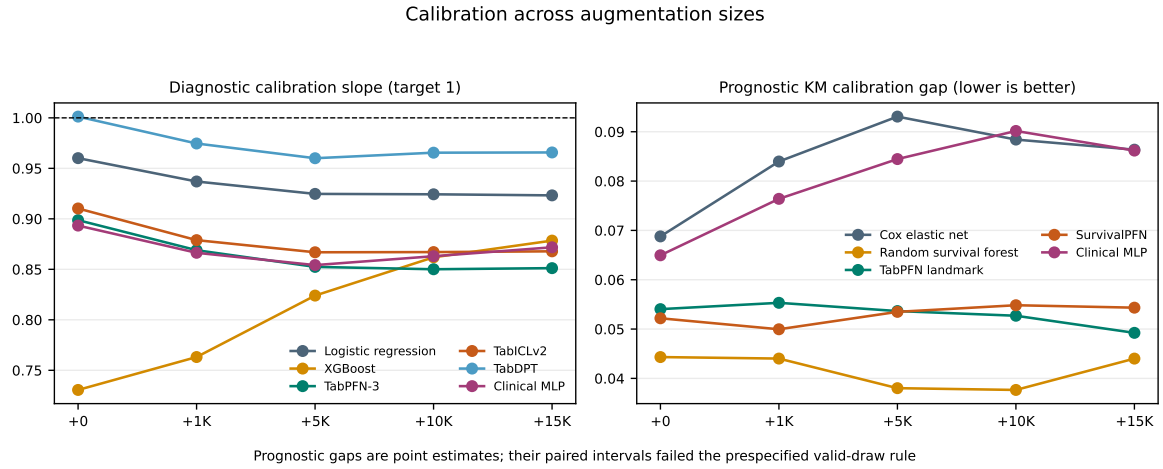


Figure 31: Diagnostic calibration slope and prognostic mean absolute KM calibration gap across augmentation sizes. The target is one for diagnostic slope and zero for prognostic gap. Prognostic curves are descriptive because the planned bootstrap intervals did not meet the valid-resample requirement.

Table 65: Descriptive change in diagnostic net benefit from `+0` to `+15K`. Positive values favour augmentation.

Model	Threshold 0.10	Threshold 0.20	Threshold 0.30
Logistic regression	+0.0011	+0.0001	+0.0022
XGBoost	+0.0057	+0.0060	+0.0071
TabPFN-3	-0.0013	+0.0010	-0.0001
TabICLv2	-0.0016	-0.0012	-0.0005
TabDPT	-0.0013	-0.0014	-0.0011
Clinical MLP	+0.0004	+0.0004	-0.0005

10.7 Grade-4 support, MLP seed sensitivity and computation

Augmentation did not resolve the limited grade-4 support. Grade 4 was absent from every synthetic pool, so its mean prevalence in the combined training data fell from 2.25% to 0.28% for diagnosis and from 6.93% to 0.15% for prognosis between `+0` and `+15K`. Fold-level diagnostic grade-4 metrics were estimable in only 44 of 1,100 model-by-fold conditions. When predictions were combined across the five folds in each repeat, all 220 model-by-repeat conditions could be summarized. At this repeat level, XGBoost's grade-4 AUC increased descriptively from 0.533 to 0.591 and Brier score decreased from 0.319 to 0.270 at `+15K`. However, each repeat contained only 60 unique grade-4 patients, no paired interval was available, and the models received no synthetic grade-4 examples. These estimates are therefore too limited to support a grade-4 benefit. Prognostic grade-4 metrics could not be estimated in any held-out split.

Three MLP initialization seeds were examined as a focused sensitivity analysis (Table 66). Diagnostic AUC changes were small and positive at `+5K` and `+15K` for all three seeds, while Brier-score changes varied in direction. Prognostic IBS increased, indicating worse probability accuracy, for every seed at both sizes. Concordance changes were mostly negative, with one small positive change at `+15K`. The consistent direction of the prognostic IBS result supports the primary-seed finding, although three seeds cannot represent the full range of optimization variability.

Table 66: MLP metric changes from the real-only fit for seeds 20230529, 20230530 and 20230531. Values are descriptive seed-specific changes rather than confidence intervals. Positive values favour AUC and Uno concordance but are unfavourable for Brier score and IBS.

Metric	+5K minus +0	+15K minus +0
Diagnostic ROC AUC	+0.0014 / +0.0025 / +0.0032	+0.0013 / +0.0021 / +0.0014
Diagnostic Brier	+0.0004 / +0.0000 / -0.0004	-0.0002 / +0.0002 / +0.0006
Prognostic Uno C	-0.0072 / -0.0032 / -0.0019	-0.0080 / -0.0038 / +0.0006
Prognostic IBS	+0.0122 / +0.0088 / +0.0094	+0.0136 / +0.0102 / +0.0110

Larger training conditions increased fit-and-prediction time for most models (Table 67). Among the diagnostic models, the three foundation models and the MLP took 5.4–6.5 times longer at `+15K`; for example, the median TabPFN-3 time increased from 2.82 to 16.25 seconds per cross-validation fold. In prognosis, TabPFN landmark increased from 5.81 to 35.56 seconds per split, while SurvivalPFN and the MLP increased by a factor of 1.3. All models assigned to graphics-processing-unit execution completed the largest condition without falling back to central-processing-unit execution. The largest recorded graphics-memory allocation was 2.58 GB for TabICLv2. These measurements describe the present implementation and hardware; they do not provide a controlled speed comparison between algorithms.

10.8 Interpretation and limitations

The main finding is that synthetic augmentation had selective rather than general value. It improved XGBoost's diagnostic discrimination, Brier score, calibration slope and descriptive net benefit. It did not clearly improve diagnostic logistic regression or the clinical MLP, and it slightly worsened some results for TabPFN-3 and TabICLv2. No prognostic model showed a clear overall benefit. In particular, Cox elastic net and the clinical MLP had worse probability

Table 67: Median fit-and-prediction time per evaluation partition. Times exclude synthetic-data generation, bootstrap calculations and report preparation.

Task	Model	+0 (s)	+15K (s)	Ratio
Diagnosis	Logistic regression	0.20	0.35	1.7x
Diagnosis	XGBoost	0.28	0.54	1.9x
Diagnosis	TabPFN-3	2.82	16.25	5.8x
Diagnosis	TabICLv2	1.42	8.78	6.2x
Diagnosis	TabDPT	1.33	8.71	6.5x
Diagnosis	Clinical MLP	1.44	7.76	5.4x
Prognosis	Cox elastic net	0.58	1.43	2.5x
Prognosis	Random survival forest	0.73	4.26	5.8x
Prognosis	TabPFN landmark	5.81	35.56	6.1x
Prognosis	SurvivalPFN	6.80	9.12	1.3x
Prognosis	Clinical MLP	0.64	0.84	1.3x

accuracy as synthetic records came to dominate the training data. TabPFN landmark was comparatively stable, but stability should not be interpreted as improvement.

The uncertainty intervals have several limits. They reflect variation among the held-out real patients while treating the fitted models, selected MLP epoch counts and generated pools as fixed. They therefore do not include uncertainty from retraining the models or generator. The intervals are pointwise and were not adjusted for the number of models, outcomes and augmentation sizes. In addition, increasing the synthetic prefix simultaneously changed sample size, outcome distribution, grade support, categorical encoding, scaling and computational effort. The observed curves cannot identify which of these changes caused a performance difference.

The generated data also omitted grade 4, contained small inconsistencies between linked variables and accounted for 97.9% of the rows in the largest prognostic training conditions. Finally, every result was evaluated within the same Surveillance, Epidemiology, and End Results (SEER) registry cohort. The analysis therefore addresses internal predictive performance and does not establish transportability or clinical utility in an independent population.

Taken together, the results show that synthetic augmentation must be evaluated for the specific model and clinical task in which it will be used. Close agreement in overall feature distributions did not guarantee better discrimination, calibration, probability accuracy or subgroup support. The next section combines these findings with the real-only, transformed-data, foundation-model and nonlinear-head experiments.

11 Integrated Results and Discussion

11.1 How the analyses should be compared

The analyses in this report address related but distinct questions. The reference-study reproduction examines whether the published diagnostic and prognostic procedures can be reconstructed using the available Surveillance, Epidemiology, and End Results (SEER) registry extract. External validation asks whether the resulting reference models retain predictive value when applied unchanged to patients from University Hospital Erlangen (*Universitätsklinikum Erlangen*, abbreviated as UK Erlangen). The transformation analyses examine which predictive and structural properties are retained in TabPFN representations and synthetic datasets. The matched model benchmark compares algorithms trained and evaluated on the same real-patient partitions. Finally, the augmentation analysis asks whether adding synthetic records improves each model relative to its own real-only result.

These results should not be combined into a single ranking. A numerical model comparison is most informative when the models use the same outcome, training patients, test patients, preprocessing and performance measure. Results from the reference reproduction, transformed datasets and matched benchmark differ in one or more of these respects. They can be interpreted together, but their performance values are not directly interchangeable.

Table 68: Purpose and main limitation of each analysis used in this integrated discussion.

Analysis	Evaluation data	What the analysis can show	Main limitation
Reference-study reproduction	Available SEER cohort	Whether the published workflow and performance range can be reconstructed	Diagnosis uses the development cohort; prognosis uses one 70/30 split; the paper's SEER release was unavailable
Reference-model external validation	Independent UK Erlangen hospital patients	Whether the unchanged reproduced models retain predictive value in another cohort	Only 30 diagnostic and 11 prognostic patients
TabPFN transformations	Embeddings of observed patients and complete synthetic datasets	Which predictive and structural properties are retained after transformation	Transformations were created from the complete cohort before later evaluation splits
Matched real-only benchmark and MLP	Held-out real SEER patients	Direct comparison of model families using the same patient partitions	Internal SEER evaluation only
Synthetic augmentation	Held-out real SEER patients	Change within each model after adding training-partition-specific synthetic records	Internal evaluation; generated records are not independent patients

Table 68 separates the five types of evidence used below. A successful reproduction shows that a published workflow can be reconstructed, but does not show how well it will perform in new patients. A transformation may retain predictive information without improving a downstream model. Synthetic records may resemble the real data in their overall distributions without adding independent clinical information. Independent hospital patients provide stronger evidence of transportability, but the UK Erlangen sample is too small for precise estimates or reliable subgroup conclusions. These distinctions guide the integrated interpretation.

11.2 Integrated diagnostic findings

The diagnostic reproduction found substantial predictive signal in the available SEER cohort. When each model was fitted and evaluated in the complete 2,672-patient development cohort, the area under the receiver operating characteristic curve (ROC AUC) was 0.8795 for the fixed 5-variable nomogram, 0.8947 for the 6-variable specification described in the reference-paper text, 0.9028 for the 8-variable data-driven model and 0.7628 for the American Joint Committee on Cancer (AJCC) T/N-stage comparator. The reference study reported an AUC of 0.877 in its SEER development cohort and 0.893 in its external hospital cohort. The similar magnitudes support the reconstruction of the published approach, but the present within-cohort values are not estimates of performance in new patients and were obtained from a different SEER release.

The UK Erlangen analysis provides independent evaluation of the unchanged reference models. In the common 29-patient complete-case set, the 5-variable nomogram achieved an AUC of 0.934 (95% confidence interval 0.809–1.000), the 8-variable model achieved 0.942 (0.837–1.000), and the T/N-stage comparator achieved 0.739 (0.521–0.927). These findings show that the multivariable models retained diagnostic ranking ability in this hospital cohort. However, only 10 patients had the diagnostic outcome, producing wide intervals and unstable calibration and decision-curve estimates. The analysis therefore supports preliminary transportability, not a precise ranking of the models or a validated clinical threshold.

The matched real-patient benchmark provides the most reliable comparison of model families within SEER (Table 69). TabPFN-3, TabICLv2 and TabDPT formed a narrow leading group, with real-only ROC AUC values of 0.902–0.903. Relative to logistic regression, their paired AUC differences were 0.0088 (95% interval 0.0034–0.0145), 0.0075 (0.0021–0.0132) and 0.0074 (0.0016–0.0133), respectively. TabPFN-3 and TabICLv2 also improved average precision by approximately 0.03 and reduced Brier score by 0.0043. TabDPT reduced Brier score by 0.0033, while its average-precision interval included zero. The differences are small, but they consistently favour the three foundation models for diagnostic ranking under this internal evaluation design.

Table 69: Integrated diagnostic results under the matched clinical-variable design. AUC, average precision (AP) and Brier score are real-only estimates. Paired changes compare maximum augmentation (`+15K`) with the same model trained on real data alone (`+0`); positive changes favour augmentation. Intervals are pointwise 95% patient-bootstrap intervals conditional on the fitted models.

Model	AUC	AP	Brier	Δ AUC at +15K	Brier improvement at +15K
Logistic regression	0.895	0.635	0.090	+0.0009 [-0.0013, 0.0033]	-0.0003 [-0.0012, 0.0006]
XGBoost	0.883	0.615	0.096	+0.0126 [0.0074, 0.0180]	+0.0060 [0.0035, 0.0086]
TabPFN-3	0.903	0.665	0.086	-0.0027 [-0.0051, -0.0001]	-0.0016 [-0.0027, -0.0007]
TabICLv2	0.902	0.664	0.086	-0.0015 [-0.0043, 0.0017]	-0.0014 [-0.0025, -0.0003]
TabDPT	0.902	0.652	0.087	-0.0015 [-0.0037, 0.0008]	-0.0005 [-0.0014, 0.0004]
Clinical MLP	0.897	0.652	0.088	+0.0013 [-0.0023, 0.0051]	+0.0002 [-0.0013, 0.0018]

The clinical multilayer perceptron (MLP) reached an AUC of 0.897, an AP of 0.652 and a Brier score of 0.088. Its paired AUC and AP differences from logistic regression included zero. The MLP adds nonlinear modelling capacity without foundation-model pretraining, so this result shows that nonlinearity alone did not reproduce the small diagnostic advantage of the foundation models under the tested protocol. It does not prove that pretraining caused the difference, because the architectures, regularization, optimization and model-specific assumptions also differ.

Synthetic augmentation changed performance selectively. XGBoost, which had the lowest real-only diagnostic AUC, improved from 0.883 to 0.895 and reduced its Brier score from 0.096 to 0.090 at `+15K`; both paired intervals excluded zero. Logistic regression and the MLP showed no clear change. TabPFN-3 had small AUC and Brier-score deteriorations, TabICLv2 had a small Brier-score deterioration, and TabDPT's maximum-size changes were inconclusive. Synthetic augmentation therefore helped one diagnostic estimator but did not act as a general increase in effective sample size.

11.3 Integrated prognostic findings

The reproduced 5-variable prognostic model also retained survival ordering in the held-out 30% SEER subset. Its time-dependent AUCs were 0.886, 0.873 and 0.844 at 12, 24 and 36 months. These results come from one 70/30 split and should not be compared directly with the matched benchmark, which averages five held-out splits and uses additional measures such as Uno's concordance index and the integrated Brier score (IBS).

In the independent UK Erlangen cohort, the primary prognostic nomogram achieved a Harrell concordance index of 0.833 (95% confidence interval 0.625–1.000) in all 11 liver-metastasis-positive patients. In the 10-patient set shared with the T/N-stage model, the corresponding values were 0.824 for the nomogram and 0.676 for T/N stage. Nomogram AUC estimates were 0.954, 0.936 and 0.833 at 12, 24 and 36 months, but every interval was wide. These results are consistent with retained prognostic ranking in independent patients, while remaining too imprecise to establish external performance with confidence.

Under the matched SEER design, discrimination was tightly clustered (Table 70). TabPFN landmark had the highest real-only Uno concordance point estimate at 0.717, followed by random survival forest at 0.712 and Cox elastic net at 0.711. None of the planned paired comparisons established a concordance advantage over its conventional reference. Random survival forest provided the best probability-error result: its IBS of 0.153 improved on Cox elastic net by 0.0071 (95% interval 0.0012–0.0128). It also had the smallest mean absolute Kaplan–Meier (KM) calibration gap, 0.044. Thus an established survival model gave the strongest probability-accuracy result, while no prognostic foundation-model advantage was demonstrated.

Table 70: Integrated prognostic results under the matched clinical-variable design. Uno's concordance index (Uno C), IBS and the mean absolute KM calibration gap are real-only estimates. Paired changes compare `+15K` with `+0`; positive changes favour augmentation. The KM gap averages the 15 estimable split–horizon combinations, each based on three risk groups.

Model	Uno C	IBS	KM gap	Δ Uno C at +15K	IBS improvement at +15K
Cox elastic net	0.711	0.160	0.069	-0.0250 [-0.0477, -0.0010]	-0.0214 [-0.0335, -0.0098]
Random survival forest	0.712	0.153	0.044	-0.0099 [-0.0284, 0.0116]	-0.0045 [-0.0099, 0.0011]
TabPFN landmark	0.717	0.155	0.054	-0.0006 [-0.0080, 0.0080]	+0.0020 [-0.0020, 0.0062]
SurvivalPFN	0.710	0.164	0.052	-0.0056 [-0.0158, 0.0042]	-0.0021 [-0.0098, 0.0057]
Clinical MLP	0.709	0.162	0.065	-0.0080 [-0.0249, 0.0085]	-0.0136 [-0.0216, -0.0058]

The clinical prognostic MLP showed no general advantage over the matched Cox or random-survival-forest models. Its real-only Uno concordance was 0.709 and its IBS was 0.162. Augmentation did not reverse this result. At `+15K`, Cox elastic net lost 0.0250 in Uno concordance and

worsened by 0.0214 on the favourably oriented IBS scale. The MLP's concordance change was inconclusive, but its IBS worsened at every augmentation size and by 0.0136 at `+15K`. Changes for random survival forest, TabPFN landmark and SurvivalPFN included zero for both main paired outcomes at the maximum size.

The prognostic training partitions contained a mean of only 323 real patients. At `+15K`, synthetic records accounted for 97.9% of the combined training rows. This dominance of the generated distribution occurred together with outcome shifts, absent grade-4 support and small inconsistencies between linked variables. These observations are compatible with the lack of prognostic benefit, but they do not identify a single causal explanation.

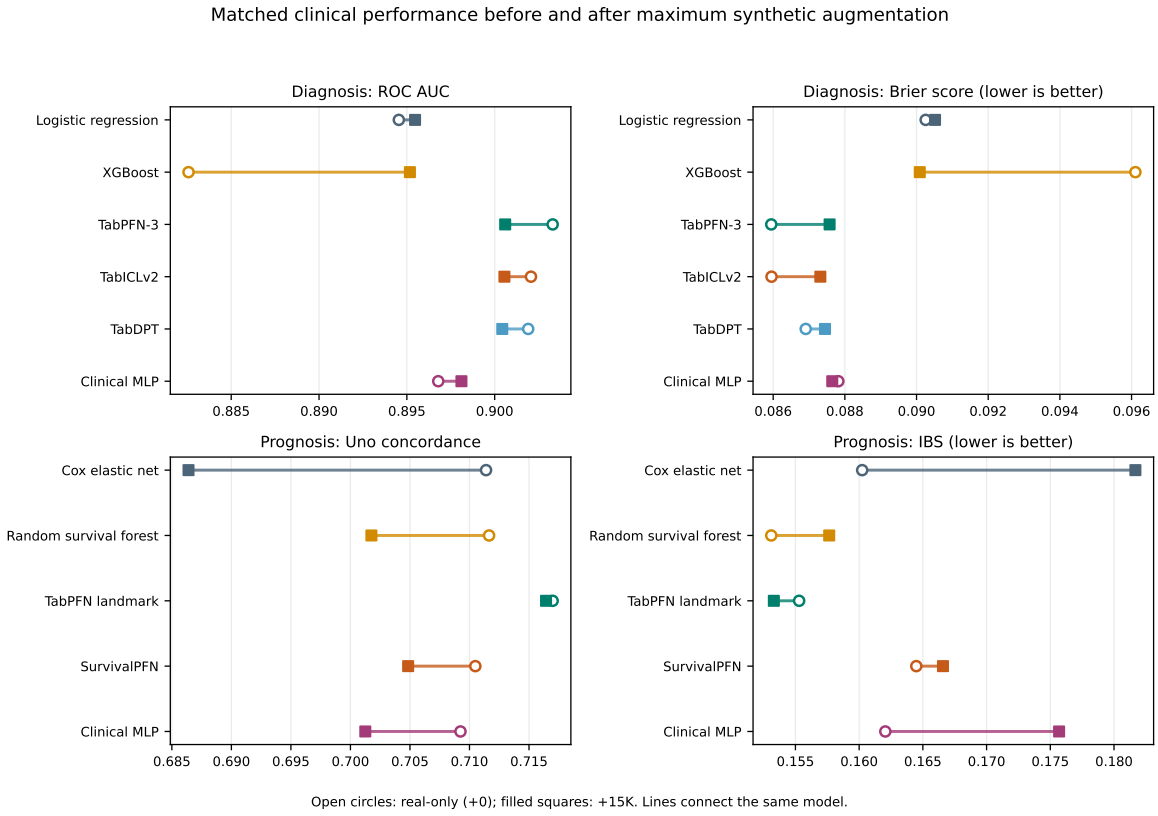


Figure 32: Matched real-only performance (open circles) and performance after adding 15,000 training-partition-specific synthetic records (filled squares). Lines connect results for the same model and show direction, not statistical significance. Movement to the right is favourable for AUC and Uno concordance; movement to the left is favourable for Brier score and IBS. Results from the reference reproduction and full-cohort transformations are excluded because they use different evaluation designs.

11.4 What the transformations preserved

The TabPFN embeddings retained substantial predictive information. A regularized diagnostic model trained on the 512-dimensional representation reached an AUC of 0.900. The prognostic representation produced AUCs of 0.861, 0.858 and 0.843 at 12, 24 and 36 months, close to the corresponding clinical- model values of 0.886, 0.873 and 0.844. Local-neighbour preservation was high: trustworthiness and continuity were approximately 0.95 for diagnosis and 0.84–0.88 for prognosis. Correlations between the complete clinical and embedding distance matrices were more moderate, ranging from 0.400 to 0.558. The embeddings therefore retained local patient

relationships more strongly than the overall geometry of the clinical feature space.

These representation results remain exploratory because the embeddings were created from the complete cohort before the later training and test partitions were defined, and supervised outcomes were used during their construction. Resampling only the prediction model fitted on those representations does not show how an embedding generated exclusively from training patients would perform in unseen patients.

Table 71: Integrated interpretation of the TabPFN transformations and the training-partition-specific augmentation experiment. Performance values remain within their original evaluation designs and should not be treated as a common model ranking.

Analysis	Predictive result	Structural result	Evaluation design	Conclusion
TabPFN embeddings	Diagnostic AUC 0.900; prognostic AUC 0.861/0.858/0.843	High local-neighbour preservation; moderate global distance agreement	Created from the complete cohort with outcomes used in the transformation	Predictive information is retained; performance in unseen patients remains uncertain
Complete synthetic datasets	High AUC on generated diagnostic data; prognostic pipeline completed	Real and synthetic records remain distinguishable; grade 4 is absent; survival and dependence shift	Generator trained on the complete cohort; evaluation used generated data or unequal transfer directions	Suitable for fidelity analysis, not external validation
Training-partition-specific augmentation	Diagnostic XGBoost improves; no general diagnostic or prognostic gain	Grade 4 remains absent; linked-variable and outcome/time shifts remain	Generator uses only real training patients; evaluation uses held-out real patients	Utility depends on the downstream model and task

The synthetic datasets generated from the complete SEER cohorts reproduced many individual-variable distributions and allowed the diagnostic and prognostic pipelines to run. However, a classifier could distinguish real from synthetic records; grade 4 was absent; survival distributions shifted; and several relationships between variables were incompletely preserved. Because the generator learned from all observed patients and evaluation was performed on generated data or with unequal real-to-synthetic directions, these experiments describe fidelity and pipeline behaviour rather than performance in unseen real patients.

The augmentation experiment provides the more direct test of practical utility. For every evaluation partition, synthetic records were generated using only the real training patients, and performance was measured only in the held-out real patients. Its mixed results show why distributional similarity and predictive utility are different properties. Small Jensen–Shannon or standardized Wasserstein distances did not guarantee improved prediction, while XGBoost benefited despite known differences in grade support and outcome prevalence. Utility depended on the task, model, real sample size and proportion of synthetic training records.

11.5 What the model comparisons suggest

For diagnosis, all three pretrained tabular models showed small matched improvements over logistic regression, whereas the clinical MLP did not. This pattern is compatible with useful information from pretraining or context-dependent inference, but the study was not designed to isolate the effect of pretraining. The MLP architecture, optimization procedure and regularization differed from those of the foundation models, and only one prespecified MLP design was tested. The defensible conclusion is therefore empirical: under the tested protocol, adding generic nonlinear capacity did not match the strongest diagnostic results.

Prognosis was more difficult. It combined censoring, multiple time horizons and a substantially smaller sample. The models also represented survival in different ways: Cox regression estimated proportional hazards, random survival forest modelled survival nonlinearly, TabPFN landmark used separate binary predictions at each horizon, and SurvivalPFN used survival-specific pretraining. These differences make small point-estimate gaps difficult to interpret as a general advantage of one model family. The results favour larger, event-rich cohorts and survival-specific model development rather than assuming that gains in binary diagnosis will transfer to prognosis.

11.6 Calibration, subgroup support and clinical meaning

Discrimination does not describe probability accuracy or clinical usefulness on its own. In the real-only diagnostic benchmark, TabDPT had a calibration slope closest to one at 1.001, followed by logistic regression at 0.960. TabPFN-3, TabICLv2 and the MLP had slopes of 0.899, 0.910 and 0.893. Augmentation improved XGBoost's slope from 0.731 to 0.878 but moved TabPFN-3 and TabICLv2 farther from one. In prognosis, random survival forest had the lowest IBS and KM calibration gap despite not having the highest concordance. The model favoured by discrimination was therefore not always the model favoured by calibration or probability error.

Decision-curve results describe net benefit across selected risk thresholds, but they do not identify a clinically validated threshold or demonstrate patient benefit. Rare-group support is an additional limitation. Diagnostic grade-4 performance could be estimated in only 44 of 1,100 fold-level model conditions, although combining predictions across folds allowed all 220 repeat-level conditions to be summarized. No prognostic split supported a grade-4 estimate. Grade 4 was absent from every synthetic pool and became progressively diluted as the synthetic fraction increased. The report cannot therefore support conclusions about equal performance or benefit in this rare group.

11.7 Answers to the project objectives

Table 72 summarizes the findings for the nine objectives defined in Section 1.3. The data, reproduction and dataset-construction objectives were completed. The empirical objectives produced qualified findings: predictive information was retained after transformation, but the embedding and complete-synthetic-dataset results have design limitations; three diagnostic foundation models showed modest matched gains; the tested MLP did not show a general advantage; and synthetic augmentation helped diagnostic XGBoost without improving prognosis.

Table 72: Integrated answers to the nine project objectives.

ID	Objective	Main finding	Interpretation and limitation
O1	SEER data preparation	Diagnostic and prognostic cohorts were constructed with documented eligibility, attrition, coding and outcomes	Completed; registry measurement and predictor-timing limits remain
O2	Reference-study reproduction	The main diagnostic and prognostic workflows were reconstructed, with several results close in magnitude to the publication	The available SEER release differs from the unavailable reference-study dataset
O3	Reference-model external validation	The unchanged reproduced models retained diagnostic and prognostic ranking in UK Erlangen	Independent patients, but only 30 diagnostic and 11 prognostic records
O4	TabPFN-derived datasets	512-variable diagnostic and 1,024-variable prognostic representations and complete synthetic records were created	Dataset construction completed
O5	Transformed-data reproduction	The pipelines ran on embeddings and synthetic datasets and retained substantial predictive ordering	Exploratory because the transformation and evaluation designs differ from held-out real-patient validation
O6	Similarity and synthetic-data fidelity	Local embedding structure was retained, but synthetic grade, survival and dependence were incompletely preserved	Mixed fidelity; grade 4 was not generated
O7	Multi-model benchmark	Three diagnostic foundation models modestly outperformed logistic regression; no prognostic foundation-model advantage was established	Internal SEER results under a matched real-only design
O8	Nonlinear prediction with MLPs	Clinical and embedding MLPs showed no general advantage over the relevant simpler models	One prespecified architecture; embedding comparison remains exploratory
O9	Synthetic augmentation	Diagnostic XGBoost benefited; other diagnostic models did not consistently benefit, and prognosis did not improve	Within-model internal effects; no external evaluation of augmented models

11.8 Overall interpretation

The project completed an end-to-end analysis from SEER cohort construction to reference-model reproduction, external evaluation, tabular foundation models, learned representations, synthetic data, nonlinear MLPs and synthetic-data augmentation. The clearest model-comparison result was a small diagnostic advantage for three tabular foundation models under matched real-patient validation. The clearest augmentation result was more specific: synthetic records improved diagnostic XGBoost, but did not provide a general benefit and slightly worsened some already strong models. No corresponding foundation-model, MLP or augmentation benefit was demonstrated for prognosis.

The overall findings do not support a simple conclusion that more complex models or more generated data necessarily improve clinical prediction. Performance depended on the task, evaluation design, estimator, real sample size and which relationships were preserved by the transformation. Future work should generate representations separately within training partitions, preserve rare clinical categories explicitly, evaluate synthetic data in larger independent cohorts and develop survival-specific foundation models using more event-rich data. The present results remain research findings and do not establish readiness for clinical deployment. The

next section consolidates the study's remaining limitations.

12 Limitations and Scope of Interpretation

12.1 Purpose of this section

This section brings together the limitations that affect the interpretation of the study. They fall into three broad groups: limitations of the available clinical data, limitations of the evaluation designs and limitations on how far the results can be applied beyond the studied patients. These issues do not make the analyses invalid, but they determine which conclusions the evidence can support.

Table 73 summarises the main limitations, their effect on interpretation and the additional evidence needed to address them. The proposed next steps describe future work; they were not part of the present study.

12.2 Data source, cohort selection and measurement

The study is retrospective and uses one release of the Surveillance, Epidemiology, and End Results (SEER) programme. The diagnostic cohort contains 2,672 patients after applying the eligibility and preprocessing procedure based on the reference study. This represents 46.4% of the 5,761 exported records. The largest exclusion removed 1,903 records with at least one unknown required field. Because the analysis uses complete records, the retained patients may differ systematically from those who were excluded. The models describe the available cohort; they cannot correct for patients or clinical information that are absent from it.

Several variables are simplified versions of the underlying clinical information. Stage values from different diagnosis years and assessment types were combined into common categories. Histology and pancreatic site were also grouped, age was top-coded, and some treatment codes combine "no treatment" with "unknown". The data do not establish whether every treatment variable would have been known at the intended prediction time. Associations involving treatment should therefore be interpreted as predictive relationships, not as evidence that a treatment caused an outcome. A model intended for prospective use would need clearly time-stamped predictors available before the prediction is made.

The outcomes are also specific. The diagnostic task predicts liver metastasis recorded at the initial diagnosis; it does not predict metastasis that develops later. The prognostic task evaluates all-cause survival in the 462 patients who already had liver metastasis at diagnosis. It does not estimate cancer-specific survival and does not represent all patients with pancreatic neuroendocrine tumours. Twelve prognostic records have zero recorded survival months. They were retained as recorded, with a positive-time adjustment used only where required by a model. This convention permits the analyses to run but cannot resolve possible uncertainty in the recorded timing.

The SEER release used here is newer than the dataset used in the reference study, because the earlier release was no longer available. Differences between the published and reproduced results may therefore arise from changes in the patients, calendar period, variable definitions or implementation. The work is best understood as a reconstruction of the published methodology on the available SEER data, rather than an exact replication on the original cohort.

Table 73: Main study limitations, their effect on interpretation and the additional evidence needed to address them.

Area	Main limitation	Effect on interpretation	What future work should add
Data source and selection	One retrospective SEER release; substantial exclusion of records with unknown required fields; original reference-study cohort unavailable	The retained cohort may not represent excluded patients, another registry period or another healthcare setting	Larger external cohorts, analysis of missing-data mechanisms and prospectively defined data collection
Measurement and timing	Grouped registry variables, combined stage sources and ambiguous treatment codes; predictor availability at the intended prediction time is not always known	Measurement error and use of information recorded after the prediction time remain possible	Time-stamped clinical data and analyses restricted to predictors known when prediction is made
Task and outcomes	Diagnosis concerns liver metastasis recorded at presentation; prognosis concerns all-cause survival only among patients already liver-positive	Results do not address later metastasis, cancer-specific survival or all patients with pancreatic neuroendocrine tumours	Prospective cohorts with task-specific, clinically adjudicated outcomes
External validation	Only the reproduced reference models were tested externally; UK Erlangen included 30 diagnostic and 11 prognostic patients	Estimates are imprecise and cannot establish subgroup safety, stable thresholds or broad transportability	Larger, temporally distinct and multicentre external validation for every model family
Statistical uncertainty	Most intervals resample evaluation patients while holding fitted models fixed and are not adjusted for multiple comparisons	Model-development variability is omitted; individual intervals do not prove clinical superiority or equivalence	Repetition of the complete development process and prespecified rules for multiplicity and equivalence
Learned representations	Supervised embeddings were created from the complete cohorts; the prognostic time target did not account directly for censoring	Embedding results show retained information but may overstate performance in previously unseen patients	Representations learned only from training patients with methods designed for censored survival data
Synthetic data	Grade 4 was not generated; some linked fields, outcomes and survival times shifted; formal privacy risk was not measured	Generated records are not independent clinical evidence, do not restore rare-group support and are not certified anonymous	Constraint-aware generation plus external utility, fidelity and privacy evaluation
Model scope	One MLP design and specific foundation-model versions and procedures were tested	Results do not cover every neural architecture, checkpoint, training dataset or survival-specific model	Broader prespecified model development and external comparison using the same evaluation design
Subgroups and clinical use	Brain metastasis and grade 4 are sparse; no clinical threshold or prospective impact study was completed	Rare-group performance, fairness, patient benefit and deployment readiness remain unknown	Subgroup-enriched cohorts, predefined clinical thresholds and prospective impact evaluation

12.3 External validation and statistical uncertainty

The reproduced reference models were evaluated without modification in the independent University Hospital Erlangen (UK Erlangen) cohort. This is an important external test, but the cohort is small. It contains 30 patients for the diagnostic task and 11 patients with liver metastasis for the prognostic task. The four diagnostic models could be compared in a common set of 29 patients because one patient had missing nodal stage. The prognostic nomogram could be evaluated in all 11 patients, whereas its comparison with the tumour- and nodal-stage model used 10. Confidence intervals were consequently wide, survival-risk groups contained only three to seven patients, and the prognostic sample was too small for decision-curve analysis. These results show whether the unchanged models can be applied to the available hospital data, but they cannot establish stable clinical thresholds, subgroup safety or performance across different hospitals.

The foundation-model, multilayer perceptron (MLP), embedding and synthetic-data experiments were evaluated only within SEER. Repeated partitions reduce dependence on one arbitrary train–test division, but all partitions still come from the same registry release. They do not replace evaluation in patients from a different time period, institution or country.

The report also contains several evaluation designs that answer different questions. The reference-study reconstruction, external evaluation, complete- cohort embedding analyses, synthetic-cohort analyses and repeated held-out SEER comparisons do not use the same patients or the same validation procedure. Their performance values should therefore be interpreted within their own designs and should not be combined into a single ranking of all models.

Most 95% confidence intervals were calculated by resampling the evaluated patients while keeping the fitted models unchanged. They quantify uncertainty associated with the available evaluation patients, but not the additional variation that could arise from repeating feature selection, model fitting, representation learning, synthetic-data generation or foundation-model pretraining. Most intervals are also pointwise and were not adjusted for the number of models, metrics and augmentation sizes examined. An interval that excludes zero supports a difference for that particular comparison; it does not by itself establish clinical superiority. Likewise, an interval that includes zero does not prove that two methods are equivalent. The three MLP initialisations provide a limited sensitivity check rather than a complete description of optimisation variability.

Some secondary results have less statistical support. The planned confidence intervals for the prognostic Kaplan–Meier calibration comparisons in the augmentation experiment could not be reported because too few bootstrap samples retained adequate numbers of patients at risk. Decision-curve results were descriptive and no risk threshold was prospectively selected for clinical action. The study therefore does not demonstrate that using any model would improve patient outcomes.

12.4 Learned representations and synthetic data

The Tabular Prior-Data Fitted Network (TabPFN) embeddings were created from each complete task cohort using outcome information. A patient's own outcome was excluded when its representation was generated, but the transformation itself was not rebuilt separately within

each later training partition. The embedding analyses show that the representations retain predictive information, but they do not establish how an embedding learned only from training patients would perform in previously unseen patients. The prognostic time representation also used observed follow-up time as a regression target and did not model censoring directly, which further limits its interpretation as a survival representation.

The synthetic cohorts generated from the complete SEER datasets have the same dependency on the full observed cohort. They are useful for studying whether distributions and modelling pipelines can be reproduced, but they are not independent validation data. In the augmentation experiment, this dependency was reduced by training a new generator only on the real training patients in each evaluation partition and testing the models only on held-out real patients. Even under this stronger design, generated records do not add new clinical observations. At the largest prognostic augmentation size, synthetic records formed 97.9% of the combined training data, so the fitted model was dominated by the generator's approximation of a small real cohort.

The fidelity results identify clinically relevant differences between real and synthetic records. Grade 4 was absent from every generated dataset, some continuous tumour sizes disagreed with their recorded size groups, and outcome frequency, survival time and relationships between variables were not always preserved. Similar distributions for individual variables therefore do not guarantee preservation of the combinations that matter to a prediction model. Conversely, the improvement observed for diagnostic XGBoost does not establish that the synthetic data were faithful in every respect.

No formal privacy analysis was performed. The absence of exact duplicate rows and the use of new identifiers are useful integrity checks, but they do not measure whether information about individual patients could be inferred from the generated data. The synthetic records should therefore not be considered anonymous clinical substitutes or released on the basis of this study alone.

12.5 Model scope, subgroups and clinical applicability

The evaluated model set is substantial but not exhaustive. The MLP experiment tested one prespecified architecture and training procedure; it was not a search over all possible neural networks. The foundation-model results apply to the specific model versions, preprocessing choices and inference procedures used in this study. Their pretraining datasets were not independently reconstructed, so the analysis cannot determine how much of the observed performance arose from pretraining or rule out overlap with similar data sources. The absence of a prognostic advantage in this study does not prove that all future survival-focused foundation models will perform similarly.

Rare clinical groups remain particularly uncertain. The prognostic cohort contains only three patients with brain metastasis and 33 with grade 4. The diagnostic cohort contains 60 patients with grade 4, but many individual evaluation partitions had too few such patients for a stable estimate. No prognostic partition met the prespecified support requirement for grade-4 performance, and synthetic augmentation could not improve this because grade 4 was absent from the generated data. A missing subgroup estimate is not evidence of equal performance, and strong overall results do not establish fairness or safety in rare groups.

Finally, the study is predictive rather than causal. It does not estimate the effect of treatment, define a clinical intervention, compare healthcare workflows or measure prospective patient benefit. Discrimination, calibration, prediction error and decision curves are useful components of model evaluation, but they are not sufficient for deployment. Clinical use would require larger external cohorts, validation across time and centres, predefined decision thresholds, assessment of rare groups and a prospective study of clinical impact.

12.6 Implications for the conclusions

Within these limits, the matched internal comparisons support small diagnostic advantages for the three evaluated tabular foundation models over logistic regression. They also support a specific diagnostic augmentation benefit for XGBoost. The analyses did not establish a general advantage for the tested MLP, synthetic augmentation or foundation models in prognosis. The independent Erlangen results provide encouraging evidence that the reproduced reference models retained predictive ranking, but the sample is too small to determine their performance precisely.

The study does not establish external performance for the foundation-model and synthetic-data experiments, causal benefit, formal privacy, reliable performance in rare subgroups or readiness for clinical deployment. Future work should evaluate all model families in larger and temporally distinct multicentre cohorts, learn every representation and synthetic-data generator using training patients only, recruit enough patients from rare clinical groups, repeat the complete model-development process when estimating uncertainty and assess both privacy risk and prospective clinical benefit.

13 Conclusions and Future Work

13.1 Answer to the central research question

This project examined whether published diagnostic and prognostic workflows for liver metastasis in pancreatic neuroendocrine tumours could be reconstructed using the available Surveillance, Epidemiology, and End Results (SEER) data. It also asked whether Tabular Prior-Data Fitted Network (TabPFN) representations, synthetic data, multilayer perceptrons (MLPs) and other tabular foundation models could improve prediction.

The reference workflows could be reconstructed and produced strong diagnostic discrimination and meaningful prognostic risk ordering in the available SEER cohorts. Exact replication of the published results was not possible because the original SEER release and external cohorts were no longer available. The independent University Hospital Erlangen (UK Erlangen) cohort provided an external evaluation of the reproduced reference models. Their predictive ranking remained encouraging, but the cohort of 30 diagnostic and 11 prognostic patients was too small to estimate performance precisely or support claims of clinical utility.

The additional modelling experiments produced a more selective answer. Three tabular foundation models achieved small diagnostic improvements over logistic regression when they were evaluated on the same SEER test patients. The tested MLP did not provide a general improvement, and none of the foundation models or the MLP improved prognostic discrimination reliably. Synthetic augmentation benefited diagnostic XGBoost, but the improvement did not extend consistently to the other diagnostic models or to prognosis.

The central conclusion is therefore that foundation models and synthetic data can provide useful gains in specific settings, but their benefit is not automatic. Performance depends on the prediction task, the model, the amount and quality of real data, the evaluation design and the clinical relationships preserved by the transformation. Generated records and more complex models did not replace the need for appropriate survival methods or independent patients.

13.2 Main scientific conclusions

Reconstruction of the reference study. The eligibility criteria and preprocessing procedure based on the reference study produced a diagnostic cohort of 2,672 patients and a prognostic cohort of 462 patients with liver metastasis. The reconstructed diagnostic and prognostic analyses recovered the main predictive patterns reported in the publication. Numerical differences remain expected because the available SEER release, patients and some variable definitions differ from those of the original study.

External evaluation of the reproduced models. In the common set of 29 UK Erlangen diagnostic patients, the primary five-variable nomogram achieved an area under the receiver operating characteristic curve (ROC AUC) of 0.934, with a 95% confidence interval from 0.809 to 1.000. In all 11 prognostic patients, the primary five-variable Cox nomogram achieved a Harrell concordance index of 0.833, with a confidence interval from 0.625 to 1.000. These results show that the unchanged SEER-trained models retained predictive ranking in the available hospital cohort. The wide intervals, small calibration groups and survival groups of only four

and seven patients prevent conclusions about stable performance or clinical benefit.

Diagnostic model comparison. TabPFN-3, TabICLv2 and TabDPT achieved ROC AUCs of 0.902–0.903 in the repeated SEER evaluation. Their improvements over logistic regression ranged from 0.0074 to 0.0088 when the models were compared on the same patients. These are small internal gains rather than evidence of clinically important superiority. The clinical MLP performed competitively but did not improve consistently over the simpler comparison models. Adding a generic nonlinear prediction model was therefore not sufficient to reproduce the strongest diagnostic results.

Prognostic model comparison. No foundation model or MLP showed a reliable concordance improvement in prognosis. Random survival forest achieved the lowest integrated Brier score (IBS) of 0.153 and improved IBS by 0.0071 relative to Cox elastic net when both models were evaluated on the same patients. This modest probability-error advantage favoured an established survival method. More generally, the prognostic results show that improvements in binary diagnosis should not be assumed to transfer to a smaller, censored survival task.

Representations and synthetic data. The TabPFN embeddings retained predictive information and much of the local patient-neighbourhood structure. However, they were created from the complete cohorts using outcome information, so their downstream results remain exploratory. The complete synthetic datasets reproduced many individual variable distributions and allowed the diagnostic and prognostic pipelines to run, but grade 4 was not generated, survival distributions shifted and several relationships between clinical variables were not fully preserved. These datasets are useful for studying representation and fidelity; they are not independent clinical evidence.

Synthetic augmentation. For each evaluation partition, synthetic records were generated using only the real training patients, and performance was measured only in held-out real patients. At the largest augmentation size of 15,000 records, diagnostic XGBoost improved ROC AUC by 0.0126 and reduced Brier score by 0.0060 compared with training on real data alone. Other diagnostic models did not improve consistently, and prognostic augmentation provided no benefit. Some strong models performed worse as synthetic records formed a larger share of the training data. Synthetic sample size should therefore not be interpreted as an equivalent increase in the number of observed patients.

13.3 Contributions of the project

Within the limitations described in Section 12, the project makes five main contributions:

C1. It constructs clearly defined diagnostic and prognostic cohorts from the SEER registry using eligibility, coding and preprocessing decisions based on the reference study.

C2. It reconstructs the reference diagnostic and prognostic workflows on the closest available SEER population and evaluates the resulting models without modification in an independent UK Erlangen cohort.

C3. It creates and evaluates TabPFN embeddings and complete synthetic datasets for predictive information, structural similarity, category support and preservation of clinically relevant relationships.

C4. It compares conventional models, tabular foundation models and MLPs on the same real-patient evaluation partitions, enabling direct comparison between model families.

C5. It tests synthetic augmentation by generating records only from the real training patients and evaluating models in held-out real patients, separating practical utility from descriptive similarity.

These contributions do not constitute a deployed clinical system, a causal analysis of treatment effects or a formal privacy guarantee.

13.4 Priorities for future research

Future work should first strengthen the clinical evidence.

P1. Larger external validation. Evaluate the most promising approaches in larger multicentre and temporally distinct cohorts, with outcomes and predictor timing specified before the models are applied.

P2. Training-only transformations and complete uncertainty. Learn embeddings, select variables, train synthetic-data generators and calibrate predictions using training patients only. Repeat the complete development process to measure both construction and evaluation uncertainty.

P3. Clinically constrained synthetic data. Preserve rare categories such as grade 4, enforce agreement between linked fields and use methods designed for censored outcomes. Evaluate utility, fidelity and privacy risk together, preferably in independent data.

P4. Survival-specific foundation models. Develop and test models that account for censoring and produce valid survival probabilities across time. Use larger cohorts with more observed deaths and retain strong conventional survival methods as comparisons.

P5. Prospective clinical evaluation. After adequate external validation, define thresholds in advance, examine calibration and subgroup performance, and test whether model use improves clinical decisions or patient-relevant outcomes.

13.5 Final conclusion

Modern tabular foundation models and synthetic data can support rare-tumour registry research, but greater complexity does not guarantee clinical value. The strongest findings are small internal diagnostic gains, a selective augmentation benefit for XGBoost and encouraging but imprecise external results for the reproduced reference models. No general prognostic, MLP or augmentation advantage was demonstrated. Larger independent clinical cohorts are now essential; until then, the models and generated records should remain research tools rather than validated support for clinical decisions.

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